

Measuring Information Frictions in Migration Decisions: A Revealed-Preference Approach

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Abstract

We investigate the role of information frictions in migration. Using novel moment inequalities and data on internal migration in Brazil, we estimate worker preferences and migration costs while allowing for unobserved worker-specific information sets. We find that common estimation procedures overestimate migration costs and underestimate the importance of expected wages in migration decisions. Model specification tests indicate that workers have limited and heterogeneous information on location-specific wages. According to our estimated model, workers' limited wage information plays a quantitatively important role in reducing migration flows and worker welfare and in limiting the effect of policies that reduce migration costs.

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1 Introduction

Migration is among the most effective ways for workers to improve their economic conditions; yet even within countries, migration rates are low. Understanding why is central to assessing the effects of policies aimed at improving the spatial allocation of workers. Spatial models use observed mobility patterns to identify workers' responsiveness to destination payoffs and the migration frictions they face. Such inferences, however, depend on what models assume workers know about opportunities elsewhere. When information is imperfect or heterogeneous across workers, standard procedures conflate information frictions with other sources of immobility, distorting not only the estimated migration costs but also the implied wage elasticities, welfare estimates, and the predicted effects of policies.

Is the main obstacle to migration high moving costs or workers' limited information about payoffs elsewhere? Disentangling these two channels is difficult because researchers rarely observe what workers know. Yet, they operate differently: moving costs reduce the actual payoff to migration, whereas limited information leaves the payoff unchanged but distorts workers' choices. Increasing access to information can thus improve workers' location decisions and, in doing so, alter the effects of policies that reduce migration costs.

We introduce a moment inequality procedure to identify the role of information frictions and migration costs in workers' location choices. Crucially, our procedure is valid even if workers' information differs in ways unobserved to the researcher. We apply it to data on Brazil to answer four questions. What do workers know about wages in different locations? How does allowing for unobserved workers' information change what observed location choices imply about how workers trade off expected wage gains against migration costs when choosing their location? How would workers' location choices change if their information changed? How does workers' information mediate the impact of changes in migration costs?

We obtain four main results. First, workers generally only have coarse information on location-specific wages, but information is more precise for nearby markets, for markets connected by larger past migration flows, and for high-population or high-internet-access origins and destinations. Second, accounting for workers' limited information changes the estimates of the model parameters: our estimates of the migration elasticity to expected wages are three times larger than those obtained using common estimation procedures, which place stronger assumptions on workers' information, whereas our migration cost estimates are 29% lower on average. Third, migration rates would increase significantly if workers had full information on location-specific wages. Fourth, models that assume workers' information is better than it truly is overestimate the welfare gains from reductions in migration costs.

Our analysis is based on a static model that incorporates expectations about wages, mi-

gration costs, amenities, and prices, as well as idiosyncratic preferences, as drivers of workers' location choices.¹ We impose no restriction on wage expectations beyond assuming these are rational; as a result, wage expectations may vary across workers with different information sets, which we treat as unobserved to the researcher and potentially heterogeneous across workers. Conversely, for each destination, we assume expectations about migration costs, amenities, and prices are common across workers sharing a prior location of residence, but allow them to vary flexibly across workers' prior locations, and impose no restriction on their rationality. Thus, while wage expectations are potentially worker-specific, expected migration costs, amenities, and prices are captured by origin-by-destination fixed effects.² A second source of unobserved heterogeneity in worker choices is the presence of idiosyncratic preferences. As in workhorse migration models (Caliendo et al., 2021; Morten and Oliveira, 2024) and models of sectoral or occupational choice (Dix-Carneiro, 2014; Traiberman, 2019), we assume these are independent across choices and follow an extreme value distribution.

Our modeling of expected migration costs, amenities, and prices as captured by origin-by-destination fixed effects implies that the number of preference parameters increases in the square of the number of locations considered in the analysis. When studying migration decisions, workers' feasible set is often large and, thus, the number of fixed effects to estimate is also large. Estimating high-dimensional parameter vectors using moment inequalities is computationally challenging when using standard procedures. We introduce a moment inequality procedure to calculate confidence intervals on each parameter in models featuring potentially large choice sets, choice-specific fixed effects, and information sets that may vary between any two agents in ways unobserved to the researcher.

Key for our procedure is a new type of moment inequality that we call a *bounding* inequality. To derive it, we compare workers' expected utility in any two locations in their choice set, obtaining as a result a conditional moment *equality* that depends on a concave function of the worker's expected utility difference in those two locations.³ The conditioning set in this moment equality is a covariate vector assumed to belong to the worker's information set. From this equality, we derive moment *inequalities* by bounding the concave function from above by its tangent at any point. Importantly, the resulting inequalities are linear in the worker's expected utility difference between the two locations. We then substitute workers' unobserved wage expectations with the ex post realized wages. While this introduces workers' expectational errors in the moment, the rational expectations assumption implies these

¹In Appendix F, we extend our analysis to models with sunk migration costs and forward-looking workers.

²Our model may be estimated separately for worker populations defined by gender, race, or education level, thus allowing expected amenities, migration costs, and prices to vary flexibly across those populations.

³To derive our inequalities, we require the relative probability of choosing the two locations to be convex in the worker's expected utility difference between them. This requirement holds in multinomial logit models, but also in nested logit models if the locations being compared belong to the same nest.

are mean zero conditional on any variable in the worker’s information set. This property of expectational errors, combined with the linearity of the inequality, leaves the inequality unaffected. The resulting moment inequality thus depends on the difference in the fixed effects between the two locations being compared as well as on the utility difference that arises from any wage variation between the two locations.

We use our bounding inequality in a novel two-step estimation procedure. The first step provides bounds on the wage preference parameter. These are computed by combining the bounding inequality described above for pairs of workers that share the same origin location but have distinct observed wages in any given destination; e.g., because they are employed in different sectors. The resulting inequality does not depend on workers’ expected migration costs, amenities, and prices, which are differenced out when comparing the utilities of two workers of the same origin, but it depends on the wage difference between the two workers. This inequality can thus be used to compute a confidence interval for the wage preference parameter. In the second step, we bound *one at a time* each of the origin-by-destination fixed effects that capture workers’ expected migration costs, amenities, and prices. Thus, instead of estimating a joint confidence set for all fixed effects, which is infeasible in settings with many choices, we estimate separate confidence intervals for each fixed effect. To compute these, we use the bounds on the wage coefficient estimated in the first step and, in the second step, combine the bounding inequalities described above with the type of odds-based moment inequalities introduced in [Dickstein et al. \(2023\)](#).

We show theoretically that our inequalities provide bounds on all parameters when the researcher correctly specifies a subset of workers’ information sets, and point identify them when this subset coincides with the true information sets. Point identification is thus achieved precisely when maximum likelihood estimators are consistent, implying no loss of identification power may be incurred in this case when using our inequalities. When the researcher only observes a proper subset of workers’ information sets, we show in simulations that the maximum likelihood estimates are not only biased but also often outside of the bounds defined by our inequalities. Our simulations also illustrate that, when the researcher misspecifies the content of the worker’s information set by assuming that a variable belongs to it when it truly does not, the identified set defined by our moment inequalities may be empty. We use this result to test how accurate workers’ wage information is.⁴

We employ our estimator to study internal migration in Brazil. We use data from the *Relação Anual de Informações Sociais* (RAIS), which records the wage, the sector, and region of work of all formal workers. We estimate our model for the population of white male

⁴In some misspecified models, the identified set is non-empty while not including the true parameter value. See [Molinari \(2020\)](#) for a discussion of this phenomenon, and [Andrews and Kwon \(2024\)](#) and [Kaido and Molinari \(2024\)](#) for inference procedures in partially identified models robust to misspecification.

workers aged 25-64 with at least a high school degree. These are, respectively, the largest race, gender, and education categories in RAIS. We define a labor market as a sector-region pair, and study the information workers in our population have about market-by-period wage shifters. These shifters account for all demand and supply factors having a common impact on the wages of all workers in our population in a market and period. To estimate them, we regress wages on sector-by-region-by-period fixed effects (which equal our shifters of interest) while controlling for worker-by-sector fixed effects (which account for unobserved worker cross-sectoral comparative advantage) and time-varying worker characteristics (e.g., sector-specific experience) with sector-specific coefficients. With those shifters in hand, we estimate the wage preference parameter using cross-sectoral variation in the probability that workers in a given origin choose to migrate to a given destination. Because identification comes from comparing migration rates across sectors within an origin–destination pair, unobserved destination-specific amenities, prices, and migration costs are not a source of potential omitted variable bias in our estimation of the wage preference parameter. This contrasts with the standard approach in the migration literature, which relies on cross-destination variation in total migration probabilities.

Our analysis yields four conclusions. First, workers face substantial information frictions. We reject the assumption that workers have perfect information about prevailing wages in every labor market. Furthermore, when exploring how finely workers can classify markets on the basis of the previous year’s wage shifters, we conclude they can only classify markets into four bins. That is, we cannot reject that workers can classify each market as being in the top 25% by its previous year’s sector-by-region wage shifter, in the 50-75% bracket, in the 25-50% bracket, or in the bottom 25%, but we reject that *all* workers can classify *all* markets into finer partitions. We also find workers’ wage information is heterogeneous and that geography plays a key role in driving that heterogeneity. Specifically, we cannot reject that information is more precise for nearby markets, for destinations linked to the origin by larger past migration flows, and for locations with larger populations or better internet access.

Second, relative to our moment inequalities, estimators common in the migration literature yield smaller estimates of the migration wage elasticity and larger migration cost estimates. Specifically, our approach yields a 95% confidence interval for the elasticity of migration to expected wages centered at 1.5, and does not include the Poisson Pseudo-Maximum Likelihood estimate, which is around 0.5. In addition, our migration cost estimates (measured in utility terms) are centered around values 29% lower than the PPML estimates. Thus, in our setting, standard assumptions on the worker information set drive the researcher to

overestimate the role that non-wage factors play in workers’ location choices.⁵

Third, we quantify workers’ individual welfare gains from improved wage information. Under the preferred quartile-information specification, the 95% confidence interval for the welfare gain from giving workers perfect wage information is [1.7%, 3.9%]. Importantly, this welfare change is partly driven by an increase in migration. When perfectly informed about wages, the average worker changes locations in two consecutive years with probability between 8.9 and 15.4%; when they can only discern quartiles of lagged wage shifters, this probability is between 4.8 and 8.3%.

Fourth, migration costs and information frictions interact in rich ways to determine workers’ migration rates. As a result, model-implied welfare gains from reductions in migration costs are sensitive to the researcher’s assumptions on worker information. The welfare gain from a 10% decline in migration costs ranges from 4.3% to 6.7% if the worker is fully informed about current sector-by-region wage shifters, but only from 2.6% to 4.2% under the preferred quartile-information specification.

Our paper relates to three strands of literature. First, it relates to work studying labor mobility across locations, sectors, or occupations. Our static model incorporates choice-specific idiosyncratic preferences and fixed costs as in [Tombe and Zhu \(2019\)](#) or [Morten and Oliveira \(2024\)](#). In the dynamic extension of our model, we allow for forward-looking workers and one-time switching costs as in [Kennan and Walker \(2011\)](#), [Dix-Carneiro \(2014\)](#), or [Traiberman \(2019\)](#). Our contribution is to show how to estimate static and dynamic migration models without fully specifying workers’ information, and to quantify the impact on model estimates and counterfactual predictions of misspecifying workers’ information sets. However, our analysis has two limitations: we do not allow workers’ location choices to depend on unobserved wage determinants and our counterfactual predictions are partial equilibrium.⁶

Second, we contribute to a literature on information frictions. Recent work has used randomized or natural experiments to evaluate the impact of workers’ information on their location choices; e.g., [Bryan et al. \(2014\)](#), [Bergman et al. \(2020, 2023\)](#), [Wilson \(2021\)](#), or [Baseler \(2023\)](#). Related work combines elicited beliefs with information interventions to study the impact of information on other economic decisions such as college choice; e.g., [Zafar \(2011\)](#) or [Wiswall and Zafar \(2015a,b\)](#). In the absence of exogenous shifts in information, structural work has modeled information acquisition explicitly: [Kaplan and Schulhofer-Wohl \(2017\)](#) introduce Bayesian learning about location characteristics, [Porcher \(2022\)](#) endogenizes this

⁵The PPML estimator we implement is consistent if all workers employed in the same sector have a common information set. The difference between the PPML and moment inequality estimates is thus consistent with our finding that information sets are heterogeneous by worker location.

⁶Additionally, our model maintains the assumption that workers’ expectations are rational; see [Moll \(2025\)](#) for a critical discussion of this assumption in the context of heterogeneous-agent macroeconomic models.

through rational inattention, and [Fan et al. \(2023\)](#) incorporate heterogeneous expectations into a model à la [Artuç et al. \(2010\)](#) or [Caliendo et al. \(2019\)](#). Our contribution is to measure information frictions while neither observing exogenous shifters of agents’ information sets nor imposing parametric restrictions on the stochastic process generating them.

Finally, our paper relates to studies using choice data to identify agents’ preferences when their expectations of choice characteristics are rational but unobserved. In the absence of measures of agents’ expectations ([Manski, 2004](#)), it is common to assume that the researcher observes agents’ full information sets ([Manski, 1991](#)). A recent approach allows the content of these sets to be partly unobserved by the researcher, but rules out heterogeneity in such content between agents of the same observable type ([Traiberman, 2019](#)). Building on [Pakes \(2010\)](#), [Ho and Pakes \(2014\)](#), and [Pakes et al. \(2015\)](#), we allow information sets to vary across agents in unobservable ways, and use moment inequalities to partially identify agents’ preferences in models, static or dynamic, with large choice sets and choice-specific fixed effects. We combine our bounding inequality with the odds-based inequality in [Dickstein and Morales \(2018\)](#) and [Dickstein et al. \(2023\)](#) to obtain tighter bounds than if each of these two inequalities were used separately. When studying dynamic models, we combine the bounding inequality with the Euler approach in [Morales et al. \(2019\)](#).

The paper is organized as follows. Section 2 presents a model of worker location choices. Section 3 describes our moment inequality estimator, and Section 4 illustrates its properties using simulated data. Section 5 discusses our empirical application. Section 6 concludes.

2 Model of Migration with Incomplete Information

We model the choice over locations of individuals in a *population* defined as, e.g., workers of a demographic group sharing an origin location in a given year. Within each population, we partition workers into S observable *types*, which correspond to sectors in our empirical application. For expositional clarity, we treat types as fixed here, but we allow in our empirical application for workers to determine their type jointly with their location; see Section 5.2. We index types by s or r and workers within a type by i or j . Defining a dummy variable y_{is}^l that equals one if worker i of type s chooses location l , we assume

$$y_{is}^l \equiv \mathbb{1}\{l = \operatorname{argmax}_{l'=1,\dots,L} \mathbb{E}[\mathcal{U}_{is}^{l'} | \mathcal{J}_{is}]\} \quad \text{for } l = 1, \dots, L, \quad (1)$$

where $\mathbb{1}\{A\}$ is an indicator function that equals 1 if A is true, $\mathcal{U}_{is}^l \in \mathbb{R}$ denotes the worker’s utility of choosing l , $\mathcal{J}_{is} \in \mathbb{R}^{d_{is}}$ with $d_{is} \geq 0$ is the worker’s information set, and $\mathbb{E}[\cdot | \mathcal{J}_{is}]$ is a

conditional expectation operator reflecting the worker's beliefs.⁷

We impose five assumptions on workers' expected utilities. First, workers' expectations are rational; i.e., for any $\mathcal{X}_{is} \in \mathbb{R}$, denoting by $F(\cdot|\mathcal{J}_{is})$ the cumulative distribution function of $\mathcal{X}_{is}$ conditional on $\mathcal{J}_{is}$, it holds that

$$\mathbb{E}[\mathcal{X}_{is}|\mathcal{J}_{is}] = \int_x x dF(x|\mathcal{J}_{is}). \quad (2)$$

Second, the utility of choosing location l for worker i of type s is

$$\mathcal{U}_{is}^l = u_{is}^l + \varepsilon_{is}^l, \quad (3a)$$

$$u_{is}^l = \kappa^l + \alpha w_{is}^l, \quad (3b)$$

where w_{is}^l is the natural logarithm of the nominal wage that worker i of type s would earn if they chose location l and α captures the relative importance of wages in workers' utility. The terms κ^l and ε_{is}^l are the common and idiosyncratic components of all other determinants of utility. When a population is defined by a demographic group, an origin location, and year (as in our application), the term κ^l flexibly captures group-by-origin-by-destination-by-year non-wage components affecting the utility of choosing location l . This term thus accounts for destination-by-year amenities and (log) prices as well as bilateral migration costs that vary by year. For simplicity, we henceforth refer to κ^l as location l 's *amenity*, though it has this broader interpretation. Equation (3) illustrates the role types play in our model: while the log wage w_{is}^l is allowed to vary by type within a location, the wage coefficient α and the amenity component κ^l are not.

Third, defining $\varepsilon_{is} = (\varepsilon_{is}^1, \dots, \varepsilon_{is}^L)$, we assume that

$$\varepsilon_{is} \subseteq \mathcal{J}_{is}, \quad (4)$$

where, for vectors $\mathcal{X}$ and $\mathcal{X}'$, we use $\mathcal{X} \subseteq \mathcal{X}'$ to indicate that the distribution of $\mathcal{X}$ conditional on $\mathcal{X}'$ is degenerate. Equation (4) imposes that, when making their location choice, worker i of type s knows ε_{is} . It does not restrict which other variables belong to $\mathcal{J}_{is}$.

Fourth, for worker i of type s , worker j of a type $r \neq s$, and locations l and l' , defining $\Delta w_{is}^{ll'} \equiv w_{is}^l - w_{is}^{l'}$, it holds

$$\mathbb{E}[\Delta w_{is}^{ll'}|\mathcal{J}_{is}, \mathcal{J}_{jr}] = \mathbb{E}[\Delta w_{is}^{ll'}|\mathcal{J}_{is}] = \mathbb{E}[\Delta w_s^{ll'}|\mathcal{J}_{is}] = \mathbb{E}[\Delta w_s^{ll'}|\mathcal{W}_{is}], \quad (5)$$

⁷Equation (1) assumes a common choice set for all workers. The estimator described in Section 3 can be adapted to allow for heterogeneous choice sets, if these are partly observed by the researcher and independent of workers' information sets and location-specific utilities.

with $\mathcal{W}_{is}$ including all elements of $\mathcal{J}_{is}$ other than ε_{is} . The first equality in equation (5) imposes that a type- s worker has at least as much information as any worker of a different type r about the differences in the wage a type- s worker would earn in different locations. The second equality imposes that the worker's expected wage difference between locations l and l' only depends on the expected difference between type-specific terms. Finally, the third equality imposes that, once we condition on all other elements of the worker's information set, idiosyncratic preferences do not help the worker forecast wages.

The first equality in equation (5) naturally holds if all workers in the population have the same information; i.e., if $\mathcal{J}_{is} = \mathcal{J}_{jr}$ for any i, j, s , and r . It also holds if workers know more about their type-specific wage differences than workers of a different type.⁸ Importantly, this equality does not restrict the variation in information across workers of the same type.

The second equality in equation (5) is imposed by data limitations. Our moment inequality procedure allows us to flexibly model the information workers have about payoff-relevant variables whose ex-post (or realized) value the researcher either observes or can consistently estimate. Generally, one cannot estimate, for every worker and location, a wage component that is location- and worker-specific. Hence, we must impose that workers ignore their own idiosyncratic location-specific wage shifters when making their location choices. In contrast, equation (5) does not impose any assumption about what workers know about worker-by-type or type-by-location wage shifters. Specifically, when types correspond to sectors, we impose no assumption on what workers know about their own sectoral comparative advantage or about sector-by-location specific shocks driving labor demand or supply.⁹

The third equality in equation (5) is an implication of the exogeneity of idiosyncratic shocks often assumed in discrete choice models.

Fifth, and last, denoting by $F_\varepsilon(\cdot)$ the cumulative distribution function of ε_{is} , it holds

$$F_\varepsilon(\varepsilon_{is}|\mathcal{W}_{is}, \mathcal{J}_{jr}) = F_\varepsilon(\varepsilon_{is}|\mathcal{W}_{is}) = F_\varepsilon(\varepsilon_{is}) = \exp\left(-\sum_{l=1}^L \exp(-\varepsilon_{is}^l)\right), \quad (6)$$

for any worker i of type s and worker j of type r . The first equality imposes that a worker's idiosyncratic preferences are independent of all other workers' information sets, including their own idiosyncratic preferences. The second equality imposes that a worker's idiosyncratic

⁸When types correspond to sectors, equation (5) imposes that, e.g., real estate workers know more about differences across locations in real estate wages than healthcare workers, and vice versa.

⁹By assuming that unobserved (to the researcher) wage shifters are unknown to workers when choosing locations, we rule out the selection mechanism in Roy (1951). We do, however, allow wages to vary by worker. In our application, worker-sector effects account for much of the variation in worker wages, thus limiting the role worker-location effects play in determining wages. In a binary choice setting, Section 8.2 in Dickstein and Morales (2018) allows agents to make choices on the basis of unobserved alternative-specific payoffs; this approach, however, does not extend directly to our two-step inequality estimator in a multinomial setting.

preferences are independent of all other elements of their own information sets. The third equality imposes a standard assumption in migration models (e.g. [Caliendo et al., 2021](#); [Morten and Oliveira, 2024](#)): that ε_{is}^l is *iid* across locations and follows a type I extreme value distribution with location parameter equal to zero and scale parameter equal to one.¹⁰

Equations (1) to (6) are the only model assumptions we impose. Hence, not only do we allow for unobserved heterogeneity in workers' information sets and, as a result, in workers' wage expectations, but we also leave the wage data-generating process unrestricted. Furthermore, equations (2) to (5) imply $\mathbb{E}[\mathcal{U}_{is}^l | \mathcal{J}_{is}] = \kappa^l + \alpha \mathbb{E}[w_{is}^l | \mathcal{W}_{is}] + \varepsilon_{is}^l$ and, thus, we can interpret κ^l as capturing the expectation that all workers in the population have about amenities in l . Consequently, we allow workers to have irrational expectations about amenities, but restrict these expectations to be common across workers in the population. Conversely, equation (2) restricts workers' wage expectations to be rational, but the flexible modeling of information sets allows expected wages in any given location to differ across all workers.

We assume the researcher observes a random sample of workers by type. For all workers, the researcher observes the location choice $y_{is} = (y_{is}^1, \dots, y_{is}^L)$. Additionally, for all types, the researcher observes wage shifters $w_s = (w_s^1, \dots, w_s^L)$ and a vector $z_s = (z_s^1, \dots, z_s^L)$, with z_s^l a predictor of w_s^l . We do not assume the researcher observes $\mathcal{W}_{is}$ for any sampled worker.

Only differences between the elements of κ are identified. We thus normalize $\kappa^1 = 0$. The goal of estimation is to recover α and $(\kappa^2, \dots, \kappa^L)$, and to learn about workers' information. We denote by $\theta \equiv (\theta_\alpha, \theta_2, \dots, \theta_L)$ the parameter vector with true value $\theta^* \equiv (\alpha, \kappa^2, \dots, \kappa^L)$. To infer workers' information, we test the null hypothesis that, for a given set of locations, certain wage predictors belong to the information set of all workers in a given group.

3 Estimation Through Moment Inequalities

If the number of choices L is large, θ will be high dimensional. Common moment inequality inference procedures rely on inverting a test at each point in a grid covering the parameter space, complicating their applicability in models with large parameter vectors. We propose a two-step procedure that circumvents these computational challenges and produces a confidence interval for each element of θ individually. In the first step, we compute a confidence interval for θ_α using inequalities that difference out the parameters $\theta_2, \dots, \theta_L$. In the second step, for each $l = 2, \dots, L$, we derive inequalities that depend only on θ_α and θ_l , which we combine with the first-step confidence interval for θ_α to obtain a confidence interval for θ_l .¹¹

¹⁰Migration models often assume idiosyncratic preference shocks follow a Fréchet distribution under a multiplicative utility formulation, which is equivalent to our assumption under a log-additive formulation.

¹¹If u_{is}^l included worker-by-location covariates other than the wage, the first-step inequalities could also be used to compute bounds on the coefficients on those covariates.

The moment inequalities used in the first step combine those used in the second step. Thus, for exposition purposes, we first describe the second-step inequalities in Section 3.1. We then describe in Section 3.2 how we build the first-step inequalities. Section 3.3 explains how we use these inequalities to estimate confidence intervals for the elements of θ .

3.1 Second-Step Moment Inequalities

We use two types of inequalities to partially identify θ_l for each $l = 2, \dots, L$. In Section 3.1.1, we introduce a new type of inequality that we call a *bounding* inequality. In Section 3.1.2, we describe how we apply the odds-based inequality in Dickstein et al. (2023) to our setting.

Both the bounding and the odds-based inequality exploit the same implication of the model described in Section 2. Specifically, equation (1) implies that, for any worker i of type s and any two locations l and l' , it holds that

$$(y_{is}^l + y_{is}^{l'}) (\mathbb{1}\{\mathbb{E}[\mathcal{U}_{is}^l - \mathcal{U}_{is}^{l'} | \mathcal{J}_{is}] \geq 0\} - y_{is}^l) = 0. \quad (7)$$

This equation indicates that, for any worker who chooses location l or location l' (for whom $y_{is}^l + y_{is}^{l'} = 1$), they would choose l if and only if their expected utility of choosing l is larger than that of choosing l' ; i.e., $y_{is}^l = 1$ if and only if $\mathbb{E}[\mathcal{U}_{is}^l - \mathcal{U}_{is}^{l'} | \mathcal{J}_{is}] \geq 0$. As equation (7) holds for every worker, it must also hold for the average worker with a particular information set $\mathcal{W}_{is}$ who effectively chooses l or l' . Equations (2) to (6) then imply the following *equality*:

$$\mathbb{E} \left[\frac{\exp(\mathbb{E}[u_{is}^l - u_{is}^{l'} | \mathcal{W}_{is}])}{1 + \exp(\mathbb{E}[u_{is}^l - u_{is}^{l'} | \mathcal{W}_{is}])} - y_{is}^l \middle| \mathcal{W}_{is}, y_{is}^l + y_{is}^{l'} = 1 \right] = 0,$$

which we can rewrite as

$$\mathbb{E}[y_{is}^{l'} + y_{is}^l (-\exp(-\mathbb{E}[u_{is}^l - u_{is}^{l'} | \mathcal{W}_{is}])) | \mathcal{W}_{is}] = 0, \quad (8)$$

where $\exp(-\mathbb{E}[u_{is}^l - u_{is}^{l'} | \mathcal{W}_{is}])$ equals the probability of choosing l' relative to that of choosing l . This equality cannot be used to identify θ due to the weak restrictions our model imposes on the content of $\mathcal{W}_{is}$ for any worker i of type s . However, as shown in Section 3.1.1 and 3.1.2, the convexity of $\exp(-x)$ in x can be exploited to derive inequalities that do not depend on $\mathcal{W}_{is}$ and provide non-trivial bounds on θ .¹² Specifically, the bounding inequalities

¹²The inequalities in Section 3.1.1 and 3.1.2 can be derived if the relative probability of choosing locations l and l' is convex (or concave) in the agent's expected utility difference between them: the relative probability of choosing l' over choosing l becomes more (or less) sensitive to $\mathbb{E}[u_{is}^l - u_{is}^{l'} | \mathcal{W}_{is}]$ as it increases. This property holds in multinomial logit models and also in nested logit models when l and l' belong to the same nest.

in Section 3.1.1 exploit the fact that any convex function is bounded from below by any first-order approximation to it, regardless of the approximation point. Conversely, the odds-based inequalities in Section 3.1.2 exploit the fact that, according to Jensen's inequality, the expectation of a convex function is larger than the function of the corresponding expectation.

3.1.1 Bounding Moment Inequalities

Given equation (8) and the convexity of $\exp(-x)$ in x , we can use the first-order approximation to this function around any point $e_{is}^{ll'}$ to derive the following moment inequality:

$$\mathbb{E}[y_{is}^{l'} - y_{is}^l \exp(-e_{is}^{ll'}) (1 + e_{is}^{ll'} - \mathbb{E}[u_{is}^l - u_{is}^{l'} | \mathcal{W}_{is}]) | \mathcal{W}_{is}] \geq 0. \quad (9)$$

Since y_{is}^l is measurable in $\mathcal{J}_{is}$, we show in Appendix A.1 that, for any $e_{is}^{ll'}$ also measurable in $\mathcal{J}_{is}$, we can apply the Law of Iterated Expectations (LIE) to obtain:

$$\mathbb{E}[y_{is}^{l'} - y_{is}^l \exp(-e_{is}^{ll'}) (1 + e_{is}^{ll'} - (u_{is}^l - u_{is}^{l'})) | \mathcal{W}_{is}] \geq 0. \quad (10)$$

Finally, given any $z_s \subseteq \mathcal{W}_{is}$, applying again the LIE, we obtain the following inequality:

$$\mathbb{E}[y_{is}^{l'} - y_{is}^l \exp(-e_{is}^{ll'}) (1 + e_{is}^{ll'} - (u_{is}^l - u_{is}^{l'})) | z_s] \geq 0, \quad (11)$$

which no longer depends on the unobserved information set $\mathcal{W}_{is}$ of any worker. The remainder of this section explains how we use this inequality to derive bounds on θ_l for all $l = 1, \dots, L$.

Given locations l and l' , we denote by $\Delta\theta_{ll'} \equiv \theta_l - \theta_{l'}$ the unknown parameter whose true value is $\Delta\kappa^{ll'} \equiv \kappa^l - \kappa^{l'}$. Then, for any two locations l and l' in the worker's choice set, a random vector z_s , and a scalar random variable $e_{is}^{ll'}$, we define the moment

$$\mathfrak{m}^{ll'}(z_s, \Delta\theta_{ll'}) \equiv \mathbb{E}[y_{is}^{l'} - y_{is}^l \exp(-e_{is}^{ll'}) (1 + e_{is}^{ll'} - (\Delta\theta_{ll'} + \alpha \Delta w_s^{ll'})) | z_s], \quad (12)$$

which equals the left-hand side of equation (11) but written as a function of $\Delta\theta_{ll'}$. Theorem 1 establishes a property of this moment when $\Delta\theta_{ll'} = \Delta\kappa^{ll'}$.¹³

Theorem 1 *Assume equations (1) to (6) hold. Then, $\mathfrak{m}^{ll'}(z_s, \Delta\kappa^{ll'}) \geq 0$ if $e_{is}^{ll'} \subseteq \mathcal{J}_{is}$ and $z_s \subseteq \mathcal{W}_{is}$ for every worker i of any type s .*

The proof of Theorem 1 is in Appendix A.1. This theorem implies that, if its conditions hold, the set of values of $\Delta\theta_{ll'}$ for which

$$\mathfrak{m}^{ll'}(z_s, \Delta\theta_{ll'}) \geq 0 \quad (13)$$

¹³Note that, given the normalization $\kappa^1 = 0$, it holds that $\Delta\theta_{11} = \kappa^1$.

includes $\Delta\kappa^{ll'}$ regardless of the value of z_s , of the locations l and l' being compared, and of how the approximation points $e_{is}^{ll'}$ are chosen. Ideally, we would like $m^{ll'}(z_s, \Delta\theta_{ll'})$ to be negative for all values of $\Delta\theta_{ll'}$ different from $\Delta\kappa^{ll'}$. We show in Appendix B.1 that the value of $e_{is}^{ll'}$ that minimizes the moment $m^{ll'}(z_s, \Delta\theta_{ll'})$ for any given instrument z_s and parameter value $\Delta\theta_{ll'}$ is given by the following expression:

$$e_{is}^{ll'} = \Delta\theta_{ll'} + \alpha \mathbb{E}[\Delta w_s^{ll'} | z_s, y_{is}^l = 1]. \quad (14)$$

To provide intuition on how the inequality implied by equations (12) to (14) may be used to partially identify $\Delta\theta_{ll'}$, we show in Appendix B.1 that this inequality can be written as

$$\frac{\mathbb{E}[y_{is}^l | z_s]}{\mathbb{E}[y_{is}^{l'} | z_s]} \exp(-\alpha \mathbb{E}[\Delta w_s^{ll'} | z_s, y_{is}^l = 1]) \leq \exp(\Delta\theta_{ll'}). \quad (15)$$

As this inequality holds for any two locations, we can swap the identity of l and l' and obtain

$$\frac{\mathbb{E}[y_{is}^l | z_s]}{\mathbb{E}[y_{is}^{l'} | z_s]} \exp(-\alpha \mathbb{E}[\Delta w_s^{ll'} | z_s, y_{is}^{l'} = 1]) \geq \exp(\Delta\theta_{ll'}). \quad (16)$$

Equations (15) and (16) provide bounds on the amenity difference $\Delta\theta_{ll'}$. These bounds are based on the relative probability with which workers who know z_s choose l over l' , $\mathbb{E}[y_{is}^l | z_s] / \mathbb{E}[y_{is}^{l'} | z_s]$. As in most spatial models, the mapping between relative choice probabilities and amenity differences is not direct, as a location l may be preferred over a location l' not because of a higher amenity value in l but because of higher expected wages in l . That is, to infer $\Delta\theta_{ll'}$, we must first net out the effect of the *expected* wage difference between l and l' . This step depends on the relationship between the worker's information set $\mathcal{W}_{is}$ and the vector z_s . When $z_s = \mathcal{W}_{is}$, the decision to move to l or l' does not provide extra information about workers' wage expectations. Therefore, $\mathbb{E}[\Delta w_s^{ll'} | z_s, y_{is}^l = 1] = \mathbb{E}[\Delta w_s^{ll'} | z_s, y_{is}^{l'} = 1] = \mathbb{E}[\Delta w_s^{ll'} | z_s]$ and the bounds in equations (15) and (16) coincide, point-identifying amenities. We formalize this intuition in the following corollary, which we prove in Appendix B.2.

Corollary 1 *Assume equations (1) to (6) hold. Then, the bounds in equations (15) and (16) imply $\Delta\theta_{ll'} = \Delta\kappa^{ll'}$ if $\mathbb{E}[\Delta w_s^{ll'} | z_s] = \mathbb{E}[\Delta w_s^{ll'} | \mathcal{W}_{is}]$ for every worker i of any type s .*

When the researcher only observes a subset z_s of workers' information sets $\mathcal{W}_{is}$, the decision to move to l or l' provides additional information about workers' wage expectations. In particular, by revealed preferences, $\mathbb{E}[\Delta w_s^{ll'} | z_s, y_{is}^l = 1] \geq \mathbb{E}[\Delta w_s^{ll'} | z_s, y_{is}^{l'} = 1]$, reflecting that workers choosing location l received more positive signals about wages in l relative to l' . This is the key intuition behind our inequalities in equations (15) and (16). By applying

these inequalities, even though we do not know the exact information sets of workers, we obtain bounds on the amenities.

3.1.2 Odds-based Moment Inequalities

As discussed in [Dickstein et al. \(2023\)](#), equation (8), the convexity of $\exp(-x)$ in x , and the rationality of workers' expectations imposed in equation (2) imply the following inequality:

$$\mathbb{E}[y_{is}^l \exp(-(u_{is}^l - u_{is}^{l'})) - y_{is}^{l'} | \mathcal{W}_{is}] \geq 0. \quad (17)$$

Thus, given any $z_s \subseteq \mathcal{W}_{is}$, the LIE implies

$$\mathbb{E}[y_{is}^l \exp(-(u_{is}^l - u_{is}^{l'})) - y_{is}^{l'} | z_s] \geq 0. \quad (18)$$

The remainder of this section explains how we use this inequality to derive bounds on $\Delta\theta_{ll'}$ for any l and l' . For any two locations l and l' and a random vector z_s , define the moment

$$m_o^{ll'}(z_s, \Delta\theta_{ll'}) \equiv \mathbb{E}[y_{is}^l \exp(-(\Delta\theta_{ll'} + \alpha\Delta w_s^{ll'})) - y_{is}^{l'} | z_s], \quad (19)$$

which equals the left-hand side of equation (18) but written as a function of the unknown parameter $\Delta\theta_{ll'}$. Theorem 2, from [Dickstein et al. \(2023\)](#), establishes a key property of this moment when evaluated at $\Delta\theta_{ll'} = \Delta\kappa^{ll'}$.

Theorem 2 *Assume equations (1) to (6) hold. Then, $m_o^{ll'}(z_s, \Delta\kappa^{ll'}) \geq 0$ if $z_s \subseteq \mathcal{W}_{is}$ for every worker i of any type s .*

The proof of Theorem 2, which follows [Dickstein et al. \(2023\)](#), is in Appendix A.2. This theorem implies that, if its conditions hold, the set of values of $\Delta\theta_{ll'}$ that satisfies

$$m_o^{ll'}(z_s, \Delta\theta_{ll'}) \geq 0 \quad (20)$$

includes $\Delta\kappa^{ll'}$. Appendix B.3 shows that this inequality can be rewritten as

$$\frac{\mathbb{E}[y_{is}^l | z_s]}{\mathbb{E}[y_{is}^{l'} | z_s]} \mathbb{E}[(\exp(\alpha\Delta w_s^{ll'}))^{-1} | z_s, y_{is}^l = 1] \geq \exp(\Delta\theta_{ll'}). \quad (21)$$

By swapping the identity of locations l and l' in equation (21), we obtain the inequality

$$\frac{\mathbb{E}[y_{is}^l | z_s]}{\mathbb{E}[y_{is}^{l'} | z_s]} (\mathbb{E}[\exp(\alpha\Delta w_s^{ll'}) | z_s, y_{is}^{l'} = 1])^{-1} \leq \exp(\Delta\theta_{ll'}). \quad (22)$$

The inequalities in equations (21) and (22) identify bounds on $\Delta\theta_{ll'}$. Due to Jensen’s inequality, these inequalities generally only partially identify $\Delta\theta_{ll'}$. However, the following corollary describes a case where these bounds coincide, point identifying $\Delta\theta_{ll'}$.

Corollary 2 *Assume equations (1) to (6) hold. Then, the bounds in equations (21) and (22) imply $\Delta\theta_{ll'} = \Delta\kappa^{ll'}$ if $\Delta w_s^{ll'} = \mathbb{E}[\Delta w_s^{ll'} | \mathcal{W}_{is}]$ for every worker i of any type s .*

We prove Corollary 2 in Appendix B.4. Corollary 2 strengthens the assumptions in Theorem 2 by requiring that workers be fully informed about $\Delta w_s^{ll'}$, $\Delta w_s^{ll'} = \mathbb{E}[\Delta w_s^{ll'} | \mathcal{W}_{is}]$. In this case, the inequality in equation (20) only holds if $\Delta\theta_{ll'}$ equals its true value $\Delta\kappa^{ll'}$. Thus, the odds-based inequality point identifies $\Delta\theta_{ll'}$ if workers forecast wages without error. This is true regardless of the extent to which the researcher observes workers’ information sets.

3.1.3 Combining Bounding and Odds-based Moment Inequalities

There are advantages in using our bounding inequality jointly with the odds-based inequality in Dickstein et al. (2023). As stated in Theorems 1 and 2, the identified sets defined by these two types of inequalities always contain the true parameter value. However, as exemplified by corollaries 1 and 2, these identified sets may differ. As a result, there may be gains from combining both types of inequalities in estimation; that is, the intersection of both identified sets may be smaller than each of them individually.

The identified set defined by the odds-based inequality increases in the relevance of the error affecting workers’ expectations. Intuitively, these inequalities are convex in the agent’s expectational error. Consequently, Jensen’s inequality implies that the odds-based moment at any parameter value is larger in expectation the larger the relevance of the expectational error, making the resulting moment inequality weaker. Thus, if workers are poorly informed about wages, the identified set defined by the odds-based inequality will be large, including parameter values quite distinct from the true one. Importantly, this is the case even if the researcher observes the worker’s true information set. Conversely, the identified set defined by the bounding inequality is not affected by the worker’s expectational error. Intuitively, the bounding inequalities are linear in this error, making it irrelevant for the identified set defined by these inequalities. Thus, even if workers are poorly informed about wages, the resulting identified set will be tight as long as the researcher observes the key variables used by workers to form their expectations. Moreover, as indicated in Corollary 1, the bounding inequality point identifies the parameter of interest if the researcher observes the worker’s information set. This is important because full observability of the worker’s information set is generally necessary for standard estimators (e.g., maximum likelihood) to be consistent;

thus, a researcher who uses bounding moment inequalities in her analysis will not suffer from any loss of identification power whenever those standard estimators are consistent.

The identified set defined by the bounding inequality increases in the relevance of the variables that enter the worker's information set but the researcher does not observe. Intuitively, the optimal approximation point in equation (14) is the best approximation to the worker's true expected utility difference between locations l and l' that the researcher can build with the variables she observes. The larger the mismatch between this point and the agent's true expected utility difference between l and l' , the weaker the bounding inequality is. Conversely, the identified set defined by the odds-based inequality is not affected by the extent to which the worker's information set is observed by the researcher. Even if workers have access to information the researcher cannot observe, the identified set defined by the odds-based inequality will be tight if the worker's true information set allows them to forecast wages with little error. Moreover, as indicated in Corollary 2, the odds-based inequality point identifies the parameter of interest if workers do not experience expectational errors.

In sum, the bounding inequality being robust to workers' expectational errors and the odds-based inequality being robust to the presence of unobserved variables that belong to the worker's true information set imply there are often gains from simultaneously using both types of inequality. We illustrate this in our simulation in Section 4. In settings such as ours in which the choice set is large and the worker's utility depends on choice-specific fixed effects, the bounding inequality has the extra advantage that it is linear in the utility difference between any two locations; see equation (11). As shown in Section 3.2, this allows us to combine multiple bounding inequalities in order to difference out the choice-specific fixed effects, making our two-step estimation procedure feasible.

3.2 First-Step Moment Inequalities

Proceeding as in the derivation of the bounding inequality in equation (11), we derive

$$\mathbb{E}[y_{jr}'(y_{is}' - y_{is}^l \exp(-e_{is}^{ll'})(1 + e_{is}^{ll'} - (u_{is}^l - u_{is}^{l'}))) | z_s, z_r] \geq 0, \quad (23a)$$

$$\mathbb{E}[y_{is}^l(y_{jr}^l - y_{jr}' \exp(-e_{jr}^{ll'})(1 + e_{jr}^{ll'} - (u_{jr}^{l'} - u_{jr}^l))) | z_s, z_r] \geq 0, \quad (23b)$$

for any locations l and l' , types s and r , random vectors z_s and z_r , and approximation points $(e_{is}^{ll'}, e_{jr}^{ll'}) \subseteq \mathcal{J}_{is} \cup \mathcal{J}_{jr}$. Loosely speaking, the moment in equation (23a) compares the utility of choosing l over l' for a worker i of type s , and that in equation (23b) compares the utility of choosing l' over l for a worker j of type r . If the approximation points in these two moments

coincide, the sum of the inequalities in equations (23a) and (23b) results in

$$\mathbb{E}[y_{is}^l y_{jr}^l + y_{is}^{l'} y_{jr}^{l'} - y_{is}^l y_{jr}^{l'} \exp(-e_{isjr}^{l'}) (2 + 2e_{isjr}^{l'} - \theta_\alpha (\Delta u_{is}^{l'} + \Delta u_{jr}^{l'})) | z_s, z_r] \geq 0, \quad (24)$$

where $e_{isjr}^{l'}$ denotes the common approximation point. As all workers, regardless of their type, have a shared valuation of the amenities in every location, the amenity differences between locations l and l' cancel when adding both moments in equation (23); thus, we can substitute $\Delta u_{is}^{l'} + \Delta u_{jr}^{l'} = \alpha (\Delta w_s^{l'} + \Delta w_r^{l'})$ in equation (24).¹⁴ The remainder of this section explains how we use the inequality in equation (24) to derive bounds on θ_α . For any locations l and l' , types s and r , vectors z_s and z_r , and scalar variable $e_{isjr}^{l'}$, we define the moment

$$\begin{aligned} \mathbb{M}^{l'}(z_s, z_r, \theta_\alpha) \equiv \\ \mathbb{E}[y_{is}^l y_{jr}^l + y_{is}^{l'} y_{jr}^{l'} - y_{is}^l y_{jr}^{l'} \exp(-e_{isjr}^{l'}) (2 + 2e_{isjr}^{l'} - \theta_\alpha (\Delta w_s^{l'} + \Delta w_r^{l'})) | z_s, z_r], \end{aligned} \quad (25)$$

which equals the left-hand side of equation (24) but written as a function of θ_α . Theorem 3 establishes a property of the moment in equation (25) when evaluated at $\theta_\alpha = \alpha$.

Theorem 3 *Assume equations (1) to (6) hold. Then, $\mathbb{M}^{l'}(z_s, z_r, \alpha) \geq 0$ if $e_{isjr}^{l'} \subseteq \mathcal{J}_{is} \cup \mathcal{J}_{jr}$ and $(z_s, z_r) \subseteq \mathcal{W}_{is} \cup \mathcal{W}_{jr}$ for all workers i and j of any types s and r , respectively.*

The proof of Theorem 3 is in Appendix A.3. This theorem implies that, if its conditions hold, the set of values of θ_α for which

$$\mathbb{M}^{l'}(z_s, z_r, \theta_\alpha) \geq 0 \quad (26)$$

includes α regardless of (z_s, z_r) , of the locations l and l' , and of how the approximation points $e_{isjr}^{l'}$ are chosen. However, the set of values of θ_α other than α that satisfy equation (26) depends on these approximation points. Appendix B.5 shows that the value of $e_{isjr}^{l'}$ that minimizes the moment $\mathbb{M}^{l'}(z_s, z_r, \theta_\alpha)$ for any given instruments z_s and z_r and parameter value θ_α is given by the following expression:

$$e_{isjr}^{l'} = \theta_\alpha \mathbb{E}[0.5(\Delta w_s^{l'} + \Delta w_r^{l'}) | z_s, z_r, y_{is}^l y_{jr}^{l'} = 1]. \quad (27)$$

As shown in Appendix B.5, equations (26) and (27) jointly imply that

$$\frac{\mathbb{E}[y_{is}^l y_{jr}^{l'} | z_s, z_r]}{\mathbb{E}[0.5(y_{is}^l y_{jr}^{l'} + y_{is}^{l'} y_{jr}^{l'}) | z_s, z_r]} \leq \exp(\theta_\alpha \mathbb{E}[0.5(\Delta w_s^{l'} + \Delta w_r^{l'}) | z_s, z_r, y_{is}^l y_{jr}^{l'} = 1]). \quad (28)$$

¹⁴Ho and Pakes (2014) apply a similar strategy to difference out choice-specific fixed effects in a model without individual-by-choice idiosyncratic preferences.

This inequality yields a lower bound on θ_α if its right-hand side is increasing in this parameter, and vice versa. Intuitively, if type- s workers are likely to choose location l whereas type- r workers are likely to choose l' , as represented by a high value of the ratio on the left-hand side of equation (28), and type- s workers expect their wage to be higher in l than in l' , and vice versa, as represented by a positive value of $\mathbb{E}[0.5(\Delta w_s^{ll'} + \Delta w_r^{ll'})|z_s, z_r, y_{is}^l y_{jr}^{l'} = 1]$, then θ_α cannot be too low. Conversely, if both worker types are still likely to choose locations l and l' , respectively, but now type- s workers expect their wage to be lower in l than in l' , and vice versa, as represented by a negative value of $\mathbb{E}[0.5(\Delta w_s^{ll'} + \Delta w_r^{ll'})|z_s, z_r, y_{is}^l y_{jr}^{l'} = 1]$, then θ_α cannot be too high. Generally, the parameter θ_α is partially identified. The following corollary describes when moment inequalities of the type in equation (28) point identify θ_α .

Corollary 3 *Assume equations (1) to (6) hold, $e_{isjr}^{ll'} \subseteq \mathcal{J}_{is} \cup \mathcal{J}_{jr}$, $(z_s, z_r) \subseteq \mathcal{W}_{is} \cup \mathcal{W}_{jr}$, $\Delta \kappa^{ll'} = 0$ and $\mathbb{E}[\Delta w_s^{ll'}|z_s] = \mathbb{E}[\Delta w_s^{ll'}|\mathcal{W}_{is}] = \mathbb{E}[\Delta w_r^{ll'}|z_r] = \mathbb{E}[\Delta w_r^{ll'}|\mathcal{W}_{jr}]$ for all workers i and j of any types s and r , respectively. Then, the moment inequalities in equation (28) can point identify θ_α .*

The proof of Corollary 3 is in Appendix B.6. To understand this corollary, it is useful to compare it to Corollary 1. The conditions listed in Corollary 3 are more restrictive. The reason is that, unless these conditions are satisfied, there is a loss of identification power coming from having to impose the common approximation point in equation (27) on each of the inequalities in equation (23). When the conditions in Corollary 3 are satisfied, the approximation point in equation (27) is optimal for each of the two inequalities in equation (23) (i.e., it coincides with the approximation point in equation (14) for both moments) and thus no loss of identification power occurs when deriving the inequality in equation (24).

Although identification of the parameter α requires instruments for wages z_s and z_r , their role here differs from that in standard gravity-based settings. In these, instruments address the correlation between wages and unobserved amenities that also shape location choices. In our framework, amenities are likewise unobserved but are differenced out by comparing workers across types, so they do not confound the relationship between wages and location choices in equation (26). Instruments are still necessary, however, because workers' expectations are unobserved and observed wages may be correlated with expectational errors. Thus, valid instruments must be uncorrelated with expectational errors, regardless of their correlation with amenities. Under rational expectations, this condition is satisfied by any variable contained in workers' information sets.

3.3 Using the Inequalities for Estimation

For estimation, we use a set of unconditional moment inequalities that we derive from the conditional ones introduced in Sections 3.1 and 3.2. We describe here how we derive unconditional bounding moment inequalities from the conditional ones described in Section 3.1.1. In practice, when computing a confidence interval for the amenity term $\Delta\theta_{ll'}$, we combine these unconditional bounding inequalities with unconditional odds-based moment inequalities derived in a similar way from the conditional ones in Section 3.1.2. In Appendix B.7, we describe how we rely on the conditional inequalities in Section 3.2 to derive the unconditional inequalities we use to compute a confidence interval for θ_α .

We implement the following steps to derive $k = 1, \dots, K$ unconditional moment inequalities from the conditional one described in Section 3.1.1. First, we choose a scalar $\Delta z_s^{ll'} \subseteq z_s$ that is correlated with $\Delta w_s^{ll'}$ and that we will use in all K inequalities. Second, for each k , we choose a subset $[\underline{z}_k, \bar{z}_k]$ of the support of $\Delta z_s^{ll'}$ and an integer d_k . We then build

$$\mathbb{E}[(y_{is}^{ll'} - y_{is}^l \exp(-e_{is}^{ll'})(1 + e_{is}^{ll'} - (\Delta\theta_{ll'} + \alpha\Delta w_s^{ll'})))g_k(\Delta z_s^{ll'})] \geq 0, \quad (29)$$

where

$$g_k(\Delta z_s^{ll'}) = \mathbb{1}\{\underline{z}_k < \Delta z_s^{ll'} \leq \bar{z}_k\} |\Delta z_s^{ll'}|^{d_k}. \quad (30)$$

The inequality in equation (29) is implied by that in equation (13) regardless of the choice of predictor $\Delta z_s^{ll'}$, interval limits $\underline{z}_k$ and $\bar{z}_k$, and exponent d_k . In practice, we fix a $q \in \mathbb{N}$ and map $\underline{z}_k$ and $\bar{z}_k$ to consecutive elements of the vector of q -quantiles of the distribution of $\Delta z_s^{ll'}$ across types and location pairs. E.g., if $q = 2$ and $d_k = 0$ for all k , we use $K = 2$ inequalities with instruments $g_1(\Delta z_s^{ll'}) = \mathbb{1}\{\Delta z_s^{ll'} \leq \text{med}(\Delta z_s^{ll'})\}$ and $g_2(\Delta z_s^{ll'}) = \mathbb{1}\{\text{med}(\Delta z_s^{ll'}) > \Delta z_s^{ll'}\}$, which split all observations depending on whether $\Delta z_s^{ll'}$ is above or below median. If $q = 4$, similar instruments will split observations according to the quartile $\Delta z_s^{ll'}$ belongs to. More generally, the larger q is, the larger the number of unconditional inequalities we use.

In practice, computing a confidence interval (CI) for $\Delta\theta_{ll'}$ requires first computing a CI for θ_α . To compute a 95% CI for $\Delta\theta_{ll'}$, we first compute 96% CIs for $\Delta\theta_{ll'}$ conditional on each value of θ_α in a 99% CI for this parameter. We denote these as $\hat{\Theta}_{.96}^l(\theta_\alpha)$. We then compute the 95% CI for θ_l as the union of $\hat{\Theta}_{.96}^l(\theta_\alpha)$ for each value of θ_α in its 99% CI. All our CIs are computed following the moment selection procedure in Andrews and Soares (2010).¹⁵

¹⁵An alternative to our estimator is to apply to each element of $(\theta_\alpha, \theta_2, \dots, \theta_L)$ a procedure that yields CIs for projections of partially identified vectors: Bugni et al. (2016); Kaido et al. (2019); Andrews et al. (2023).

4 Properties of Moment Inequalities: Simulation

This section uses simulations to illustrate properties of the moment inequalities introduced in Section 3. Three insights emerge. First, consistent with Theorems 1 to 3, when the conditions in those theorems are satisfied, we obtain intervals that contain the true parameter values even when workers imperfectly forecast wages and the researcher imperfectly observes agents' information sets. Second, when the researcher misspecifies agents' information sets, the maximum likelihood (ML) estimator is inconsistent and sometimes lies outside the bounds defined by our moment inequalities. Third, our inequalities may yield empty confidence sets when the researcher incorrectly assumes that certain covariates belong to agents' information sets, demonstrating the potential of our estimator to test for the true content of such sets.

4.1 Simulation Setup

Workers choose between three locations $l = \{1, 2, 3\}$ according to the model in Section 2. We simulate data for 6,000,000 workers, each of a different type, allowing us to index observations by s . We set the wage coefficient to $\alpha = 1$ and the location-specific amenities to $\kappa^1 = \kappa^2 = 0$ and $\kappa^3 = 1$. While the estimator described in Section 3 is valid for any stochastic process for wages and any specification of workers' information sets, we need to specify these aspects of the model to generate the model-implied choice for all sampled workers.¹⁶

We assume the wage of worker s in location l is determined by

$$w_s^l = z_{1s}^l + z_{2s}^l + z_{3s}^l, \quad (31)$$

with z_{ks}^l independent across l , s , and k , and distributed uniformly with mean μ_k^l and support of length $2\sigma_k$. We assume worker s observes (z_{1s}^l, z_{2s}^l) for all l . Thus, for all s and l , $\mathbb{E}[w_s^l | \mathcal{W}_s] = z_{1s}^l + z_{2s}^l$, and z_{3s}^l equals the expectational error worker s makes when forecasting w_s^l .¹⁷

We assume the researcher observes (y_s^l, w_s^l, z_{2s}^l) for every worker s and location l . Therefore, for all s and l , z_{1s}^l is a variable used by worker s when forming their expectations about w_s^l but not observed by the researcher.

¹⁶We do not aim to illustrate the statistical properties of our inference procedure but to characterize the bounds defined by the inequalities introduced in Section 3. To this end, the large number of workers is useful, as it limits the impact of simulation noise on our results. We set $L > 2$ to illustrate that our inequalities apply to multinomial settings; larger values of L result in tighter bounds for θ_α (as the number of inequalities of the type in equation (26) increases in the number of possible location pairs) at the expense of more time needed to compute CIs for the larger number of parameters $(\theta_2, \dots, \theta_L)$.

¹⁷We set $\mu_1^l = \mu_3^l = 0$ for all l and $(\mu_2^1, \mu_2^2, \mu_2^3) = (0, -0.5, -1)$; thus, mean wages decline in order from $l = 1$ to $l = 3$. In terms of the dispersion, we set $\sigma_2 = 4$ and present results for different values of σ_1 and σ_3 .

4.2 Moment Inequality Estimates

Table 1 presents the main simulation results. We consider cases that differ in the value of σ_1 and σ_3 . The former determines the dispersion in z_{1s}^l and, thus, the relevance of unobserved (to the researcher) variables that belong to the worker’s information set. The latter determines the dispersion in z_{3s}^l and, thus, the relevance of payoff-relevant variables that workers do not observe. Section 4.2.1 discusses our CIs for the wage coefficient θ_α , displayed in Table 1 in the column labeled *First Step* and computed using the inequalities introduced in Section 3.2. Section 4.2.2 describes CIs for θ_2 and θ_3 , displayed in the columns labeled *Second Step* and computed using the inequalities introduced in Section 3.1. Appendix C.1 describes the unconditional moment inequalities we use in this simulation exercise.

4.2.1 Confidence Intervals for the Wage Coefficient

Cases 1 and 2 share the feature that $\sigma_1 = 0$ and, thus, the researcher observes the agent’s full information set—as $z_{1s}^l = 0$ for every l , the agent’s information set only includes z_{2s}^l , which is the wage predictor used by the researcher. In these cases, we obtain CIs for θ_α that are tight around its true value. This is related to two aspects of our setting. First, we build first-stage inequalities separately for each pair of locations; thus, as $\kappa^1 = \kappa^2$ in our simulation, the inequality with location indices equal to $l = 1$ and $l' = 2$ (or vice versa) verifies that $\Delta\kappa^{ll'} = 0$. Furthermore, when building the inequality corresponding to any location pair l and l' , we match each worker s with a worker r such that $\mathbb{E}[\Delta w_s^{ll'} | \Delta z_{2s}^{ll'}]$ is close to $\mathbb{E}[\Delta w_r^{l'l} | \Delta z_{2r}^{l'l}]$. As a result, consistent with Corollary 3, the inequalities used to compute the bounds on θ_α in Table 1 are close to point-identifying this parameter whenever $\mathbb{E}[w_s^l | \mathcal{W}_s] = \mathbb{E}[w_s^l | \Delta z_{2s}^{ll'}]$ for all l and s . This extra condition is met precisely when $\sigma_1 = 0$.¹⁸

Cases 3 and 4 share the feature that $\sigma_1 > 0$. Thus, the researcher only observes part of the agent’s information set—the true information set is (z_{1s}^l, z_{2s}^l) for every l , but the researcher only observes z_{2s}^l . By relying on only a subset of the agent’s information set when building the moment inequalities, the CI for θ_α becomes wider.

In case 5, the researcher wrongly assumes the agent has perfect information on wages. The resulting CI is very tight but does not include the true parameter value; it only includes $\theta_\alpha = 0.87$. This illustrates a problematic situation for the researcher using moment inequalities, as such a researcher may wrongly conclude that $\alpha = 0.87$ and that the inequalities are very tight around the truth (see Molinari, 2020; Andrews and Kwon, 2024). In our setting, as shown in Table C.2 in Appendix C.3, the resulting confidence set becomes empty as we increase the

¹⁸In Table C.3 in Appendix C.4, we present results for other values of $(\kappa^1, \kappa^2, \kappa^3)$. The CI for θ_α is tight if $\Delta\kappa^{ll'} = 0$ for at least two locations l and l' . In Table C.1 in Appendix C.2, we show that the CI becomes wider as the mean difference between $\mathbb{E}[\Delta w_s^{ll'} | z_{2s}]$ and $\mathbb{E}[\Delta w_r^{l'l} | z_{2r}]$ for matched workers s and r increases.

Table 1: Simulation Results - Moment Inequality Confidence Intervals

Case	σ_1	σ_3	z_s^l	First Step	Second Step		
				θ_α	Mom. Ineq.	θ_2	θ_3
1	0	0	z_{2s}^l	[1, 1.02]	Bounding Odds-based Both	[0, 0] [0, 0] [0, 0]	[1, 1] [1, 1] [1, 1]
2	0	1	z_{2s}^l	[1, 1.01]	Bounding Odds-based Both	[0, 0] [-0.33, 0.32] [0, 0]	[1, 1] [0.68, 1.33] [1, 1]
3	1	0	z_{2s}^l	[0.82, 1.29]	Bounding Odds-based Both	[-0.31, 0.31] [0, 0] [0, 0]	[0.70, 1.30] [1, 1.01] [1, 1.01]
4	1	1	z_{2s}^l	[0.82, 1.31]	Bounding Odds-based Both	[-0.31, 0.31] [-0.38, 0.39] [-0.31, 0.31]	[0.69, 1.31] [0.68, 1.45] [0.69, 1.31]
5	0	1	w_s^l	[0.87, 0.87]	Bounding Odds-based Both	[-0.05, -0.01] $\emptyset$ $\emptyset$	[0.85, 0.88] $\emptyset$ $\emptyset$

The true parameter values are $\alpha = 1$, $\kappa^2 = 0$, and $\kappa^3 = 1$. The column θ_α contains 95% CIs based on the estimator described in Section 3.2. The columns θ_2 and θ_3 contain 95% CIs based on the estimators described in Section 3.1. The rows labeled *Bounding* use the inequalities in Section 3.1.1; those labeled *Odds-based* use the inequalities in Section 3.1.2; and those labeled *Both* combine both inequalities. CIs are computed using the moment selection procedure in Andrews and Soares (2010). See Appendix C.1 for more details.

number of instruments used to form our inequalities.

4.2.2 Confidence Intervals for Amenities

Consistent with Corollary 1, the bounding inequalities point identify θ_2 and θ_3 when the agent's information set is fully observed by the researcher; i.e., when $\sigma_1 = 0$, as in cases 1 and 2. When $\sigma_1 > 0$, as in cases 3 and 4, there are variables the agent knows but the researcher does not observe and, as a result, the CIs built using the bounding inequalities alone include parameter values in addition to the true ones. A comparison of cases 1 and 2, or cases 3 and 4, shows this holds regardless of how large workers' expectational errors are.¹⁹

The length of the CIs on θ_2 and θ_3 defined by the odds-based inequalities increases in the importance of the worker's expectational errors; i.e., increases in σ_3 , as a comparison of cases 1 and 2, or cases 3 and 4, illustrates. When workers make no expectational errors (i.e., when $\sigma_3 = 0$) and the first-step CI for θ_α equals its true value α , the odds-based inequalities

¹⁹All CIs in Table 1 are computed using the approximation points in equations (14) and (27). In Table C.4 in Appendix C.5, we show the CIs become wider when using other approximation points. How we compute these points thus does not affect the validity of our inequalities, but it may affect how tight these are.

point identify the amenity parameters θ_2 and θ_3 , as predicted by Corollary 2. When $\sigma_3 = 0$ but the CI for θ_α includes values other than α , the odds-based moment inequalities may still point identify the amenities (as in cases 1 and 3), but will not do so always.²⁰

Since the bounding inequalities are insensitive to agents' expectational errors (i.e., insensitive to σ_3) and the odds-based inequalities are partially insensitive to agents having information the researcher does not observe (i.e., partially insensitive to σ_1), there are advantages from combining both types of inequalities in estimation (see Section 3.1.3). Cases 2 and 3 show that, when either $\sigma_1 = 0$ or $\sigma_3 = 0$, the combination of bounding and odds-based inequalities point-identifies θ_2 and θ_3 , although, when considered in isolation, neither of these two inequalities point-identifies these parameters in both cases.

Case 4 is the most empirically relevant: the worker's information set is partly unobserved (i.e., $\sigma_1 > 0$) and workers predict wages with error (i.e., $\sigma_3 > 0$). Our estimator yields CIs that contain the true parameter value. In this case, the odds-based inequalities are redundant: the combined CIs are equal to those generated by the bounding inequalities alone.

Case 5 shows the bounding inequalities may produce non-empty confidence intervals when the researcher wrongly assumes workers have complete information. Conversely, the odds-based inequalities alone, or when used jointly with the bounding inequalities, produce empty CIs for θ_2 and θ_3 even when the CI for θ_α is non-empty.

4.3 Maximum Likelihood Estimates

Table 2 reports ML estimates. Given a wage predictor z_s^l , we compute the ML estimator of $(\theta_\alpha, \theta_2, \theta_3)$ assuming that z_s^l is all the information that worker s has about w_s^l ; that is,

$$\operatorname{argmax}_{(\theta_\alpha, \theta_2, \theta_3)} \left\{ \sum_{s=1}^S \sum_{l=1}^3 \mathbb{1}\{y_s^l = 1\} \ln \left(\frac{\exp(\theta_l + \theta_\alpha \mathbb{E}[w_s^l | z_s^l])}{\sum_{l'=1}^3 \exp(\theta_{l'} + \theta_\alpha \mathbb{E}[w_s^{l'} | z_s^{l'}])} \right) \right\} \quad \text{with } \theta_1 = 0. \quad (32)$$

If the researcher's wage predictor is z_{2s}^l , the ML estimator is consistent if and only if $\sigma_1 = 0$, as in cases 1 and 2, because only then the worker's wage expectation coincides with the researcher's assumed one; i.e., only then $\mathbb{E}[w_s^l | \mathcal{W}_s] = \mathbb{E}[w_s^l | z_s^l]$ for every s and l . Conversely, if $z_s^l = z_{2s}^l$ and $\sigma_1 > 0$, as in cases 3 and 4, the worker's expectation and the researcher's assumed one do not coincide, and the ML estimator is biased; in particular, it underestimates the importance of expected wages in workers' utility. In case 5, the researcher assumes workers have perfect information (i.e., $z_s^l = w_s^l$ and, thus, $\mathbb{E}[w_s^l | z_s^l] = w_s^l$) but, contrary to that assumption, workers make forecasting errors (i.e., $\sigma_3 > 0$), and the ML estimator is

²⁰For example, in unreported results, we observe that the confidence intervals for θ_2 and θ_3 defined by the odds-based inequalities include values other than the true ones when $\sigma_1 = 2$ and $\sigma_3 = 0$.

Table 2: Simulation Results - Maximum Likelihood Estimator

Case	σ_1	σ_3	z_s^l	θ_α	θ_2	θ_3
1	0	0	z_{2s}^l	1	0	1
2	0	1	z_{2s}^l	1	0	1
3	1	0	z_{2s}^l	0.91	0	0.92
4	1	1	z_{2s}^l	0.91	0	0.92
5	0	1	w_s^l	0.87	-0.03	0.87

The true parameter values are $\alpha = 1$, $\kappa^2 = 0$, and $\kappa^3 = 1$. Estimates are computed according to equation (32). Given the large sample size, unreported standard errors are always smaller than 0.001.

biased.

A comparison of the estimates in Tables 1 and 2 yields three conclusions. First, when the ML estimator identifies the true parameter values (as in cases 1 and 2), the CIs defined by the bounding inequalities either include only the true parameter values (as is the case for θ_2 and θ_3) or are very tight around them (as is the case for θ_1). Thus, little identification power is lost when using our moment inequality estimator instead of the ML estimator in cases in which the latter is consistent. Second, when the ML estimator does not identify the true parameter values (as in cases 3 to 5), our moment inequality estimator still yields CIs that contain those values. Moreover, as illustrated by case 3, the CIs produced by our moment inequality estimator may not include the corresponding ML estimates of some of the parameters; e.g., for θ_3 .

4.4 Estimates in Misspecified Models

We discuss here how the moment inequality and maximum likelihood estimates are affected when the data-generating process deviates from the setup described in Section 4.1 and the researcher does not account for these deviations when specifying the relevant moment inequalities or likelihood function. We summarize the main lessons from this analysis here; all underlying estimates and a detailed discussion of the results are in Appendix C.6.

First, we consider a setting in which both the worker’s choice and the data available to the researcher follow patterns common in models à la Roy (1951). In these settings, workers’ location choices are partly driven by idiosyncratic wage components unobserved by the researcher, and the researcher observes direct wage measures exclusively for the locations chosen by each worker. In our simulations, we vary the relative weight in workers’ choices of this idiosyncratic wage component and the idiosyncratic preference shifter; see equation (3a). As discussed in Appendix C.6.1, we find that a researcher who ignores the role that idiosyncratic wage shocks may play in workers’ location choices will generally obtain either

empty confidence sets (if using odds-based inequalities) or upward-biased estimates of the wage coefficient α (if using either bounding inequalities or a maximum likelihood estimator). However, as Table C.6 illustrates, this bias is small except when the idiosyncratic wage shocks carry more weight in workers' choices than the idiosyncratic preference shocks.

Second, we consider a setting in which workers choose their preferred location according to the model described in Section 4.1, but the data on both the wage measure w_s^l and predictor z_s^l for every location l and worker are not measured at the worker level. Instead, they are aggregated across workers within groups; that is, the researcher observes group-level averages of wages and wage shifters rather than the corresponding worker-level variables. As discussed in Appendix C.6.2, we find that the aggregation of the wage measure w_s^l biases the moment inequality estimates of the wage coefficient downward. However, this bias only materializes when we aggregate the six million observations in our simulated data into 20 (or fewer) groups; that is, for the moment inequality estimates of α to be substantially downward biased, it must be the case that the aggregated wage measure is coarse. In contrast, the aggregation of the wage predictor z_s^l causes no bias in the moment inequality estimates, although the resulting confidence intervals become wider as the aggregated predictor becomes coarser.

Third, we consider a setting in which workers have heterogeneous information sets. In the setup described in Section 4.1, we assume the information set of every worker s includes the wage predictor z_{2s}^l for every alternative l . In Appendix C.6.3, we consider a setting in which a fraction of simulated workers do not have access to detailed information on z_{2s}^l , and instead only observe whether its value for each alternative l is above or below its median. As the share of less informed workers increases, the confidence intervals for the wage coefficient defined by our moment inequalities are initially non-empty but downward biased; once the share of uninformed workers exceeds a threshold value (30% in the results shown in Table C.8), the confidence set becomes empty. Thus, generally, our inequalities yield empty confidence sets only when a sufficiently large fraction of the population of interest does not have the instrument used to build the inequalities in its information set; see Section 3.3.

5 Empirical Application

In our empirical application, we study internal migration in Brazil. We describe our data in Section 5.1, extend the worker model to allow for sectoral choice in Section 5.2, discuss our estimation approach and results in Section 5.3, and present tests of the content of workers' information sets in Section 5.4. In Section 5.5, we evaluate the effect of counterfactual changes in information and migration costs.

5.1 Data

Our main data source is the *Relação Anual de Informações Sociais* (RAIS), an administrative dataset that includes information on workers and establishments in the Brazilian formal labor market. We use the establishment’s location (microregion, the closest equivalent to a commuting zone in the Brazilian administrative map) and sector (industry) to define labor markets. We measure workers’ annual wages; thus, our wage measure includes information on work hours as well as on the number of days during the year each worker is employed.²¹

We restrict our sample to workers with similar demographic characteristics. Specifically, we study workers aged 25-64, identified as male and white, with at least a high school degree. Since RAIS only covers formal employment, workers who are not employed or hold informal jobs are absent from the dataset. Hence, our conclusions are limited to formal workers, and we accordingly restrict our sample to individuals with a persistent attachment to the formal labor market, selecting only those recorded in RAIS for at least seven years during our sample period, which spans 2002 to 2011.

To ensure we observe a large number of individuals per market, we focus on 1,000 labor markets consisting of all combinations of the 50 microregions (out of 558) and 20 sectors (out of 51) with the largest total employment reported in RAIS. We obtain the data we use in estimation by randomly sampling one million individuals per year among those employed in the 1,000 labor markets of interest. Appendix D provides more details on the RAIS data and the construction of our sample, and reports summary statistics on migration rates.²²

5.2 Empirical Model: Allowing for Choice of Sectors

The model in Section 2 treats the worker type s as fixed and focuses on workers’ migration choices. Here, we equate a worker’s type to their sector of employment, and study workers’ choices over labor markets, defined as sector-location pairs. We write our empirical model below for workers in a population defined as a demographic group sharing an origin location and prior sector of employment, that is, a prior labor market.

We index workers by i , periods by t , locations by l and sectors by s . Defining a dummy

²¹RAIS is the only Brazilian panel dataset that provides yearly individual migration choices and wages. We exploit the panel dimension to compute microregion-sector-year wages net of worker-sector fixed effects. Cross-sectional household surveys cover the informal sector but provide data at more aggregated geographic levels (e.g., PNAD), or do not provide past location and wages for the same period (e.g., census).

²²Our moment inequalities are valid if the researcher observes a subset of the markets workers choose from; see Footnote 7. Thus, the informal sector and all microregion-sector pairs not included in the analysis may still belong to the worker’s choice set. However, our estimates are based on how workers located in one of the 50 largest microregions compare two regions among these 50: when interpreting our estimates, one should think mostly of urban migrants evaluating other high-density regions.

variable y_{it}^{sl} that equals one if worker i chooses labor market (s, l) at period t , we assume that

$$y_{it}^{sl} \equiv \mathbb{1}\left\{(s, l) = \underset{\substack{s'=1, \dots, S \\ l'=1, \dots, L}}{\operatorname{argmax}} \mathbb{E}[\mathcal{U}_{it}^{s'l'} | \mathcal{J}_{it}]\right\}, \quad s = 1, \dots, S, \quad l = 1, \dots, L, \quad (33)$$

where $\mathcal{J}_{it}$ is worker i 's information set at period t , and the period- t utility of choosing market (s, l) for worker i is

$$\mathcal{U}_{it}^{sl} = u_{it}^{sl} + \varepsilon_{it}^{sl}, \quad (34a)$$

$$u_{it}^{sl} = \kappa_t^l + \tau_t^s + \alpha w_{it}^{sl}, \quad (34b)$$

where w_{it}^{sl} is the natural logarithm of the nominal wage that worker i would earn if they chose market (s, l) and the terms κ_t^l , τ_t^s and ε_{it}^{sl} are the common and idiosyncratic components of all other determinants of utility. As equation (34) captures the utility of workers in a given population defined by a demographic group, a prior labor market, and a year, the terms κ_t^l and τ_t^s capture migration and sector-switching costs, respectively, that can vary flexibly across demographic groups, origin-destination location pairs, and prior-current sector pairs. The parameter κ_t^l also captures location-specific prices and amenities, while τ_t^s captures sector-specific compensating differentials, both of which may vary over time. The key restriction that equation (34) imposes is that κ_t^l and τ_t^s are additive.

In addition to equations (33) and (34), we maintain here assumptions analogous to those in our baseline model. First, as in equation (2), we assume workers' expectations are rational. Second, similarly to equation (4), we assume the vector of idiosyncratic shocks $\varepsilon_{it} \equiv \{\varepsilon_{it}^{sl}\}_{sl}$ belongs to the information set $\mathcal{J}_{it}$. Third, analogously to equation (5), we assume that, for any workers i and j that choose sectors s and $r \neq s$, respectively, wage expectations satisfy

$$\mathbb{E}[\Delta w_{it}^{sl'} | \mathcal{J}_{it}, \mathcal{J}_{jt}] = \mathbb{E}[\Delta w_{it}^{sl'} | \mathcal{J}_{it}] = \mathbb{E}[\Delta w_t^{sl'} | \mathcal{J}_{it}] = \mathbb{E}[\Delta w_t^{sl'} | \mathcal{W}_{it}], \quad (35)$$

where $\Delta w_{it}^{sl'} = w_{it}^{sl} - w_{it}^{sl'}$ and $\Delta w_t^{sl'} = w_t^{sl} - w_t^{sl'}$, with w_t^{sl} a common component of wages in market (s, l) and $\mathcal{W}_{it}$ a vector that includes all variables in $\mathcal{J}_{it}$ other than ε_{it} . Fourth, as in equation (6), we assume that ε_{it} is independent of both $\mathcal{W}_{it}$ and $\mathcal{J}_{jt}$ for any worker $j \neq i$ and has a parametric distribution such that the following moment equality holds

$$\mathbb{E}\left[\frac{\exp(\mathbb{E}[u_{it}^{sl} - u_{it}^{sl'} | \mathcal{W}_{it}]/\psi)}{1 + \exp(\mathbb{E}[u_{it}^{sl} - u_{it}^{sl'} | \mathcal{W}_{it}]/\psi)} - y_{it}^{sl} \middle| \mathcal{W}_{it}, y_{it}^{sl} + y_{it}^{sl'} = 1\right] = 0, \quad (36)$$

for a constant $\psi > 0$. Equation (36) holds if ε_{it} follows a generalized extreme-value distribution such that the worker's choice over labor markets corresponds to a nested logit model

with nests indexed by sectors; that is, if the cumulative distribution function of ε_{it} is

$$F_\varepsilon(\varepsilon_{it}) = \exp \left(- \sum_{s=1}^S \left(\sum_{l=1}^L \exp(-\varepsilon_{it}^{sl}) \right)^\psi \right). \quad (37)$$

To derive the moment inequalities, we start from an equality analogous to that in equation (7). That is, for any worker i , period t and labor markets (s, l) and (s, l') , we write

$$(y_{it}^{sl} + y_{it}^{sl'}) (\mathbb{1}\{\mathbb{E}[\mathcal{U}_{it}^{sl} - \mathcal{U}_{it}^{sl'} | \mathcal{J}_{it}] \geq 0\} - y_{it}^{sl}) = 0. \quad (38)$$

Then, following the same steps described in Section 3, we reach a moment equality analogous to that in equation (8):

$$\mathbb{E}[y_{it}^{sl'} + y_{it}^{sl} (-\exp(-\mathbb{E}[u_{it}^{sl} - u_{it}^{sl'} | \mathcal{W}_{it}]/\psi)) | \mathcal{W}_{it}] = 0. \quad (39)$$

Given this moment equality, we can follow the steps outlined in Section 3.1 to derive second-step bounding and odds-based moment inequalities. We can also analogously derive first-step moment inequalities similar to those in Section 3.2. Importantly, the moment equality in equation (39) depends on preference parameters through the difference in the expected utility between two labor markets that differ exclusively in their location. Thus the sector-specific term τ_t^s in equation (34) differences out, and the within-sector location comparison has the same structure as in the baseline model of Section 2, up to a scaling term $1/\psi$ inherited from the assumed distribution of idiosyncratic preferences ε_{it} .

5.3 Estimation of Model Parameters

In Section 5.3.1, we detail the implementation of the moment inequality estimator. In Section 5.3.2, we discuss our estimates and compare them to those obtained using other estimators.

5.3.1 Implementation of Moment Inequalities

As discussed in Section 5.2, key variables that enter the moment inequalities we use to identify model parameters are the differences between any two locations l and l' in the location-by-sector-by-time wage shifters w_t^{sl} . We estimate these using data on observed wages. Specifically, we assume the log wage a worker would have obtained if employed in a particular sector, location, and year can be expressed as:

$$w_{it}^{sl} = w_t^{sl} + \underbrace{\gamma_i^s + \beta_\varepsilon^s \exp_{it}^s + \beta_{ee}^s (\exp_{it}^s)^2 + \beta_a^s \text{age}_{it} + \beta_{aa}^s \text{age}_{it}^2}_{\text{sector-specific skill}} + \nu_{it}^{sl}, \quad (40)$$

where exp_{it}^s denotes the number of years of employment of worker i in sector s prior to year t , age_{it} denotes the age of worker i at t and ν_{it}^{sl} is an unobserved term. Thus, in addition to the labor market-specific term w_t^{sl} , we allow wages to depend on a worker-by-sector-by-year term and a residual that varies by sector, location, worker and year. We model the worker-by-sector-by-year term as the sum of a worker-by-sector fixed effect and a function of the worker's age and sector-specific experience.²³

Given equation (40), it holds that, for any sector s and locations l and l' , $\Delta w_{it}^{sl'} = \Delta w_{it}^{sl} + \Delta \nu_{it}^{sl'}$ and, thus, equation (35) implies that $\mathbb{E}[\Delta \nu_{it}^{sl'} | \mathcal{J}_{it}] = 0$. We impose no assumption on what workers know about shifters w_t^{sl} , which account for supply and demand factors that impact the wages of all workers in a labor market, or on their sector-specific skill. In particular, at the time of choosing their labor market of employment for a year t , we allow workers to have information on a worker-by-sector specific term that the researcher cannot observe: the term γ_i^s . This selection of *sectoral* labor markets based on unobserved determinants of wages is important in, e.g., [Dix-Carneiro \(2014\)](#). In contrast, we impose the assumption that the selection of *local* labor markets is not driven by worker-by-location terms unobserved to the researcher. This assumption is consistent with the findings in [Kennan and Walker \(2011\)](#), who allow for a permanent worker-specific location-match component in wages, but conclude that the estimated effect of this component is negligible.²⁴

In addition to having a measure of wage shifters $w_t^s = (w_t^{s1}, \dots, w_t^{sL})$, which we denote in the following simply as *wages*, our inequalities require a wage predictor z_t^{sl} for each sector, region, and year. To build these predictors, we implement the following procedure. First, we fix $b \in \mathbb{N}$ and calculate the vector of b -quantiles of the distribution of w_{t-1}^{sl} across sectors and locations. Second, we build one bin for each interval constructed using two consecutive elements of this vector of quantiles. Third, we identify the bin to which each market belongs. Fourth, and finally, we compute z_t^{sl} as the average wage in year $t - 1$ across all labor markets that belong to the same bin as w_{t-1}^{sl} . Note that, as b increases, the wage predictors become closer on average to the $t - 1$ wages. Thus, a larger b can be interpreted as workers having more precise wage information. Specifically, if $b = 2$, our wage predictor assumes workers can determine if lagged wages in a labor market are above or below the median. If $b = 4$, workers can determine the quartile of the $t - 1$ wage distribution to which a labor market

²³We measure workers' sector-specific experience using information from 1993 onward. As all coefficients in equation (40) are indexed by s , we estimate them by running separate regressions for each of the 20 sectors in our analysis. These regressions are run on our sample prior to extracting the set of 1 million individuals per year that we use in our moment inequality estimation. The median R^2 across these regressions is 0.83.

²⁴Some location choices may be driven by worker-by-location-by-year shocks; e.g., job offers. In Appendix sections C.6.1 and C.6.2, we present simulations that illustrate the impact on the confidence sets implied by our inequalities of using the market-level variable $\Delta w_{it}^{sl'}$ instead of the worker-level variable $\Delta w_{it}^{sl'}$ when $\mathbb{E}[\Delta \nu_{it}^{sl'} | \mathcal{J}_{it}] \neq 0$.

belongs. As $b \rightarrow \infty$, z_t^{sl} and w_{t-1}^{sl} coincide and, thus, workers know every market’s lagged wages. We provide additional implementation details in Appendix E.1.

5.3.2 Estimation Results

First-step estimates: wage coefficient. Panel (a) in Figure 1 reports 95% CIs for α under different informational assumptions. When we set $b = 2$, and thus assume workers can at least determine whether lagged wages in any given labor market are above or below the median of the distribution of wages across all labor markets (the “2 bins” case), we obtain a 95% CI that equals [1.09, 2.21]. The large width of this interval reflects that the dummy variable indicating whether lagged wages are below or above the median is only loosely correlated with current wages. When we increase the assumed precision of workers’ information and impose that workers can at least determine the quartile to which lagged market-specific wages belong (“4 bins”), we obtain a tighter interval equal to [1.21, 1.84]. Assuming workers can classify locations according to more detailed quantiles of the wage distribution, or that they know the actual value of lagged or current wages, yields empty CIs. Thus, we reject the hypothesis that workers know lagged location-specific wages with a level of precision above quartiles. Below, we use [1.21, 1.84] as our preferred set estimator of α .²⁵

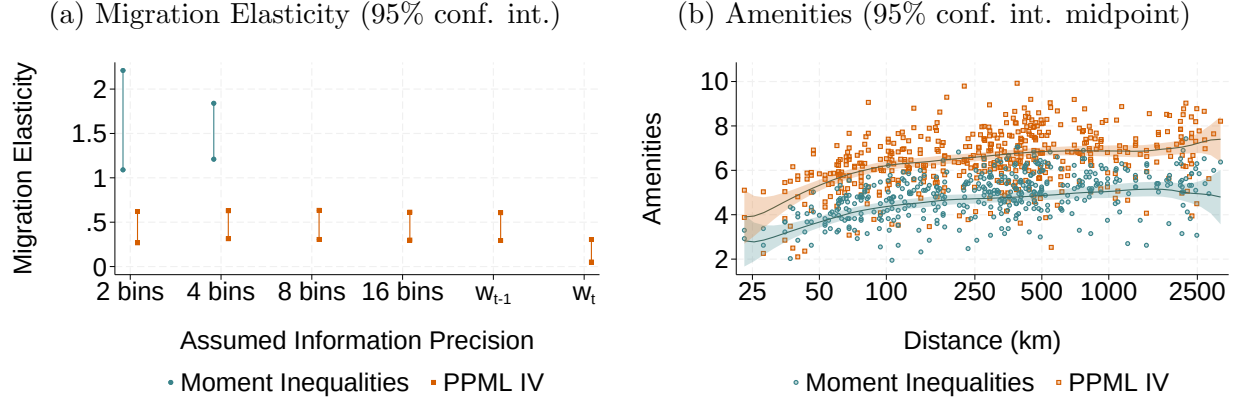
For comparison, we also include 95% CIs computed using a two-step PPML-IV estimator; see Artuç and McLaren (2015). As discussed in Appendix E.3, this estimator yields point estimates of α at the expense of assuming that all workers in the same sector in a period t (regardless of their location of residence) have the same information set and, consequently, the same wage expectations. This is a stronger assumption than the one required for our moment inequalities to be valid: the researcher must specify a (possibly different) variable that belongs to every worker’s information set, but the additional information each worker has may vary flexibly across workers and labor markets.²⁶ The PPML-IV CIs based on lagged-wage proxies lie roughly between 0.27 and 0.63, while the contemporaneous-wage PPML CI is [0.05, 0.31]. These intervals do not overlap with our preferred moment inequality CI, whose lower bound is 1.21. Relative to our moment inequality estimator, the PPML-IV estimators thus understate the value workers assign to expected monetary returns from migration.

Appendix E.4 provides two additional checks on our preferred moment inequality specification. First, this specification logically implies that moment inequalities that use coarser

²⁵As discussed in Section 3, our moment inequalities hold at the true parameter value when the instrument vector z_s belongs to the information set of every worker in the population of interest. Appendix C.6.3 presents simulation results that illustrate the behavior of our moment inequalities when a fraction of workers in the population do not have information on z_s ; we summarize these results in Section 4.4.

²⁶E.g., the moment inequality CI that uses quartiles of lagged wages as wage predictor is valid if workers, even within the same sector, location, and period, have different information, as long as all workers can at least classify labor markets into quartiles on the basis of lagged market wages.

Figure 1: Migration Elasticity and Amenities from Moment Inequalities vs. PPML-IV



Panel (a) reports 95% CIs for α . The blue circles delimit the moment inequality CIs. The absence of circles reflects empty CIs. The orange squares mark the PPML-IV CIs for the proxy-based specifications and the PPML CI for current wages. In panel (b), the blue circles indicate the midpoints of the moment inequality 95% CIs for κ_{nt}^l in 2011, and the orange squares indicate the corresponding PPML estimates. The fit lines are kernel-weighted local polynomial estimates, with shaded areas representing 95% CIs.

wage information than quartiles of lagged wages as instruments should also be valid. Indeed, when we use terciles of lagged wages as instruments, and when we stack tercile and quartile moments, we obtain similar CIs as when using quartiles alone. Second, since an implication of our baseline results is that we cannot reject the null hypothesis that all workers can classify lagged wages into quartiles, our quartile specification must yield moment inequalities valid for any subset of workers. We check this implication by re-estimating the quartile specification on five random 20% subsamples of workers; all five CIs remain non-empty and similar to those obtained when using the full sample of workers.

Second-step estimates: amenities. Panel (b) in Figure 1 illustrates the moment inequality estimates of the bilateral non-wage component κ_{nt}^l ; specifically, it displays midpoints of the 95% CI for each term κ_{nt}^l . For comparison, we also display PPML-IV estimates.²⁷ Although we estimate each parameter κ_{nt}^l without imposing any restriction on its variability, our estimates tend to increase in the distance between locations n and l , consistent with these parameters accounting for migration costs in our model. The differences in levels between the PPML-IV and the moment inequality estimates are substantial, the latter being on average 29% smaller than the former. Moreover, if we convert migration costs into their log-wage equivalents by dividing them by the estimates of the wage coefficient α obtained by each

²⁷We report estimates for 2011 and all 477 origin-destination pairs with both PPML-IV and moment inequality estimates. For the sake of clarity, we make explicit here the dependency of the amenity term on the origin location, which we index by n . Appendix E.2 displays the moment inequality CIs. Appendix E.5 shows that the estimates of κ_{nt}^l exhibit significant variation over time, but within-pair over-time variation is smaller than persistent cross-pair dispersion.

estimation method, we find that the moment inequality estimates are 74% smaller. In sum, estimation procedures commonly used in the migration literature yield estimates of migration costs or, more generally, of the relative importance of non-wage variables in migration choices that are larger than our moment inequality estimates.

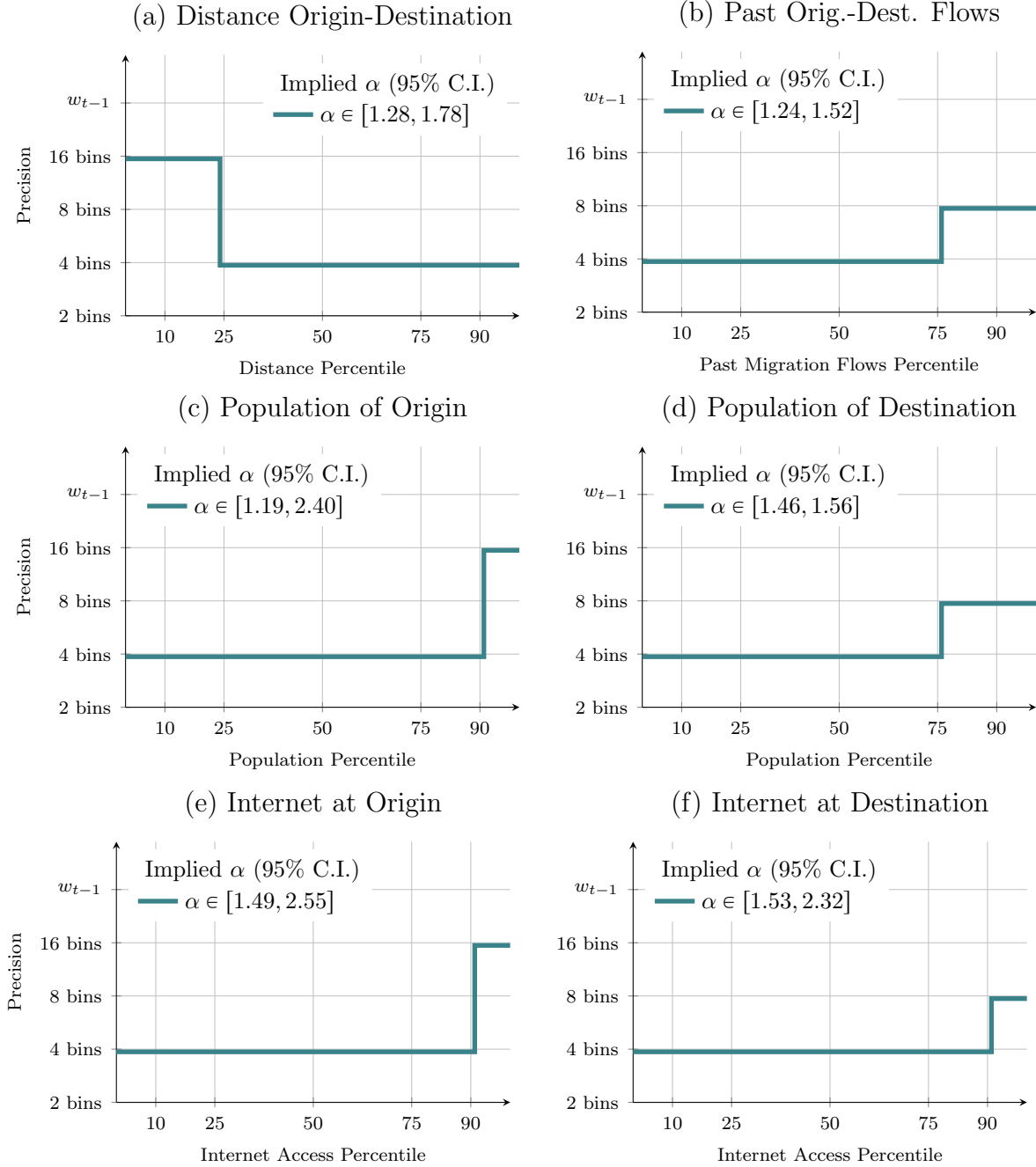
5.4 Tests of Information Heterogeneity

As panel (a) in Figure 1 shows, the moment inequality 95% CI for α is nonempty when we assume workers know the quartile to which lagged wages belong, but is empty when we assume *all* workers can classify *all* markets into eight (or more) bins. However, it is possible that some workers are more informed than others or that workers have more information about some markets than others. We explore here whether observed migration choices are consistent with certain workers having more precise information about wages in some markets. We do so by checking whether the moment inequality 95% CI for the wage coefficient α remains non-empty when we assume that a group of workers has extra wage information about a group of markets. In particular, we consider worker groups defined by the population and internet penetration in their prior location of residence, and groups of markets defined by their distance and past migration flows from the worker’s location and by their population and internet penetration. A key assumption underlying these tests is that the wage coefficient α is common across all groups of workers and markets; that is, for example, we impose that α does not vary with the internet penetration in the worker’s prior location.²⁸

We implement the same testing procedure for each dimension of heterogeneity. First, we classify workers (or markets) into six intervals delimited by the 10th, 25th, 50th, 75th, and 90th percentiles of the distribution of workers (or markets) along the corresponding dimension. We then order these intervals according to the direction along which we hypothesize information may be more precise. For example, we classify workers into intervals depending on the internet penetration in the location of residence, and order these from higher to lower internet penetration. Consistent with our finding in panel (a) in Figure 1, we start from a baseline information set according to which all workers can classify all markets into quartiles, and test whether workers in the first interval can further classify markets into eight bins. In practice, this translates into wage predictors with different levels of precision in different combinations of workers and markets, affecting the values of the instruments in equation (30) and thus changing our inequalities. If the resulting 95% CI for α is empty, we reject that

²⁸The distance between markets equals the geodesic distance between their centroids. Past migration flows are measured as the total number of workers recorded in RAIS as having migrated between any two locations in the three years prior to our sample period (1999-2001). The population of each location is computed as the total employment in RAIS in 1999-2001. The measure of internet penetration in each location equals the average share of households with broadband internet access between 2007 and 2011. See Appendix D.

Figure 2: Testing for Heterogeneous Information Sets



This figure displays the information-precision schedules that cannot be rejected in the data. Each panel studies heterogeneity along one dimension: distance, past migration flows, population at the origin or destination, and internet access at the origin or destination. The solid line shows the most precise schedule that is not rejected before the algorithm reaches the first rejected next test. The legend reports the 95% confidence interval for α implied by the displayed schedule. Precision is measured by the wage proxy used in the information set: 4 bins correspond to quartiles of lagged wages, 8 and 16 bins correspond to finer partitions, and w_{t-1} corresponds to lagged wages themselves.

assumption and display the previous non-rejected schedule. If it is non-empty, we increase the level of precision to 16 bins on that first interval and perform a new test. Calling B_j the maximum precision level tested and not rejected for the j th interval, the next iteration maintains B_j on that interval and searches for the maximum level of precision in interval $j + 1$, up to precision B_j .

Figure 2 displays our results. Panel (a) shows that we cannot reject that workers are better informed about wages in labor markets within the closest quartile of distance from their location of residence, roughly 390 km. For those markets, we cannot reject that workers know lagged wages with a precision equivalent to 16 bins. As discussed in Section 5.5.1, migration rates in our model increase in workers' information. Thus, the fact that migration rates decrease in distance (Beine et al., 2016) may be due less than previously thought to migration costs increasing in this dimension and more to the role information plays in migration choices. In panel (b), destinations in the top quartile of past origin-destination migration flows can be assigned 8-bin information, while extending 8-bin information to the top half of past-flow destinations is rejected. This finding may explain why workers of a particular origin tend to persistently migrate to the same destinations, providing an explanation for the impact of enclaves on migration flows (Munshi, 2020). Panels (c) and (d) show that workers from the top decile of origins by population can be assigned 16-bin information, and that all workers can be assigned 8-bin information about destinations in the top quartile of the population distribution. The information premium from living in highly populated areas adds to the benefits of cities discussed in, for example, Glaeser and Maré (2001) and De la Roca and Puga (2017). Finally, panels (e) and (f) show analogous patterns for internet access: workers from the top decile of origins by internet access can be assigned 16-bin information, while destinations in the top decile of internet access can be assigned 8-bin information. This finding is consistent with prior evidence on the informational impact of broadband internet access (Akerman et al., 2022).

5.5 Counterfactuals

To study the relevance of the estimates described above, we quantify the impact of changes in information sets and migration costs on workers' location choices, expected utility and speed of adjustment to wage shocks. Here, we omit the impact that changes in information and costs may have on migration flows through changes in wages and prices.

While we compute the moment inequality estimates above without imposing any assumption on the stochastic process of wages, specifying this element of the model is needed to determine workers' migration choices. Hence, to compute the results in this section, we use observed wages in our sample of 50 regions and 20 sectors over 2002–2011 to estimate a

common AR(1) process for wages with sector- and location-specific drifts. We estimate the coefficient of autocorrelation in wages to be 0.93.

Also, while our estimation procedure makes it computationally feasible to obtain CIs for all model parameters, computing model predictions that account for uncertainty in all parameter estimates is costly, as it requires building a multidimensional grid that spans all CIs and evaluating our model at each point in that grid. Instead, we consider all points in the CI for the wage coefficient α , but calibrate amenities by regressing the midpoint of the moment inequality CIs (see panel (b) in Figure 1) on a constant and the log distance between the origin and the destination locations.

We compute model implications analytically using multinomial-logit choice probabilities and average outcomes over 100 simulated paths of wages across the 50 locations and 20 sectors in our sample.

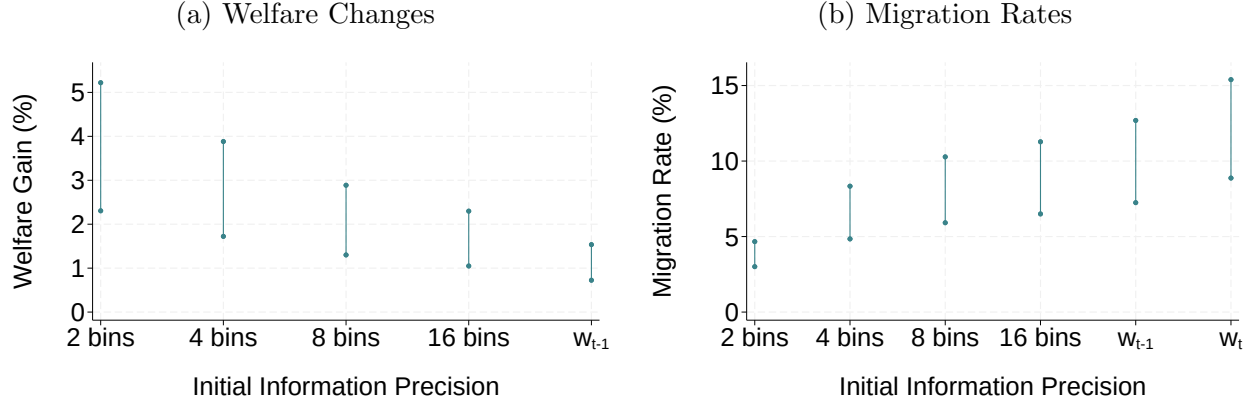
5.5.1 Changes in Information Sets

We evaluate the impact of providing workers with information on market wages in all locations. Specifically, we assume all workers have an initial level of information on wages common across all destinations, and focus on the impact of receiving perfect wage information on workers' migration probability, measured as the probability a worker changes locations in two consecutive periods, and welfare. We measure welfare as the average utility across simulated workers and periods, including the contribution of idiosyncratic tastes for locations and, importantly, using ex-post wages as the income measure. Hence, workers with perfect wage information choose locations maximizing their ex post utility, while workers with incomplete information maximize expected utility, and may thus choose locations that do not offer the highest utility ex post.

Panel (a) in Figure 3 shows that the gains from improving workers' information can be substantial. For workers whose initial information only allows them to determine whether lagged market wages are above or below the median, welfare gains are between 2.3 and 5.2%, with the highest gains in this interval corresponding to the model that sets the wage coefficient α at the higher value. The gains remain significant for workers who were initially better informed. Even if all workers observed lagged wages perfectly, a hypothesis we reject in panel (a) in Figure 1, the gains from observing contemporaneous wages would still range between 0.7 and 1.5%.

Panel (b) in Figure 3 reports migration rates for workers with different information. Migration rates increase steeply in the precision of workers' wage information. They are between 3.0 and 4.7% for workers with the coarsest information set we consider, and between

Figure 3: Effects of Providing Full Information About Wages



For the information sets indicated on the horizontal axis, panel (a) displays changes in welfare as a result of giving workers perfect wage information. Panel (b) displays migration rates for different information sets. The intervals illustrate the range of model predictions consistent with a value of α in the 95% CI [1.21, 1.84].

8.9 and 15.4% when information is complete.²⁹ While it is hardly surprising that better-informed workers have higher welfare, as better-informed workers choose more often the location that yields higher ex post utility, it is not obvious that better-informed workers will have higher migration rates. Workers in our model are rational, and thus, when acquiring additional wage information, expected wages go up for certain workers and locations and down for others in such a way that average expectations do not change. The reason why average migration rates change while average expectations do not is that a worker's migration probabilities are a nonlinear function of the worker's expectations.

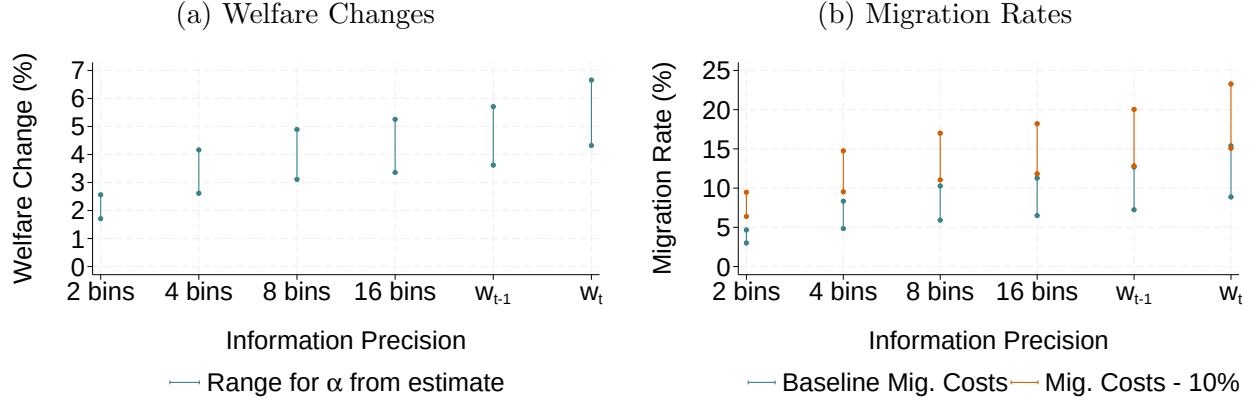
5.5.2 Reducing Migration Costs

As shown in Bryan and Morten (2019) and Morten and Oliveira (2024), reducing physical barriers to geographic mobility is an important policy lever to alleviate spatial misallocation. However, the benefits of reducing migration costs may depend on whether agents are well-informed about the economic opportunities in different regions. In this section, we evaluate how the effect of reductions in migration costs depends on workers' wage information. Specifically, for several information sets, we compute the predictions of our model for a 10% reduction in our calibrated migration costs.

Panel (a) in Figure 4 reveals that the welfare gains from a 10% reduction in migration

²⁹When we assume all workers know the quartile to which lagged wages belong, which is the strongest informational assumption we tested without rejecting it (see Section 5.3.2), our model predicts a migration rate between 4.8 and 8.3%, which includes the average migration rate of 6.8% observed in the sample; see Appendix D.2. This illustrates that our estimated model fits the baseline migration elasticity in the data.

Figure 4: Effects of Reducing Migration Costs, by Information Level



This figure displays counterfactual changes in welfare (panel (a)) and migration rates (panel (b)) from a 10% reduction in migration costs, depending on workers' information. The intervals correspond to the range of model predictions consistent with a value of α in the 95% confidence interval [1.21, 1.84].

costs increase with the precision of workers' wage information. When workers are fully informed, the welfare gains range from 4.3 to 6.7%, depending on the value of α . When workers only discern whether lagged wages in a location are above or below the median, the same reduction in migration costs yields welfare gains between 1.7 and 2.6%. Under the preferred quartile-information specification, the gains range from 2.6 to 4.2%.

Panel (b) in Figure 4 illustrates the migration-rate effects of reducing migration costs. Migration rates increase significantly for all information levels. In relative terms, the increase is largest for workers with less precise wage information, but welfare gains remain smaller for these workers, as their additional location changes are based on coarser wage information.

5.5.3 Speed of Adjustment in Response to Local Shocks

Our estimates imply two potentially countervailing forces in the dynamic adjustment of labor to location-specific wage changes. On the one hand, workers have less precise wage information than implied by standard full-information models. On the other hand, our estimate of the wage coefficient α is larger than the estimate we would have obtained under the full-information assumption. Appendix E.6 uses a simple counterfactual exercise featuring a permanent wage increase in São Paulo to study how workers' information and the wage preference parameter affect the model-predicted short- and long-run labor responses to such a wage change. The results show that incomplete information can both limit the long-run response and slow the adjustment, but a higher wage coefficient can more than offset this force, generating both a larger long-run response and faster convergence toward it.

6 Conclusion

We introduce a new moment inequality method to measure the impact of migration costs and information frictions on workers’ location choices. Our method allows workers’ information sets to be unobserved by the researcher and to vary flexibly between workers, and migration costs to vary flexibly across pairs of origin and destination locations. Applying our method to data on the internal migration of formal workers in Brazil, we obtain four main results. First, workers have heterogeneous information on location-specific wages. Geography, past migration links, population, and internet access all shape the precision of workers’ wage information. Second, accounting for this heterogeneity in information sets alters the mapping from observed location choices and wages to workers’ preferences. More specifically, our wage preference estimates are three times larger than those obtained using common estimation procedures, and our estimates of bilateral non-wage components are, on average, about 29% lower. Third, our estimated model predicts that providing wage information to workers results in increases in both migration rates and welfare. Fourth, policies that reduce migration costs by, for example, improving transportation infrastructure are more effective in improving worker welfare the better informed workers are about destination characteristics.

The two-step moment inequality estimator we introduce may be used more generally to estimate multinomial discrete choice models when the choice set is large, payoffs are parameterized with choice-specific fixed effects, and information sets are unobserved to the researcher and potentially heterogeneous across agents. This type of model may be suitable to study, for example, student decisions over which schools to apply to, or patient decisions over which hospital to attend.

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A Proofs

Appendix A.1 provides the proof of Theorem 1. Appendix A.2, following Dickstein et al. (2023), provides the proof of Theorem 2. Appendix A.3 contains the proof of Theorem 3.

A.1 Proof of Theorem 1

Equation (1) implies that, for any worker i of type s and locations l and l' , it holds that

$$(y_{is}^l + y_{is}^{l'}) (\mathbb{1}\{\mathbb{E}[\mathcal{U}_{is}^l - \mathcal{U}_{is}^{l'} | \mathcal{J}_{is}] \geq 0\} - y_{is}^l) = 0.$$

Equations (2) to (5) imply we can rewrite this equality as

$$(y_{is}^l + y_{is}^{l'}) (\mathbb{1}\{\Delta\kappa^{ll'} + \alpha\mathbb{E}[\Delta w_s^{ll'} | \mathcal{W}_{is}] + \Delta\varepsilon_{is}^{ll'} \geq 0\} - y_{is}^l) = 0. \quad (\text{A.1})$$

Equation (6) implies the expectation of this equality conditional on $\mathcal{W}_{is}$ and a dummy variable that equals one if worker i of type s chooses either location l or location l' equals

$$\mathbb{E} \left[\frac{\exp(\Delta\kappa^{ll'} + \alpha\mathbb{E}[\Delta w_s^{ll'} | \mathcal{W}_{is}])}{1 + \exp(\Delta\kappa^{ll'} + \alpha\mathbb{E}[\Delta w_s^{ll'} | \mathcal{W}_{is}])} - y_{is}^l \middle| \mathcal{W}_{is}, y_{is}^l + y_{is}^{l'} = 1 \right] = 0,$$

which implies the following moment equality

$$\mathbb{E}[1 - y_{is}^l - y_{is}^l \exp(-\Delta\kappa^{ll'} - \alpha\mathbb{E}[\Delta w_s^{ll'} | \mathcal{W}_{is}]) | \mathcal{W}_{is}, y_{is}^l + y_{is}^{l'} = 1] = 0.$$

Given the conditioning on the event $y_{is}^l + y_{is}^{l'} = 1$, we can further simplify this equality as

$$\mathbb{E}[y_{is}^{l'} + y_{is}^l (-\exp(-(\Delta\kappa^{ll'} + \alpha\mathbb{E}[\Delta w_s^{ll'} | \mathcal{W}_{is}]))) | \mathcal{W}_{is}] = 0. \quad (\text{A.2})$$

As $-\exp(-x)$ is concave in x , a linear approximation to it bounds it from above; that is, $-\exp(-a)(1 + a - x) \geq -\exp(-x)$ for all $a \in \mathbb{R}$. Thus, given an approximation point $e_{is}^{ll'}$ for each worker i of type s , we derive the inequality

$$\mathbb{E}[y_{is}^{l'} - y_{is}^l \exp(-e_{is}^{ll'})(1 + e_{is}^{ll'} - \Delta\kappa^{ll'} - \alpha\mathbb{E}[\Delta w_s^{ll'} | \mathcal{W}_{is}]) | \mathcal{W}_{is}] \geq 0. \quad (\text{A.3})$$

Consider the alternative moment,

$$\mathbb{E}[y_{is}^{l'} - y_{is}^l \exp(-e_{is}^{ll'})(1 + e_{is}^{ll'} - \Delta\kappa^{ll'} - \alpha\Delta w_s^{ll'}) | \mathcal{W}_{is}]. \quad (\text{A.4})$$

Given $\nu_{is}^{ll'} \equiv \Delta w_s^{ll'} - \mathbb{E}[\Delta w_s^{ll'} | \mathcal{J}_{is}]$ and equation (5), we can rewrite this moment as

$$\mathbb{E}[y_{is}' - y_{is}^l \exp(-e_{is}^{ll'}) (1 + e_{is}^{ll'} - \Delta \kappa^{ll'} - \alpha(\mathbb{E}[\Delta w_s^{ll'} | \mathcal{W}_{is}] + \nu_{is}^{ll'}) | \mathcal{W}_{is}].$$

As $\mathcal{W}_{is} \subseteq \mathcal{J}_{is}$, we use the Law of Iterated Expectations (LIE) to write this moment as

$$\mathbb{E}[\mathbb{E}[y_{is}' - y_{is}^l \exp(-e_{is}^{ll'}) (1 + e_{is}^{ll'} - \Delta \kappa^{ll'} - \alpha(\mathbb{E}[\Delta w_s^{ll'} | \mathcal{W}_{is}] + \nu_{is}^{ll'}) | \mathcal{J}_{is}) | \mathcal{W}_{is}].$$

As $(y_{is}^l, e_{is}^{ll'}) \subseteq \mathcal{J}_{is}$, we can further write this moment as

$$\mathbb{E}[y_{is}' - y_{is}^l \exp(-e_{is}^{ll'}) (1 + e_{is}^{ll'} - \Delta \kappa^{ll'} - \alpha(\mathbb{E}[\Delta w_s^{ll'} | \mathcal{W}_{is}] + \mathbb{E}[\nu_{is}^{ll'} | \mathcal{J}_{is}]) | \mathcal{W}_{is}].$$

As equation (5) implies $\mathbb{E}[\nu_{is}^{ll'} | \mathcal{J}_{is}] = 0$, the moments in equations (A.3) and (A.4) coincide. Thus, the moment inequality in equation (A.3) implies the following inequality:

$$\mathbb{E}[y_{is}' - y_{is}^l \exp(-e_{is}^{ll'}) (1 + e_{is}^{ll'} - \Delta \kappa^{ll'} - \alpha \Delta w_s^{ll'}) | \mathcal{W}_{is}] \geq 0.$$

Finally, as $z_s \subseteq \mathcal{W}_{is}$, the LIE implies that $m^{ll'}(z_s, \Delta \kappa^{ll'}) \geq 0$, proving Theorem 1. ■

A.2 Proof of Theorem 2

We start from the moment equality in equation (A.2). Consider the alternative moment

$$\mathbb{E}[y_{is}' + y_{is}^l (-\exp(-(\Delta \kappa^{ll'} + \alpha \Delta w_s^{ll'}))) | \mathcal{W}_{is}]. \quad (\text{A.5})$$

Given $\nu_{is}^{ll'} \equiv \Delta w_s^{ll'} - \mathbb{E}[\Delta w_s^{ll'} | \mathcal{J}_{is}]$ and equation (5), we can rewrite this moment as

$$\mathbb{E}[y_{is}' + y_{is}^l (-\exp(-(\Delta \kappa^{ll'} + \alpha(\mathbb{E}[\Delta w_s^{ll'} | \mathcal{W}_{is}] + \nu_{is}^{ll'})))) | \mathcal{W}_{is}].$$

As, by definition, $\mathcal{W}_{is} \subseteq \mathcal{J}_{is}$, we can use the LIE to further write this moment as

$$\mathbb{E}[\mathbb{E}[y_{is}' + y_{is}^l (-\exp(-(\Delta \kappa^{ll'} + \alpha(\mathbb{E}[\Delta w_s^{ll'} | \mathcal{W}_{is}] + \nu_{is}^{ll'})))) | \mathcal{J}_{is}] | \mathcal{W}_{is}].$$

As $-\exp(x)$ is concave in $x \in \mathbb{R}$, equation (2) and Jensen's inequality imply the inequality

$$\begin{aligned} & \mathbb{E}[\mathbb{E}[y_{is}' + y_{is}^l (-\exp(-(\Delta \kappa^{ll'} + \alpha(\mathbb{E}[\Delta w_s^{ll'} | \mathcal{W}_{is}] + \nu_{is}^{ll'})))) | \mathcal{J}_{is}] | \mathcal{W}_{is}] \\ & \leq \mathbb{E}[y_{is}' + y_{is}^l (-\exp(-(\Delta \kappa^{ll'} + \alpha \mathbb{E}[\Delta w_s^{ll'} | \mathcal{W}_{is}]))) | \mathcal{W}_{is}], \end{aligned}$$

where the left-hand side coincides with the moment in equation (A.5), and the right-hand side coincides with the moment in equation (A.2). Thus, given equation (A.2), it holds that

$$\mathbb{E}[y_{is}^{l'} + y_{is}^l(-\exp(-(\Delta\kappa^{ll'} + \alpha\Delta w_s^{ll'})))]|\mathcal{W}_{is}] \leq 0.$$

After multiplying both sides of this equation by -1 , we obtain

$$\mathbb{E}[y_{is}^l \exp(-(\Delta\kappa^{ll'} + \alpha\Delta w_s^{ll'})) - y_{is}^{l'}|\mathcal{W}_{is}] \geq 0.$$

Finally, as $z_s \subseteq \mathcal{W}_{is}$, the LIE implies $m_o^{ll'}(z_s, \Delta\kappa^{ll'}) \geq 0$, proving Theorem 2. ■

A.3 Proof of Theorem 3

Equation (A.1) implies that, for any worker i of type s , any worker j of type r , and any locations l and l' , it holds that

$$y_{jr}^{l'}(y_{is}^l + y_{is}^{l'})(\mathbb{1}\{\Delta\kappa^{ll'} + \alpha\mathbb{E}[\Delta w_s^{ll'}|\mathcal{W}_{is}] + \Delta\varepsilon_{is}^{ll'} \geq 0\} - y_{is}^l) = 0. \quad (\text{A.6})$$

Equation (6) implies the expectation of this equality conditional on $\mathcal{W}_{is}$, $\mathcal{W}_{jr}$, and a dummy variable that equals one if worker i of type s chooses either location l or location l' equals

$$\mathbb{E}\left[y_{jr}^{l'}\left(\frac{\exp(\Delta\kappa^{ll'} + \alpha\mathbb{E}[\Delta w_s^{ll'}|\mathcal{W}_{is}])}{1 + \exp(\Delta\kappa^{ll'} + \alpha\mathbb{E}[\Delta w_s^{ll'}|\mathcal{W}_{is}])} - y_{is}^l\right)\middle|\mathcal{W}_{is}, \mathcal{W}_{jr}, y_{is}^l + y_{is}^{l'} = 1\right] = 0,$$

which, after some algebra, implies

$$\mathbb{E}[y_{jr}^{l'}(1 - y_{is}^l - y_{is}^l \exp(-(\Delta\kappa^{ll'} + \alpha\mathbb{E}[\Delta w_s^{ll'}|\mathcal{W}_{is}])))|\mathcal{W}_{is}, \mathcal{W}_{jr}, y_{is}^l + y_{is}^{l'} = 1] = 0.$$

Given the conditioning on the event $y_{is}^l + y_{is}^{l'} = 1$, we can further simplify this equality as

$$\mathbb{E}[y_{is}^{l'}y_{jr}^{l'} + y_{is}^ly_{jr}^{l'}(-\exp(-(\Delta\kappa^{ll'} + \alpha\mathbb{E}[\Delta w_s^{ll'}|\mathcal{W}_{is}])))|\mathcal{W}_{is}, \mathcal{W}_{jr}] = 0.$$

As $-\exp(-x)$ is concave in $x \in \mathbb{R}$, any linear approximation to it bounds it from above. Thus, given an approximation point $e_{is}^{ll'}$ for each worker i of type s , we derive the inequality

$$\mathbb{E}[y_{is}^{l'}y_{jr}^{l'} + y_{is}^ly_{jr}^{l'}\exp(-e_{is}^{ll'})(-(1 + e_{is}^{ll'}) + \Delta\kappa^{ll'} + \alpha\mathbb{E}[\Delta w_s^{ll'}|\mathcal{W}_{is}])|\mathcal{W}_{is}, \mathcal{W}_{jr}] \geq 0. \quad (\text{A.7})$$

Consider the alternative moment,

$$\mathbb{E}[y_{is}^{l'}y_{jr}^{l'} + y_{is}^ly_{jr}^{l'}\exp(-e_{is}^{ll'})(-(1 + e_{is}^{ll'}) + \Delta\kappa^{ll'} + \alpha\Delta w_s^{ll'})|\mathcal{W}_{is}, \mathcal{W}_{jr}]. \quad (\text{A.8})$$

Given $\nu_{is}^{ll'} \equiv \Delta w_s^{ll'} - \mathbb{E}[\Delta w_s^{ll'} | \mathcal{J}_{is}]$ and equation (5), we can rewrite this moment as

$$\mathbb{E}[y_{is}' y_{jr}' + y_{is}^l y_{jr}' \exp(-e_{is}^{ll'})(-(1 + e_{is}^{ll'}) + \Delta \kappa^{ll'} + \alpha(\mathbb{E}[\Delta w_s^{ll'} | \mathcal{W}_{is}] + \nu_{is}^{ll'})) | \mathcal{W}_{is}, \mathcal{W}_{jr}].$$

As, by definition, $\mathcal{W}_{is} \subseteq \mathcal{J}_{is}$ and $\mathcal{W}_{jr} \subseteq \mathcal{J}_{jr}$, we can use the LIE to rewrite this moment as

$$\mathbb{E}[\mathbb{E}[y_{is}' y_{jr}' + y_{is}^l y_{jr}' \exp(-e_{is}^{ll'})(-(1 + e_{is}^{ll'}) + \Delta \kappa^{ll'} + \alpha(\mathbb{E}[\Delta w_s^{ll'} | \mathcal{W}_{is}] + \nu_{is}^{ll'})) | \mathcal{J}_{is}, \mathcal{J}_{jr}] | \mathcal{W}_{is}, \mathcal{W}_{jr}].$$

As $e_{is}^{ll'} \subseteq \mathcal{J}_{is} \cup \mathcal{J}_{jr}$, we can further write this moment as

$$\mathbb{E}[y_{is}' y_{jr}' + y_{is}^l y_{jr}' \exp(-e_{is}^{ll'})(-(1 + e_{is}^{ll'}) + \Delta \kappa^{ll'} + \alpha(\mathbb{E}[\Delta w_s^{ll'} | \mathcal{W}_{is}] + \mathbb{E}[\nu_{is}^{ll'} | \mathcal{J}_{is}, \mathcal{J}_{jr}])) | \mathcal{W}_{is}, \mathcal{W}_{jr}].$$

As equation (5) implies $\mathbb{E}[\nu_{is}^{ll'} | \mathcal{J}_{is}, \mathcal{J}_{jr}] = 0$, the moments in equations (A.7) and (A.8) coincide. Thus, the moment inequality in equation (A.7) implies the following inequality:

$$\mathbb{E}[y_{is}' y_{jr}' + y_{is}^l y_{jr}' \exp(-e_{is}^{ll'})(-(1 + e_{is}^{ll'}) + \Delta \kappa^{ll'} + \alpha \Delta w_s^{ll'}) | \mathcal{W}_{is}, \mathcal{W}_{jr}] \geq 0. \quad (\text{A.9})$$

This moment inequality is one of the two that we will combine to obtain the inequality in equation (26). To obtain the second moment inequality, we start from

$$y_{is}^l (y_{jr}^l + y_{jr}') (\mathbb{1}\{\Delta \kappa^{ll'} + \alpha \mathbb{E}[\Delta w_r^{ll'} | \mathcal{W}_{jr}] + \Delta \varepsilon_{jr}^{ll'} \geq 0\} - y_{jr}') = 0, \quad (\text{A.10})$$

which is analogous to that in equation (A.6). Following the same steps described above to go from equation (A.6) to equation (A.9), we can derive the following inequality

$$\mathbb{E}[y_{is}^l y_{jr}^l + y_{is}^l y_{jr}' \exp(-e_{jr}^{ll'})(-(1 + e_{jr}^{ll'}) + \Delta \kappa^{ll'} + \alpha \Delta w_r^{ll'}) | \mathcal{W}_{is}, \mathcal{W}_{jr}] \geq 0. \quad (\text{A.11})$$

As the moments in equations (A.9) and (A.11) have the same conditioning set, we can add them. If we further impose that $e_{is}^{ll'} = e_{jr}^{ll'} = e_{isjr}^{ll'}$, we obtain the following moment inequality:

$$\mathbb{E}[y_{is}^l y_{jr}^l + y_{is}' y_{jr}' + y_{is}^l y_{jr}' \exp(-e_{isjr}^{ll'})(-(1 + e_{isjr}^{ll'}) + \alpha(\Delta w_s^{ll'} + \Delta w_r^{ll'})) | \mathcal{W}_{is}, \mathcal{W}_{jr}] \geq 0.$$

As $z_s \subseteq \mathcal{W}_{is}$ and $z_r \subseteq \mathcal{W}_{jr}$, the LIE implies $\mathbb{M}^{ll'}(z_s, z_r, \alpha) \geq 0$, proving Theorem 3. ■

Supplemental Appendix for “Measuring Information Frictions in Migration Decisions: A Revealed-Preference Approach”

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B Additional Derivations

Appendix B.1 derives expressions appearing in Section 3.1.1. Appendix B.2 proves Corollary 1. Appendix B.3 derives expressions appearing in Section 3.1.2. Appendix B.4 proves Corollary 2. Appendix B.5 derives expressions appearing in Section 3.2. Appendix B.6 proves Corollary 3. Appendix B.7 shows how we use inequalities to identify the parameter θ_α .

B.1 Second-Step Bounding Inequalities: Additional Derivations

Derivation of equation (14). Consider points $e_{is}^{ll'}$ such that $e_{is}^{ll'} = h^{ll'}(z_s, \Delta\theta_{ll'})$ for some function $h^{ll'}(\cdot)$. We compute the function in equation (14) by finding the value of $h^{ll'}(z_s, \Delta\theta_{ll'})$ that minimizes the moment in equation (12) at each value of $\Delta\theta_{ll'}$. Specifically, given z_s and $\Delta\theta_{ll'}$, the first-order condition of the moment in equation (12) with respect to $h^{ll'}(z_s, \Delta\theta_{ll'})$ is

$$\mathbb{E}[y_{is}^l(h^{ll'}(z_s, \Delta\theta_{ll'}) - (\Delta\theta_{ll'} + \alpha\Delta w_s^{ll'}))|z_s] = 0,$$

or, equivalently, $\mathbb{E}[h^{ll'}(z_s, \Delta\theta_{ll'}) - (\Delta\theta_{ll'} + \alpha\Delta w_s^{ll'})|z_s, y_{is}^l = 1] = 0$. Solving for $h^{ll'}(z_s, \Delta\theta_{ll'})$, we obtain the expression in equation (14).

Derivation of equation (15). Equations (12) to (14) imply the following inequality

$$\mathbb{E}[y_{is}^{l'} - y_{is}^l \exp(-(\Delta\theta_{ll'} + \alpha\mathbb{E}[\Delta w_s^{ll'}|z_s, y_{is}^l = 1]))(1 - \alpha(\Delta w_s^{ll'} - \mathbb{E}[\Delta w_s^{ll'}|z_s, y_{is}^l = 1]))|z_s] \geq 0.$$

We can simplify this inequality as

$$\begin{aligned} \mathbb{E}[y_{is}^{l'}|z_s] &\geq \exp(-(\Delta\theta_{ll'} + \alpha\mathbb{E}[\Delta w_s^{ll'}|z_s, y_{is}^l = 1])) \\ &\quad \times (\mathbb{E}[y_{is}^l|z_s] - \alpha\mathbb{E}[y_{is}^l(\Delta w_s^{ll'} - \mathbb{E}[\Delta w_s^{ll'}|z_s, y_{is}^l = 1])|z_s]), \end{aligned}$$

or equivalently,

$$\mathbb{E}[y_{is}^{l'}|z_s] \geq \exp(-(\Delta\theta_{ll'} + \alpha\mathbb{E}[\Delta w_s^{ll'}|z_s, y_{is}^l = 1]))$$

$$\times (\mathbb{E}[y_{is}^l | z_s] - \alpha(\mathbb{E}[\Delta w_s^{ll'} | z_s, y_{is}^l = 1] \mathbb{E}[y_{is}^l | z_s] - \mathbb{E}[\Delta w_s^{ll'} | z_s, y_{is}^l = 1] \mathbb{E}[y_{is}^l | z_s])).$$

Eliminating terms that cancel each other, we obtain the inequality in equation (15); i.e.,

$$\frac{\mathbb{E}[y_{is}^l | z_s]}{\mathbb{E}[y_{is}^{l'} | z_s]} \exp(-\alpha \mathbb{E}[\Delta w_s^{ll'} | z_s, y_{is}^l = 1]) \leq \exp(\Delta \theta_{ll'}).$$

Derivation of equation (16). Swapping the indices l and l' in equation (15) we obtain

$$\frac{\mathbb{E}[y_{is}^{l'} | z_s]}{\mathbb{E}[y_{is}^l | z_s]} \exp(-\alpha \mathbb{E}[\Delta w_s^{ll'} | z_s, y_{is}^{l'} = 1]) \leq \exp(\Delta \theta_{l'l}).$$

Rearranging terms, we obtain the inequality in equation (16).

B.2 Second-Step Bounding Inequalities: Proof of Corollary 1

Equations (12) to (14) imply the following moment inequality:

$$\mathbb{E}[y_{is}^l \exp(-(\Delta \theta_{ll'} + \alpha \mathbb{E}[\Delta w_s^{ll'} | z_s, y_{is}^l = 1])) - y_{is}^{l'} | z_s] \geq 0. \quad (\text{B.1})$$

Assuming $z_s \subseteq \mathcal{W}_{is}$ and using the LIE, we can write

$$\mathbb{E}[\mathbb{E}[y_{is}^l \exp(-(\Delta \theta_{ll'} + \alpha \mathbb{E}[\Delta w_s^{ll'} | z_s, y_{is}^l = 1])) - y_{is}^{l'} | \mathcal{W}_{is}] | z_s] \geq 0.$$

Given that $z_s \subseteq \mathcal{W}_{is}$, we can further rewrite

$$\mathbb{E}[\mathbb{E}[y_{is}^l | \mathcal{W}_{is}] \exp(-(\Delta \theta_{ll'} + \alpha \mathbb{E}[\Delta w_s^{ll'} | z_s, y_{is}^l = 1])) - \mathbb{E}[y_{is}^{l'} | \mathcal{W}_{is}] | z_s] \geq 0. \quad (\text{B.2})$$

As y_{is}^l is a function of $(\mathcal{W}_{is}, \varepsilon_{is})$, equation (5) implies $\mathbb{E}[\Delta w_s^{ll'} | \mathcal{W}_{is}, y_{is}^l] = \mathbb{E}[\Delta w_s^{ll'} | \mathcal{W}_{is}]$. Since $\mathbb{E}[\Delta w_s^{ll'} | z_s] = \mathbb{E}[\Delta w_s^{ll'} | \mathcal{W}_{is}]$ according to Corollary 1, this corollary implies we can rewrite equation (B.2) as

$$\mathbb{E}[\mathbb{E}[y_{is}^l | \mathcal{W}_{is}] \exp(-(\Delta \theta_{ll'} + \alpha \mathbb{E}[\Delta w_s^{ll'} | z_s])) - \mathbb{E}[y_{is}^{l'} | \mathcal{W}_{is}] | z_s] \geq 0.$$

Given equation (6), we can further rewrite

$$\begin{aligned} \mathbb{E} \left[\exp(\Delta \kappa^{ll'} + \alpha \mathbb{E}[\Delta w_s^{ll'} | \mathcal{W}_{is}]) \left(\sum_{l''=1}^L \exp(\Delta \kappa^{l''l'} + \alpha \mathbb{E}[\Delta w_s^{l''l'} | \mathcal{W}_{is}]) \right)^{-1} \times \right. \\ \left. \exp(-(\Delta \theta_{ll'} + \alpha \mathbb{E}[\Delta w_s^{ll'} | z_s])) - \mathbb{E}[y_{is}^{l'} | \mathcal{W}_{is}] | z_s \right] \geq 0. \end{aligned}$$

Using a similar expression for the probability of choosing l' conditional on $\mathcal{W}_{is}$, we derive

$$\mathbb{E}\left[\left(\exp(\Delta\kappa^{ll'} - \Delta\theta_{ll'}) - 1\right)\left(\sum_{l''=1}^L \exp(\Delta\kappa^{l''l'} + \alpha\mathbb{E}[\Delta w_s^{l''l'}|\mathcal{W}_{is}])\right)^{-1}\middle|z_s\right] \geq 0,$$

where we have used that $\mathbb{E}[\Delta w_s^{ll'}|z_s] = \mathbb{E}[\Delta w_s^{ll'}|\mathcal{W}_{is}]$ according to Corollary 1. Then,

$$(\exp(\Delta\kappa^{ll'} - \Delta\theta_{ll'}) - 1)\mathbb{E}\left[\left(\sum_{l''=1}^L \exp(\Delta\kappa^{l''l'} + \alpha\mathbb{E}[\Delta w_s^{l''l'}|\mathcal{W}_{is}])\right)^{-1}\middle|z_s\right] \geq 0.$$

The expectation in this inequality is always strictly positive. Thus, the inequality implies

$$\exp(\Delta\kappa^{ll'} - \Delta\theta_{ll'}) - 1 \geq 0 \quad \Leftrightarrow \quad \Delta\kappa^{ll'} \geq \Delta\theta_{ll'}. \quad (\text{B.3})$$

This inequality holds for any locations l and l' . Swapping the indices l and l' , we obtain:

$$\exp(\Delta\kappa^{l'l} - \Delta\theta_{l'l}) - 1 \geq 0 \quad \Leftrightarrow \quad \Delta\kappa^{l'l} \leq \Delta\theta_{l'l}. \quad (\text{B.4})$$

Equations (B.3) and (B.4) imply $\Delta\kappa^{ll'} = \Delta\theta_{ll'}$, proving Corollary 1. ■

B.3 Second-Step Odds-Based Inequalities: Additional Derivations

Derivation of equation (21). Equations (19) and (20) imply the following inequality

$$\mathbb{E}[y_{is}^l \exp(-(\Delta\theta_{ll'} + \alpha\Delta w_s^{ll'}))|z_s] \geq \mathbb{E}[y_{is}^{l'}|z_s].$$

We can rewrite this inequality as

$$\mathbb{E}[y_{is}^l|z_s] \exp(-\Delta\theta_{ll'}) \mathbb{E}[(\exp(\alpha\Delta w_s^{ll'}))^{-1}|z_s, y_{is}^l = 1] \geq \mathbb{E}[y_{is}^{l'}|z_s].$$

Rearranging terms, we obtain the expression in equation (21); i.e.,

$$\frac{\mathbb{E}[y_{is}^l|z_s]}{\mathbb{E}[y_{is}^{l'}|z_s]} \mathbb{E}[(\exp(\alpha\Delta w_s^{ll'}))^{-1}|z_s, y_{is}^l = 1] \geq \exp(\Delta\theta_{ll'}).$$

Derivation of equation (22). Swapping the indices l and l' in equation (21), we obtain

$$\frac{\mathbb{E}[y_{is}^{l'}|z_s]}{\mathbb{E}[y_{is}^l|z_s]} \mathbb{E}[\exp(-\alpha\Delta w_s^{l'l})|z_s, y_{is}^{l'} = 1] \geq \exp(\Delta\theta_{l'l}).$$

Rearranging terms, we immediately obtain the inequality in equation (22).

B.4 Second-Step Odds-Based Inequalities: Proof of Corollary 2

Equations (19) and (20) imply the following moment inequality:

$$\mathbb{E}[y_{is}^l \exp(-(\Delta\theta_{ll'} + \alpha\Delta w_s^{ll'})) - y_{is}^{l'} | z_s] \geq 0. \quad (\text{B.5})$$

Assuming $z_s \subseteq \mathcal{W}_{is}$, the LIE implies we can write this inequality as

$$\mathbb{E}[\mathbb{E}[y_{is}^l \exp(-(\Delta\theta_{ll'} + \alpha\Delta w_s^{ll'})) - y_{is}^{l'} | \mathcal{W}_{is}] | z_s] \geq 0.$$

Since $\Delta w_s^{ll'} = \mathbb{E}[\Delta w_s^{ll'} | \mathcal{W}_{is}]$ according to Corollary 2, we can rewrite this inequality as

$$\mathbb{E}[\mathbb{E}[y_{is}^l | \mathcal{W}_{is}] \exp(-(\Delta\theta_{ll'} + \alpha\Delta w_s^{ll'})) - \mathbb{E}[y_{is}^{l'} | \mathcal{W}_{is}] | z_s] \geq 0.$$

Given the expression for the probability of choosing l conditional on $\mathcal{W}_{is}$, we rewrite

$$\mathbb{E}[\exp(\Delta\kappa^{ll'} - \Delta\theta_{ll'}) (\sum_{l''=1}^L \exp(\Delta\kappa^{ll''} + \alpha\Delta w_s^{ll''}))^{-1} - \mathbb{E}[y_{is}^{l'} | \mathcal{W}_{is}] | z_s] \geq 0.$$

Using a similar expression for the probability of choosing l' conditional on $\mathcal{W}_{is}$, we derive

$$(\exp(\Delta\kappa^{ll'} - \Delta\theta_{ll'}) - 1) \mathbb{E}[(\sum_{l''=1}^L \exp(\Delta\kappa^{ll''} + \alpha\Delta w_s^{ll''}))^{-1} | z_s] \geq 0. \quad (\text{B.6})$$

The expectation in this inequality is always strictly positive. Thus, the inequality implies

$$\exp(\Delta\kappa^{ll'} - \Delta\theta_{ll'}) - 1 \geq 0 \quad \Leftrightarrow \quad \Delta\kappa^{ll'} \geq \Delta\theta_{ll'}. \quad (\text{B.7})$$

This inequality holds for any locations l and l' . Swapping the indices l and l' , we obtain:

$$\exp(\Delta\kappa^{l'l} - \Delta\theta_{l'l}) - 1 \geq 0 \quad \Leftrightarrow \quad \Delta\kappa^{l'l} \leq \Delta\theta_{l'l}. \quad (\text{B.8})$$

Equations (B.7) and (B.8) imply $\Delta\kappa^{ll'} = \Delta\theta_{ll'}$, proving Corollary 2. ■

B.5 First-Step Moment Inequalities: Additional Derivations

Derivation of equation (27). Consider points $e_{isjr}^{ll'}$ such that $e_{isjr}^{ll'} = h^{ll'}(z_s, z_r, \theta_\alpha)$ for some function $h^{ll'}(\cdot)$. We compute the function in equation (27) by finding the value of $h^{ll'}(z_s, z_r, \theta_\alpha)$ that minimizes the moment in equation (25) at each value of θ_α . Specifically, given z_s , z_r , and θ_α , the first-order condition of the moment in equation (25) with respect to $h^{ll'}(z_s, z_r, \theta_\alpha)$

is

$$\mathbb{E}[y_{is}^l y_{jr}^{l'} (2h^{ll'}(z_s, z_r, \theta_\alpha) - \theta_\alpha(\Delta w_s^{ll'} + \Delta w_r^{ll'})) | z_s, z_r] = 0,$$

or, equivalently, $\mathbb{E}[2h^{ll'}(z_s, z_r, \theta_\alpha) - \theta_\alpha(\Delta w_s^{ll'} + \Delta w_r^{ll'}) | z_s, z_r, y_{is}^l y_{jr}^{l'} = 1] = 0$. Solving for $h^{ll'}(z_s, z_r, \theta_\alpha)$, we obtain the solution in equation (27).

Derivation of equation (28). Equations (25) to (27) imply the following inequality

$$\begin{aligned} & \mathbb{E}[y_{is}^l y_{jr}^l + y_{is}^{l'} y_{jr}^{l'} - y_{is}^l y_{jr}^{l'} \exp(-\theta_\alpha \mathbb{E}[0.5(\Delta w_s^{ll'} + \Delta w_r^{ll'}) | z_s, z_r, y_{is}^l y_{jr}^{l'} = 1]) \times \\ & (2 - \theta_\alpha((\Delta w_s^{ll'} + \Delta w_r^{ll'}) - \mathbb{E}[\Delta w_s^{ll'} + \Delta w_r^{ll'} | z_s, z_r, y_{is}^l y_{jr}^{l'} = 1])) | z_s, z_r] \geq 0, \end{aligned}$$

or, equivalently,

$$\begin{aligned} & \mathbb{E}[y_{is}^l y_{jr}^l + y_{is}^{l'} y_{jr}^{l'} | z_s, z_r] \geq \mathbb{E}[y_{is}^l y_{jr}^{l'} \exp(-\theta_\alpha \mathbb{E}[0.5(\Delta w_s^{ll'} + \Delta w_r^{ll'}) | z_s, z_r, y_{is}^l y_{jr}^{l'} = 1]) \times \\ & (2 - \theta_\alpha((\Delta w_s^{ll'} + \Delta w_r^{ll'}) - \mathbb{E}[\Delta w_s^{ll'} + \Delta w_r^{ll'} | z_s, z_r, y_{is}^l y_{jr}^{l'} = 1])) | z_s, z_r]. \end{aligned}$$

Using the LIE, we can rewrite this inequality as

$$\begin{aligned} & \mathbb{E}[y_{is}^l y_{jr}^l + y_{is}^{l'} y_{jr}^{l'} | z_s, z_r] \geq \mathbb{E}[\exp(-\theta_\alpha \mathbb{E}[0.5(\Delta w_s^{ll'} + \Delta w_r^{ll'}) | z_s, z_r, y_{is}^l y_{jr}^{l'} = 1]) \times \\ & (2 - \theta_\alpha((\Delta w_s^{ll'} + \Delta w_r^{ll'}) - \mathbb{E}[\Delta w_s^{ll'} + \Delta w_r^{ll'} | z_s, z_r, y_{is}^l y_{jr}^{l'} = 1])) | z_s, z_r, y_{is}^l y_{jr}^{l'} = 1] \times \\ & \mathbb{E}[y_{is}^l y_{jr}^{l'} | z_s, z_r] \geq 0. \end{aligned}$$

Simplifying this expression, and rearranging, we obtain the expression in equation (28).

B.6 First-Step Moment Inequalities: Proof of Corollary 3

Since $z_s \subseteq \mathcal{W}_{is}$ and $z_r \subseteq \mathcal{W}_{jr}$ according to Corollary 3, we rewrite equation (28) as

$$\begin{aligned} & \frac{\mathbb{E}[\mathbb{E}[y_{is}^l y_{jr}^{l'} | \mathcal{W}_{is}, \mathcal{W}_{jr}] | z_s, z_r]}{\mathbb{E}[0.5(\mathbb{E}[y_{is}^l y_{jr}^l | \mathcal{W}_{is}, \mathcal{W}_{jr}] + \mathbb{E}[y_{is}^{l'} y_{jr}^{l'} | \mathcal{W}_{is}, \mathcal{W}_{jr}]) | z_s, z_r]} \leq \\ & \exp(\theta_\alpha \mathbb{E}[0.5(\Delta w_s^{ll'} + \Delta w_r^{ll'}) | z_s, z_r, y_{is}^l y_{jr}^{l'} = 1]). \end{aligned}$$

Equations (5) and (6) further imply that we can rewrite this inequality as

$$\begin{aligned} & \frac{\mathbb{E}[\mathbb{E}[y_{is}^l | \mathcal{W}_{is}] \mathbb{E}[y_{jr}^{l'} | \mathcal{W}_{jr}] | z_s, z_r]}{\mathbb{E}[0.5(\mathbb{E}[y_{is}^l | \mathcal{W}_{is}] \mathbb{E}[y_{jr}^l | \mathcal{W}_{jr}] + \mathbb{E}[y_{is}^{l'} | \mathcal{W}_{is}] \mathbb{E}[y_{jr}^{l'} | \mathcal{W}_{jr}]) | z_s, z_r]} \leq \\ & \exp(\theta_\alpha \mathbb{E}[0.5(\Delta w_s^{ll'} + \Delta w_r^{ll'}) | z_s, z_r, y_{is}^l y_{jr}^{l'} = 1]). \end{aligned} \tag{B.9}$$

Given equations (1) to (6), it holds that, for any $l_1 = 1, \dots, L$ and $l_2 = 1, \dots, L$, we can write

$$\mathbb{E}[y_{is}^{l_1} | \mathcal{W}_{is}] = \frac{\exp(\Delta\kappa^{l_1 l_2} + \alpha \mathbb{E}[\Delta w_s^{l_1 l_2} | \mathcal{W}_{is}])}{\sum_{l''=1}^L \exp(\Delta\kappa^{l'' l_2} + \alpha \mathbb{E}[\Delta w_s^{l'' l_2} | \mathcal{W}_{is}])},$$

and similarly for worker j of type r . We then rewrite the inequality in equation (B.9) as

$$\begin{aligned} & \frac{\mathbb{E}[\exp(\Delta\kappa^{ll'} + \alpha \mathbb{E}[\Delta w_s^{ll'} | \mathcal{W}_{is}]) \exp(\Delta\kappa^{ll'} + \alpha \mathbb{E}[\Delta w_r^{ll'} | \mathcal{W}_{jr}]) | z_s, z_r]}{\mathbb{E}[0.5(\exp(\Delta\kappa^{ll'} + \alpha \mathbb{E}[\Delta w_s^{ll'} | \mathcal{W}_{is}]) + \exp(\Delta\kappa^{ll'} + \alpha \mathbb{E}[\Delta w_r^{ll'} | \mathcal{W}_{jr}])) | z_s, z_r]} \\ & \leq \exp(\theta_\alpha \mathbb{E}[0.5(\Delta w_s^{ll'} + \Delta w_r^{ll'}) | z_s, z_r, y_{is}^l y_{jr}^{l'} = 1]). \end{aligned}$$

Simplifying this expression, we obtain

$$\begin{aligned} & \frac{\mathbb{E}[\exp(\alpha \mathbb{E}[\Delta w_s^{ll'} | \mathcal{W}_{is}]) \exp(\alpha \mathbb{E}[\Delta w_r^{ll'} | \mathcal{W}_{jr}]) | z_s, z_r]}{\mathbb{E}[0.5(\exp(\Delta\kappa^{ll'} + \alpha \mathbb{E}[\Delta w_s^{ll'} | \mathcal{W}_{is}]) + \exp(\Delta\kappa^{ll'} + \alpha \mathbb{E}[\Delta w_r^{ll'} | \mathcal{W}_{jr}])) | z_s, z_r]} \\ & \leq \exp(\theta_\alpha \mathbb{E}[0.5(\Delta w_s^{ll'} + \Delta w_r^{ll'}) | z_s, z_r, y_{is}^l y_{jr}^{l'} = 1]). \end{aligned}$$

Since $\mathbb{E}[\Delta w_s^{ll'} | z_s] = \mathbb{E}[\Delta w_s^{ll'} | \mathcal{W}_{is}] = \mathbb{E}[\Delta w_r^{ll'} | z_r] = \mathbb{E}[\Delta w_r^{ll'} | \mathcal{W}_{jr}] = \Delta\bar{w}$, for a common constant $\Delta\bar{w} \in \mathbb{R}$, according to Corollary 3, this inequality becomes

$$\frac{\exp(\alpha \Delta\bar{w}) \exp(\alpha \Delta\bar{w})}{0.5(\exp(\Delta\kappa^{ll'} + \alpha \Delta\bar{w}) + \exp(\Delta\kappa^{ll'} + \alpha \Delta\bar{w}))} \leq \exp(\theta_\alpha \Delta\bar{w}).$$

If $\Delta\kappa^{ll'} = 0$, then it becomes

$$\frac{\exp(\alpha \Delta\bar{w}) \exp(\alpha \Delta\bar{w})}{0.5(\exp(\alpha \Delta\bar{w}) + \exp(\alpha \Delta\bar{w}))} \leq \exp(\theta_\alpha \Delta\bar{w}) \quad \Leftrightarrow \quad \exp(\alpha \Delta\bar{w}) \leq \exp(\theta_\alpha \Delta\bar{w}).$$

Thus, two inequalities of this type, one with $\Delta\bar{w} > 0$ and the other one with $\Delta\bar{w} < 0$, will only be satisfied if $\theta_\alpha = \alpha$, thereby proving Corollary 3. $\blacksquare$

B.7 Using Inequalities for Estimation of the Wage Parameter

We describe here how we use the inequalities in Section 3.2 to compute a CI for θ_α . The inequality in equation (25) is specific to locations l and l' and conditions on the vectors z_s and z_r . Given that conditional moment inequality, we implement the following steps to derive $k = 1, \dots, K$ unconditional moment inequalities. First, we choose scalars $\Delta z_s^{ll'} \subseteq z_s$ and $\Delta z_r^{ll'} \subseteq z_r$ that are correlated with $\Delta w_s^{ll'}$ and $\Delta w_r^{ll'}$, and that we use in all K inequalities. Second, for each k , we choose a subset $[z_{ks}, \bar{z}_{ks}]$ of the support of $\Delta z_s^{ll'}$, a subset $[z_{kr}, \bar{z}_{kr}]$ of the support of $\Delta z_r^{ll'}$, and an exponent $d_k \in \mathbb{Z}$. Given these choices, we build the inequality

$$\mathbb{E}[(y_{is}^l y_{jr}^l + y_{is}^{l'} y_{jr}^{l'} - y_{is}^l y_{jr}^{l'} \exp(-e_{isjr}^{ll'})(2 + 2e_{isjr}^{ll'} - \theta_\alpha(\Delta w_s^{ll'} + \Delta w_r^{ll'})))] \times g_k(\Delta z_s^{ll'}, \Delta z_r^{ll'}) \geq 0, \quad (\text{B.10})$$

where the term in parentheses coincides with the moment function in equation (25) and

$$g_k(\Delta z_s^{ll'}, \Delta z_r^{ll'}) = \mathbb{1}\{z_{ks} < \Delta z_s^{ll'} \leq \bar{z}_{ks}\} \mathbb{1}\{z_{kr} < \Delta z_r^{ll'} \leq \bar{z}_{kr}\} (|\Delta z_s^{ll'}| |\Delta z_r^{ll'}|)^{d_k}.$$

Effectively, for a pre-specified $q \in \mathbb{N}$, we choose limits z_{ks} and $\bar{z}_{ks}$ that correspond to consecutive elements of the vector of q -quantiles of the distribution of $\Delta z_s^{ll'}$ across all types and location pairs. We do the same for the limits z_{kr} and $\bar{z}_{kr}$. Concerning the exponent d_k , we set it to either -1 , 0 , or 1 . For example, when $q = 2$ and $d_k = 0$ for all k , the number of inequalities of the type in equation (29) is $K = 4$, and the corresponding instruments are

$$g_k(\Delta z_s^{ll'}, \Delta z_r^{ll'}) = \begin{cases} \mathbb{1}\{\Delta z_s^{ll'} \leq \text{med}(\Delta z_s^{ll'})\} \mathbb{1}\{\Delta z_r^{ll'} \leq \text{med}(\Delta z_r^{ll'})\} & \text{if } k = 1, \\ \mathbb{1}\{\Delta z_s^{ll'} \leq \text{med}(\Delta z_s^{ll'})\} \mathbb{1}\{\Delta z_r^{ll'} > \text{med}(\Delta z_r^{ll'})\} & \text{if } k = 2, \\ \mathbb{1}\{\Delta z_s^{ll'} > \text{med}(\Delta z_s^{ll'})\} \mathbb{1}\{\Delta z_r^{ll'} \leq \text{med}(\Delta z_r^{ll'})\} & \text{if } k = 3, \\ \mathbb{1}\{\Delta z_s^{ll'} > \text{med}(\Delta z_s^{ll'})\} \mathbb{1}\{\Delta z_r^{ll'} > \text{med}(\Delta z_r^{ll'})\} & \text{if } k = 4. \end{cases}$$

To compute the sample analogue of the moment inequality in equation (B.10), we average across worker types and workers within each type. If the cardinality L of the worker's choice set is large, it may be convenient to further average across all possible location pairs (l, l') .

C Additional Simulation Results

In Appendix C.1, we describe the inequalities we use to obtain the results in Table 1. In Appendix C.2, we explore alternative ways of building the first-step inequalities. In Appendix C.3, we present estimates analogous to those in Table 1, but using a larger set of instruments. In Appendix C.4, we explore the robustness of the results in Table 1 to different values of amenities. In Appendix C.5, we compare the estimates in Table 1 to alternative estimates computed using approximation points other than those in equations (14) and (27). In Appendix C.6, we examine how moment inequality and maximum likelihood estimates are affected when the data-generating process deviates from our baseline simulation setup and the researcher ignores these deviations.

C.1 Inequalities Used in Computing the CIs in Table 1

First step. Given a predictor z_s^l of the wage level w_s^l in every location l , a pair of locations l and l' , and a pair of indices k and k' determining the instruments we use, we compute the CIs for θ_α displayed in Table 1 using moment inequalities of the type:

$$\sum_{s=1}^S (y_s^l y_{r(s)}^l + y_s^{l'} y_{r(s)}^{l'} - y_s^l y_{r(s)}^{l'} \exp(-e_s^{ll'})) (2 + 2e_s^{ll'} - \theta_\alpha(\Delta w_s^{ll'} + \Delta w_{r(s)}^{ll'})) g_k(\Delta z_s^{ll'}) g_{k'}(\Delta z_{r(s)}^{ll'}) \geq 0, \quad (\text{C.1})$$

where $r(s)$ indexes the type matched with type s . We use this moment inequality for every pair of locations in the set $\{(l, l'); l \in \{1, 2, 3\}, l' \in \{1, 2, 3\}, l \neq l'\}$ and every pair of instrument indices in the set $\{(k, k'); k \in \{1, 2\}, k' \in \{1, 2\}\}$. Thus, we use 24 inequalities to identify θ_α . We now describe in more detail how we choose the type $r(s)$ for each s ; how we define the approximation points $e_s^{ll'}$ for each s ; and how we define the instrument $g_k(\cdot)$ for $k \in \{1, 2\}$.

First, for each $s = 1, \dots, S$, we select $r(s)$ randomly among those that satisfy

$$|\hat{\mathbb{E}}[\Delta w_s^{ll'} | \Delta z_s^{ll'}, y_s^l = 1] - \hat{\mathbb{E}}[\Delta w_{r(s)}^{ll'} | \Delta z_{r(s)}^{ll'}, y_{r(s)}^{l'} = 1]| \leq \tau, \quad (\text{C.2})$$

with $\tau = 0.002$. In this equation, for example, $\hat{\mathbb{E}}[\Delta w_s^{ll'} | \Delta z_s^{ll'}, y_s^l = 1]$ is the predicted value of $\Delta w_s^{ll'}$ computed using a linear regression of $\Delta w_s^{ll'}$ on $\Delta z_s^{ll'}$ estimated on the subset of observations with $y_s^l = 1$. To understand why we impose the restriction in equation (C.2) when selecting the type $r(s)$ to match with each given type s , one should note that, as S goes to infinity, the moment inequality in equation (C.1) will be satisfied at $\theta_\alpha = \alpha$ regardless of how the type $r(s)$ for every s is chosen. However, Corollary 3 indicates that a condition for this inequality to point identify θ_α is that, given wage predictors z_s^l and z_r^l for the types s and r combined in the inequality, these wage predictors and types satisfy

$$\mathbb{E}[\Delta w_s^{ll'} | \mathcal{W}_s] = \mathbb{E}[\Delta w_s^{ll'} | z_s] = \mathbb{E}[\Delta w_{r(s)}^{ll'} | \mathcal{W}_r] = \mathbb{E}[\Delta w_{r(s)}^{ll'} | z_r]. \quad (\text{C.3})$$

By selecting the type r according to equation (C.2), we aim to approximate the condition in equation (C.3) while taking into account that the sets $\mathcal{W}_s$ and $\mathcal{W}_r$ are generally not observed.

Second, in terms of the approximation point $e_s^{ll'}$ for each type s used in equation (C.1), we build on equation (27) and impose

$$e_s^{ll'} = \theta_\alpha 0.5 (\hat{\mathbb{E}}[\Delta w_s^{ll'} | \Delta z_s^{ll'}, y_s^l = 1] + \hat{\mathbb{E}}[\Delta w_{r(s)}^{ll'} | \Delta z_{r(s)}^{ll'}, y_{r(s)}^{l'} = 1]), \quad (\text{C.4})$$

where, as indicated above, e.g., $\hat{\mathbb{E}}[\Delta w_s^{ll'} | \Delta z_s^{ll'}, y_s^l = 1]$ is the predicted value of $\Delta w_s^{ll'}$ computed

using a linear regression of $\Delta w_s^{ll'}$ on $\Delta z_s^{ll'}$ estimated on the subset of observations with $y_s^l = 1$. Equation (27) indicates that the optimal approximation point $e_s^{ll'}$ is a function of, for example, the conditional expectation of $\Delta w_s^{ll'}$ given $\Delta z_s^{ll'}$ and $y_s^l = 1$; in equation (C.4), we compute instead a linear regression. In unreported results, we have instead approximated nonparametrically the conditional expectation of $\Delta w_s^{ll'}$ given $\Delta z_s^{ll'}$ and $y_s^l = 1$, obtaining very similar results to those attained when using linear regressions instead.

Third, and finally, in terms of the function $g_k(\cdot)$ for $k = 1, 2$, we impose:

$$g_k(\Delta z_s^{ll'}) = \begin{cases} \mathbb{1}\{\Delta z_s^{ll'} \leq 0\} & \text{for } k = 1, \\ \mathbb{1}\{0 < \Delta z_s^{ll'}\} & \text{for } k = 2. \end{cases} \quad (\text{C.5})$$

This instrument function corresponds to that in Section 3.3 with $q = 2$ and $d = 0$.

Second step. Given a location l , a predictor z_s^l of w_s^l , an index k determining the instruments we use, and a value $\check{\theta}_\alpha$ that belongs to the CI for θ_α computed in the first step of our inference procedure, we compute CIs for θ_l using either the following bounding inequalities

$$\sum_{s=1}^S (y_s^1 - y_s^l \exp(-e_s^{l1})(1 + e_s^{l1} - (\theta_l + \check{\theta}_\alpha \Delta w_s^{l1}))) g_k(\Delta z_s^{l1}) \geq 0, \quad (\text{C.6a})$$

$$\sum_{s=1}^S (y_s^l - y_s^1 \exp(-e_s^{1l})(1 + e_s^{1l} + (\theta_l + \check{\theta}_\alpha \Delta w_s^{1l}))) g_k(\Delta z_s^{1l}) \geq 0; \quad (\text{C.6b})$$

or the following odds-based moment inequalities

$$\sum_{s=1}^S (y_s^l \exp(-(\theta_l + \check{\theta}_\alpha \Delta w_s^{l1})) - y_s^1) g_k(\Delta z_s^{l1}) \geq 0, \quad (\text{C.7a})$$

$$\sum_{s=1}^S (y_s^1 \exp(\theta_l + \check{\theta}_\alpha \Delta w_s^{1l}) - y_s^l) g_k(\Delta z_s^{1l}) \geq 0; \quad (\text{C.7b})$$

or both the bounding and odds-based moment inequalities jointly. The instrument function $g_k(\cdot)$ in equations (C.6) and (C.7) is defined as in equation (C.5). The points e_s^{l1} and e_s^{1l} entering the inequalities in equation (C.6) are

$$e_s^{l1} = \theta_l + \check{\theta}_\alpha \hat{\mathbb{E}}[\Delta w_s^{l1} | \Delta z_s^{l1}, y_{is}^l = 1], \quad (\text{C.8a})$$

$$e_s^{1l} = -\theta_l + \check{\theta}_\alpha \hat{\mathbb{E}}[\Delta w_s^{1l} | \Delta z_s^{1l}, y_{is}^1 = 1], \quad (\text{C.8b})$$

where, as indicated above, e.g., $\hat{\mathbb{E}}[\Delta w_s^{ll'} | \Delta z_s^{ll'}, y_s^l = 1]$ is the predicted value of $\Delta w_s^{ll'}$ computed using a linear regression of $\Delta w_s^{ll'}$ on $\Delta z_s^{ll'}$ estimated on the subset of observations with $y_s^l = 1$.

Differences across cases. In cases 1 to 4 in Table 1, the wage predictor z_s^l introduced

in equations (C.1) to (C.8) equals the shifter z_{2s}^l in equation (31). In case 5, z_s^l equals w_s^l . In this case, equation (C.2) simplifies to $|\Delta w_s^{ll'} - \Delta w_{r(s)}^{ll'}| \leq \tau$; equation (C.4) simplifies to $e_s^{ll'} = \theta_\alpha 0.5(\Delta w_s^{ll'} + \Delta w_{r(s)}^{ll'})$; and equation (C.8) simplifies to $e_s^{l1} = \theta_l + \check{\theta}_\alpha \Delta w_s^{l1}$ and $e_s^{1l} = -e_s^{l1}$.

C.2 First-step Moment Inequalities with Loose Type Matches

In Table C.1, we present CIs for θ_α computed according to equations (C.1) to (C.5), and show how these CIs vary as we change the value of τ entering equation (C.2). Table C.1 shows that the 95% CI for θ_α becomes wider as we increase the value of τ .

Table C.1: Simulation Results - Moment Inequality Confidence Intervals With Loose Matches

Case	σ_1	σ_3	z_s^l	τ	1st Step
					θ_α
2	0	1	z_{2s}^l	8	[0.73, 1.32]
2	0	1	z_{2s}^l	4	[0.79, 1.25]
2	0	1	z_{2s}^l	2	[0.94, 1.08]
2	0	1	z_{2s}^l	1	[0.98, 1.03]
2	0	1	z_{2s}^l	0.8	[0.99, 1.03]
2	0	1	z_{2s}^l	0.08	[1, 1.02]
2	0	1	z_{2s}^l	0.008	[1, 1.02]
2	0	1	z_{2s}^l	0.002	[1, 1.01]

The column θ_α contains 95% CIs computed using the inequalities described in Appendix C.1 and the inference procedure in Andrews and Soares (2010). The CI with $\tau = 0.002$ corresponds to that in Table 1.

C.3 Two-step Moment Inequalities with Additional Instruments

In Table C.2, we present CIs analogous to those in Table 1 with the only difference that, instead of using the instrument function in equation (C.5), we use the following instrument function

$$g_k(\Delta z_s^{ll'}) = \begin{cases} \mathbb{1}\{\Delta z_s^{ll'} \leq Q_{25}(\Delta z_s^{ll'})\} & \text{for } k = 1, \\ \mathbb{1}\{Q_{25}(\Delta z_s^{ll'}) < \Delta z_s^{ll'} \leq Q_{50}(\Delta z_s^{ll'})\} & \text{for } k = 2, \\ \mathbb{1}\{Q_{50}(\Delta z_s^{ll'}) < \Delta z_s^{ll'} \leq Q_{75}(\Delta z_s^{ll'})\} & \text{for } k = 3, \\ \mathbb{1}\{Q_{75}(\Delta z_s^{ll'}) < \Delta z_s^{ll'}\} & \text{for } k = 4, \end{cases} \quad (\text{C.9})$$

where $Q_{25}(\Delta z_s^{ll'})$, $Q_{50}(\Delta z_s^{ll'})$, and $Q_{75}(\Delta z_s^{ll'})$ respectively denote the 25th, 50th, and 75th percentiles of the distribution of $\Delta z_s^{ll'}$ across all types s and pairs of locations l and l' . As a comparison of the results in Tables 1 and C.2 illustrates, the CIs for θ_α , θ_2 , and θ_3 , become tighter when we swap the instrument function in equation (C.5) for the more detailed function

in equation (C.9). This change in instrument functions increases the number of moment inequalities we use in our estimation.

Table C.2: Simulation Results - Confidence Intervals With Additional Instruments

Case	σ_1	σ_3	z_s^l	First Step	Second Step		
				θ_α	Mom. Ineq.	θ_2	θ_3
1	0	0	z_{2s}^l	$[1, 1.02]$	Bounding	$[0, 0]$	$[1, 1]$
					Odds-based	$[0, 0]$	$[1, 1]$
					Both	$[0, 0]$	$[1, 1]$
2	0	1	z_{2s}^l	$[1, 1.01]$	Bounding	$[0, 0]$	$[1, 1]$
					Odds-based	$[-0.33, 0.32]$	$[0.68, 1.33]$
					Both	$[0, 0]$	$[1, 1]$
3	1	0	z_{2s}^l	$[0.91, 1.15]$	Bounding	$[-0.31, 0.31]$	$[0.70, 1.30]$
					Odds-based	$[0, 0]$	$[1, 1.01]$
					Both	$[0, 0]$	$[1, 1.01]$
4	1	1	z_{2s}^l	$[0.91, 1.19]$	Bounding	$[-0.31, 0.31]$	$[0.70, 1.30]$
					Odds-based	$[-0.32, 0.32]$	$[0.68, 1.33]$
					Both	$[-0.31, 0.31]$	$[0.70, 1.31]$
5	0	1	w_s^l	$\emptyset$	Bounding	$\emptyset$	$\emptyset$
					Odds-based	$\emptyset$	$\emptyset$
					Both	$\emptyset$	$\emptyset$

This table contains 95% CIs computed using the inequalities described in Appendix C.1 with the only exception that the instrument functions $g_k(\cdot)$ for all $k = 1, \dots, K$ are not those defined in equation (C.5) but those defined in equation (C.9).

C.4 Amenity Differences Across All Locations

In Table C.3, we present CIs for θ_α computed according to equations (C.1) to (C.5), and show how these CIs vary as we change the value of the amenity parameters ($\kappa^1, \kappa^2, \kappa^3$) used to generate the simulated data (see Section 4 for details). Table C.3 shows how the 95% CI for θ_α changes as the minimum value of $\Delta\kappa^{ll'}$ between any two locations l and l' increases; i.e., as $\min_{l \in \{1,2,3\}, l' \neq l} \Delta\kappa^{ll'}$ increases.

C.5 Estimator with Alternative Approximation Points

Panel A in Table C.4 presents CIs computed similarly to those in Table 1. The results in Panel B incorporate two differences in the way these CIs are computed: (a) instead of using the approximation points in equation (C.4) in the first-step moment inequalities, we use

$$e_s^{ll'} = \theta_\alpha 0.5(\Delta z_s^{ll'} + \Delta z_{r(s)}^{ll'}); \quad (\text{C.10})$$

Table C.3: Simulation Results - Confidence Intervals With Amenity Differences

Case	σ_1	σ_3	z_s^l	$(\kappa^1, \kappa^2, \kappa^3)$	1st Step
					θ_α
2	0	1	z_{2s}^l	(0, 0, 1)	[1, 1.01]
2	0	1	z_{2s}^l	(0, 0.5, 1)	[0.92, 1.09]
2	0	1	z_{2s}^l	(0, 0, 2)	[1, 1.02]
2	0	1	z_{2s}^l	(0, 1, 2)	[0.73, 1.29]
2	0	1	z_{2s}^l	(0, 0, 3)	[1, 1.02]
2	0	1	z_{2s}^l	(0, 1.5, 3)	[0.55, 1.49]

The column θ_α contains 95% CIs computed using the inequalities described in Appendix C.1 and the inference procedure in Andrews and Soares (2010). The CI with $(\kappa^1, \kappa^2, \kappa^3) = (0, 0, 1)$ corresponds to that in Table 1.

and, instead of using the approximation points in equation (C.8) in the second-step bounding moment inequalities, we use

$$e_s^{l1} = \theta_l + \check{\theta}_\alpha \Delta z_s^{l1}, \quad (\text{C.11a})$$

Table C.4: Simulation Results - Confidence Intervals With Alternative Approximation Points

σ_1	σ_3	z_s^l	Mom. Ineq.	θ_α	θ_2	θ_3
<i>Panel A: Confidence Intervals Using Approximation Points in Eqs. (C.4) and (C.8)</i>						
0	0	z_{2s}^l	Bounding	[1, 1.02]	[0, 0]	[1, 1]
0.25	0	z_{2s}^l	Bounding	[0.98, 1.04]	[-0.02, 0.02]	[0.98, 1.02]
0.50	0	z_{2s}^l	Bounding	[0.95, 1.06]	[-0.09, 0.08]	[0.92, 1.08]
0.75	1	z_{2s}^l	Bounding	[0.89, 1.16]	[-0.18, 0.18]	[0.82, 1.18]
1	0	z_{2s}^l	Bounding	[0.82, 1.29]	[-0.31, 0.31]	[0.70, 1.30]
<i>Panel B: Confidence Intervals Using Approximation Points in Eqs. (C.10) and (C.11)</i>						
0	0	z_{2s}^l	Bounding	[1, 1.02]	[0, 0]	[1, 1]
0.25	0	z_{2s}^l	Bounding	[0.98, 1.04]	[-0.02, 0.02]	[0.98, 1.02]
0.50	0	z_{2s}^l	Bounding	[0.94, 1.09]	[-0.09, 0.08]	[0.92, 1.08]
0.75	1	z_{2s}^l	Bounding	[0.88, 1.24]	[-0.20, 0.20]	[0.81, 1.21]
1	0	z_{2s}^l	Bounding	[0.81, 1.50]	[-0.41, 0.42]	[0.63, 1.46]

Panel A contains 95% CIs computed using the inequalities described in Appendix C.1. Panel B contains 95% CIs computed using inequalities analogous to those in Appendix C.1, with the only exception that the approximation points in equations (C.10) and (C.11) are used instead of those in equations (C.4) and (C.8). As the choice of approximation points only affects the CIs computed using bounding moment inequalities (i.e., it does not affect the CIs computed using odds-based moment inequalities), we only report those in the table.

$$e_s^{1l} = -\theta_l + \check{\theta}_\alpha \Delta z_s^{1l}. \quad (\text{C.11b})$$

Relative to the approximation points in equations (C.4) and (C.8), those in equations (C.10) and (C.11) have the advantage that they do not involve any estimation; in particular, they do not require computing predicted values from a linear regression. Consequently, they do not affect the validity of standard moment inequality inference procedures. However, as a comparison of both panels in Table C.4 illustrates, the use of the approximation points in equations (C.10) and (C.11) results in CIs that are larger than those obtained when the approximation points in equations (C.4) and (C.8) are used. The difference between the two sets of CIs increases in the value of σ_1 .

C.6 Model Misspecification: Additional Details

In this section, we provide additional details on the exercises described in Section 4.4.

C.6.1 Choice Model à la Roy (1951)

Setup. We consider a setup that differs from that in Section 4.1 in two aspects. First, we generalize the model generating the data. In particular, instead of equation (3), we assume

$$\mathcal{U}_{is}^l = u_{is}^l + \tilde{\eta} \eta \varepsilon_{is}^l, \quad (\text{C.12a})$$

$$u_{is}^l = \kappa^l + \alpha \check{w}_{is}^l \quad (\text{C.12b})$$

$$\check{w}_{is}^l = w_{is}^l + \tilde{\eta}(1 - \eta)\xi_{is}^l, \quad (\text{C.12c})$$

with $\tilde{\eta} \equiv (\eta^2 + (1 - \eta)^2)^{-\frac{1}{2}}$. The variable ξ_{is}^l is a wage shock distinct from the preference shock ε_{is}^l . Thus, the wage measure is now $\check{w}_{is}^l$, with w_{is}^l denoting a component of wages. The parameter η determines the relative weight that preference and wage shocks have in utility. In particular, if $\eta = 1$, the utility function in equation (C.12) equals that used as the basis for the simulation exercise described in Section 4.1. The parameter $\tilde{\eta}$ guarantees that the variance of the sum of wage and preference shocks remains constant regardless of the value of η ; that is, the variance of the random variable $\tilde{\eta}(\eta \varepsilon_{is}^l + (1 - \eta)\xi_{is}^l)$ remains constant. Given $\xi_{is} = (\xi_{is}^1, \dots, \xi_{is}^L)$, we assume

$$\xi_{is} \subseteq \mathcal{J}_{is}, \quad (\text{C.13})$$

for any worker i of type s . Denoting by $F_\xi(\cdot)$ the cumulative distribution function of ξ_{is} , we impose that

$$F_\xi(\xi_{is}|\mathcal{W}_{is}, \varepsilon_{is}, \mathcal{J}_{jr}) = F_\xi(\xi_{is}|\mathcal{W}_{is}, \varepsilon_{is}) = F_\xi(\xi_{is}) = \exp\left(-\sum_{l=1}^L \exp(-\xi_{is}^l)\right), \quad (\text{C.14})$$

for any worker i of type s and worker j of type r . That is, ε_{is} and ξ_{is} are independent of each other and of $(\mathcal{W}_{is}, \mathcal{J}_{jr})$ and they are both identically distributed; see equations (6) and (C.14). All other aspects of the simulated model are as described in Section 2 and 4.1.

Second, we limit the data available to the researcher. Specifically, we assume the researcher observes $(y_{is}^l, y_{is}^l \check{w}_{is}^l, z_{2is}^l)$ for all workers and locations. That is, the researcher observes $\check{w}_{is}^l$ only for the alternative l chosen by worker i of type s . To isolate the effect of this data restriction on our estimates, we also consider the case in which the researcher observes $(y_{is}^l, w_{is}^l, z_{2is}^l)$ for all workers and locations, as in the setup studied in Section 4.1.

Except for the two aspects described above, the setup is as described in Section 4.1. In particular, we set the wage coefficient to $\alpha = 1$ and the location-specific amenities to $\kappa^1 = \kappa^2 = 0$ and $\kappa^3 = 1$ and, as the simulated data includes one worker per type, we index observations only by s below.

Estimation Approach and Simulation Results. First, we consider the case in which the researcher observes (y_s^l, w_s^l, z_{2s}^l) for every worker s and location l , as in our baseline setup. In this case, we use the moment inequalities in Appendix C.1 to compute confidence intervals for the wage coefficient α and the parameters κ^2 and κ^3 . The resulting moment inequality estimates are in Table C.5 with the corresponding maximum likelihood estimates reported in the last three columns.

Note that the equality in equation (8) holds if structural errors are preference shocks alone or wage shocks alone. As a result, if the parameter η in equation (C.12) is either 0 or 1, the proofs in Appendix A imply the bounding inequality in equation (13), the odds-based inequality in equation (20) and the first-step inequality in equation (26) hold at the true parameter values. Intuitively, if $\eta = 1$, the utility functions in equations (3) and (C.12) coincide and we are back in our baseline setting. If $\eta = 0$, the functions in equations (3) and (C.12) differ but, given equations (C.13) and (C.14), the equality in equation (8) still holds, which allows us to derive our moment inequalities. Consistent with this discussion, if $\eta \in \{0, 1\}$, the confidence sets in Table C.5 are non-empty and contain the true parameter values and the maximum likelihood estimates coincide with the true value.

If $\eta \in (0, 1)$, given the distributions in equations (6) and (C.14), the choice model is not multinomial logit. The reason is that the structural error in the utility function in equation (C.12) is $\tilde{\eta}(\eta \varepsilon_{is}^l + (1 - \eta) \xi_{is}^l)$ and a linear combination of two independent type I extreme value

Table C.5: Choice Model à la Roy (1951): Data on Wages for All Alternatives

Case	η	Moment Inequalities				Likelihood		
		First Step	Second Step					
		θ_α	Mom. Ineq.	θ_2	θ_3	θ_α	θ_2	θ_3
1	0	[1.00, 1.02]	Bounding Odds-based	$\begin{bmatrix} 0, 0 \\ 0, 0 \end{bmatrix}$	$\begin{bmatrix} 1, 1 \\ 1, 1 \end{bmatrix}$	1	0	1
2	0.1	[1.00, 1.02]	Bounding Odds-based	$\begin{bmatrix} 0, 0 \\ 0, 0 \end{bmatrix}$	$\begin{bmatrix} 1, 1 \\ 1, 1 \end{bmatrix}$	1	0	1
3	0.2	[1.00, 1.02]	Bounding Odds-based	$[-0.01, 0.00]$ $\emptyset$	$[0.99, 1.01]$ $\emptyset$	1	0	1
4	0.3	[1.00, 1.01]	Bounding Odds-based	$[-0.02, 0.02]$ $\emptyset$	$[0.98, 1.01]$ $\emptyset$	1	0	1
5	0.4	[0.99, 1.01]	Bounding Odds-based	$[-0.02, 0.02]$ $\emptyset$	$[0.98, 1.01]$ $\emptyset$	1	0	1
6	0.5	[0.99, 1.01]	Bounding Odds-based	$[-0.02, 0.02]$ $\emptyset$	$[0.97, 1.01]$ $\emptyset$	1	0	1
7	0.6	[0.99, 1.01]	Bounding Odds-based	$[-0.02, 0.02]$ $\emptyset$	$[0.98, 1.01]$ $\emptyset$	1	0	1
8	0.7	[1.00, 1.01]	Bounding Odds-based	$[-0.02, 0.01]$ $\emptyset$	$[0.98, 1.01]$ $\emptyset$	1	0	1
9	0.8	[1.00, 1.01]	Bounding Odds-based	$[-0.01, 0.01]$ $\emptyset$	$[0.99, 1.01]$ $\emptyset$	1	0	1
10	0.9	[1.00, 1.02]	Bounding Odds-based	$\begin{bmatrix} 0, 0 \\ 0, 0 \end{bmatrix}$	$\begin{bmatrix} 1, 1 \\ 1, 1 \end{bmatrix}$	1	0	1
11	1	[1.00, 1.02]	Bounding Odds-based	$\begin{bmatrix} 0, 0 \\ 0, 0 \end{bmatrix}$	$\begin{bmatrix} 1, 1 \\ 1, 1 \end{bmatrix}$	1	0	1

Note: Columns 3, 5, and 6 contain 95% CIs computed using the inequalities in Appendix C.1. The last three columns contain 95% CIs computed using these inequalities but with $\tilde{w}_s^l$ in place of w_s^l in equations (C.1) to (C.8), with $\tilde{w}_s^l = \varphi_0 + \varphi_1 z_{2s}^l$ and φ_0 and φ_1 defined in equation (C.15). All results are computed under the restriction $\sigma_1 = \sigma_3 = 0$ and with the wage shifter z_{2s}^l as the instrument z_s^l .

random variables is not a type I extreme value random variable. As a result, the equality in equation (8) does not hold and, thus, the proofs included in Appendix A no longer imply that the bounding inequality in equation (13), the odds-based inequality in equation (20), or the first-step inequality in equation (26) hold at the true parameter values. Indeed, when $\eta \in [0.2, 0.8]$, Table C.5 shows that the odds-based moment inequalities yield an empty confidence set. When $\eta \in \{0.1, 0.9\}$, the distribution of $\tilde{\eta}(\eta \varepsilon_{is}^l + (1 - \eta) \xi_{is}^l)$ is sufficiently close to that of a type I extreme value random variable and, as a result, the confidence sets defined by the odds-based moment inequalities are no longer empty. Interestingly, the bounding moment inequalities are robust to the fact that, when $\eta \in (0, 1)$, the true choice model is not exactly a multinomial logit model, and still yield confidence sets that contain the true

parameter values even when η is equal to or around 0.5. That is, although z_{2s}^l belongs to the information set of every worker s , the parametric misspecification of the distribution of the structural error in the utility function in equation (C.12) causes a researcher who uses the odds-based moment inequalities to incorrectly infer that z_{2s}^l does not belong to the worker's information set. This is not the case for the researcher using the bounding moment inequalities, who obtains a confidence set containing the true value of the parameter vector.

Moreover, we consider a setting where the researcher observes $(y_s^l, y_s^l \tilde{w}_s^l, z_{2s}^l)$ for every worker s and location l , as is typical in applications of choice models à la Roy (1951). Here, to estimate the parameters of interest, we must first construct a proxy of the potential wage of each worker s in each location l . To do so, we use data on $(\tilde{w}_s^l, z_{2s}^l)$ for the worker-location pairs with $y_s^l = 1$ to compute the value of (φ_0, φ_1) that satisfies the moment condition

$$\mathbb{E}[\tilde{w}_s^l - \varphi_0 - \varphi_1 z_{2s}^l | z_{2s}^l, y_s^l = 1] = 0. \quad (\text{C.15})$$

Then, we use the estimated values of φ_0 and φ_1 to construct a measure of the potential wage of a worker s in a location l as $\tilde{w}_s^l = \varphi_0 + \varphi_1 z_{2s}^l$. Subsequently, we obtain confidence intervals for the wage coefficient α and location-specific amenities (κ^2, κ^3) using the moment inequalities in Appendix C.1, but with $\tilde{w}_s^l$ in place of w_s^l in equations (C.1) to (C.8). Similarly, we compute the maximum likelihood estimates of α and (κ^2, κ^3) following equation (32), but with the wage measure $\tilde{w}_s^l$ acting as a proxy for agents' true wage expectations. The resulting moment inequality and maximum likelihood estimates are in Table C.6.

To understand the results, we again note that the equality in equation (8) holds if structural errors are preference shocks alone or wage shocks alone. However, since we now use $\tilde{w}_s^l$ in place of w_s^l in our inequalities, the validity of these inequalities requires additional restrictions on the relationship between the wage measures $\tilde{w}_s^l$ and w_s^l . In particular, if $\eta \in \{0, 1\}$, the inequalities in equations (C.1) to (C.8) with $\tilde{w}_s^l$ in place of w_s^l are valid if, for any two locations l and l' , we have

$$\mathbb{E}[\Delta \tilde{w}_s^{ll'} - \mathbb{E}[\Delta w_s^{ll'} | \mathcal{W}_s] | \mathcal{J}_s] = \varphi_1 \Delta z_{2s}^{ll'} - (\Delta z_{1s}^{ll'} + \Delta z_{2s}^{ll'}) = 0, \quad (\text{C.16})$$

where the first equality uses the expression $\tilde{w}_s^l = \varphi_0 + \varphi_1 z_{2s}^l$ as well as the assumptions on the distribution of w_s^l conditional on $\mathcal{W}_s$ imposed in Section 4.1. Since we focus here on the effect that idiosyncratic wage shocks have on our inequalities, we impose for simplicity that $z_{1s}^l = 0$ for all s and l . Then, equation (C.16) holds if and only if $\varphi_1 = 1$.

The value of φ_1 is determined by the moment in equation (C.15). Given the definition of $\tilde{w}_s^l$ in equation (C.12c), the restrictions on the distribution of w_s^l conditional on $\mathcal{W}_{is}$ described

Table C.6: Choice Model à la Roy (1951): Data on Wages for Chosen Alternative

Case	η	Moment Inequalities				Max. Likelihood		
		First Step	Second Step			θ_α	θ_2	θ_3
		θ_α	Mom. Ineq.	θ_2	θ_3			
1	0	[1.19, 1.75*]	Bounding Odds-based	$[-0.41, 0.42]$ $[-0.01, 0.00]$	$[0.63, 1.50]$ [1,1]	1.25	0	1
2	0.1	[1.19, 1.75*]	Bounding Odds-based	$[-0.41, 0.43]$ [0,0]	$[0.63, 1.50]$ [1,1]	1.25	0	1
3	0.2	[1.18, 1.75*]	Bounding Odds-based	$[-0.42, 0.44]$ $\emptyset$	$[0.62, 1.51]$ $\emptyset$	1.23	0	1
4	0.3	[1.16, 1.32]	Bounding Odds-based	$[-0.44, 0.46]$ $\emptyset$	$[0.59, 1.53]$ $\emptyset$	1.20	0	1
5	0.4	[1.12, 1.21]	Bounding Odds-based	$[-0.08, 0.08]$ $\emptyset$	$[0.92, 1.08]$ $\emptyset$	1.15	0	1
6	0.5	[1.08, 1.13]	Bounding Odds-based	$[-0.05, 0.05]$ $\emptyset$	$[0.95, 1.04]$ $\emptyset$	1.10	0	1
7	0.6	[1.05, 1.07]	Bounding Odds-based	$[-0.03, 0.03]$ $\emptyset$	$[0.97, 1.01]$ $\emptyset$	1.06	0	1
8	0.7	[1.02, 1.03]	Bounding Odds-based	$[-0.02, 0.02]$ $\emptyset$	$[0.98, 1.01]$ $\emptyset$	1.03	0	1
9	0.8	[1.01, 1.02]	Bounding Odds-based	$[-0.01, 0.01]$ $\emptyset$	$[0.99, 1.01]$ $\emptyset$	1.01	0	1
10	0.9	[1.00, 1.02]	Bounding Odds-based	$[0, 0]$ $\emptyset$	$\emptyset$ [1, 1]	1	0	1
11	1	[1.00, 1.02]	Bounding Odds-based	$[0, 0]$ [0, 0]	$[1, 1]$ [1, 1]	1	0	1

Note: Columns 3, 5, and 6 contain 95% CIs computed using the inequalities in Appendix C.1. The last three columns contain 95% CIs computed using these inequalities but with $\tilde{w}_s^l$ in place of w_s^l in equations (C.1) to (C.8), with $\tilde{w}_s^l = \varphi_0 + \varphi_1 z_{2s}^l$ and φ_0 and φ_1 defined in equation (C.15). All results are computed under the restriction $\sigma_1 = \sigma_3 = 0$ and with the wage shifter z_{2s}^l as the instrument z_s^l .

in Section 4.1, and the restriction $z_{1s}^l = 0$ for all s and l , we can write equation (C.15) as

$$\mathbb{E}[z_{2s}^l + \tilde{\eta}(1 - \eta)\xi_s^l - \varphi_0 - \varphi_1 z_{2s}^l | z_{2s}^l, y_s^l = 1] = 0. \quad (\text{C.17})$$

A vector (φ_0, φ_1) that verifies equation (C.17) has $\varphi_1 = 1$ if $\varphi_0 = \tilde{\eta}(1 - \eta)\mathbb{E}[\xi_s^l | z_{2s}^l, y_s^l = 1]$. However, since ξ_s^l and z_{2s}^l belong to the worker's information set $\mathcal{I}_s$, the expectation $\mathbb{E}[\xi_s^l | z_{2s}^l, y_s^l = 1]$ will generally depend on z_{2s}^l . In particular, this expectation is constant if and only if $\eta = 1$, in which case we have $(\varphi_0, \varphi_1) = (0, 1)$. Consistent with this discussion, the confidence sets in Table C.6 contain the true parameter values when $\eta = 1$ (that is, in case 11) but not when $\eta = 0$ (that is, in case 1).

If $\eta \in (0, 1)$, the distributions imposed in equations (6) and (C.14) imply the resulting choice model is not a multinomial logit. As illustrated in Table C.5, for any $\eta \in [0.2, 0.8]$, the confidence sets defined by the odds-based moment inequalities are empty, and we obtain the same result in Table C.6. The confidence set defined by the bounding inequalities is generally non-empty, consistent with the results in Table C.5, but the implied confidence interval for the wage coefficient α tends to be upward biased, with the bias larger for smaller values of η . Intuitively, the smaller the value of η , the larger the variance of the idiosyncratic wage shock $\tilde{\eta}(1 - \eta)\xi_s^l$ in equation (C.12c) and the lower the value of φ_1 implied by the moment condition in equation (C.17). In turn, this lower value of φ_1 implies a larger estimated value of the wage coefficient α .

C.6.2 Estimation with Aggregate Wage Data: Additional Details

Setup. We consider a setup that differs from that in Section 4.1 only in that we limit the data available to the researcher. In particular, we focus on the case in which $\sigma_1 = \sigma_3 = 0$, as in case 1 in Table 1, and partition the workers in the data into $g = 1, \dots, G$ groups and, denoting as $g(s)$ the group to which a worker s belongs, we build

$$w_g^l = \frac{\sum_{s=1}^S \mathbb{1}\{g(s) = g\} w_s^l}{\sum_{s=1}^S \mathbb{1}\{g(s) = g\}} \quad \text{and} \quad z_{2g}^l = \frac{\sum_{s=1}^S \mathbb{1}\{g(s) = g\} z_{2s}^l}{\sum_{s=1}^S \mathbb{1}\{g(s) = g\}}.$$

That is, w_g^l and z_{2g}^l are the average values of w_s^l and z_{2s}^l , respectively, across all workers in group g . To define groups, we partition the support of w_s^l into segments of equal length and assign two workers s and s' to the same group if w_s^l and $w_{s'}^l$ fall in the same segment.

We consider two possible sets of observed variables the researcher may have access to. First, we consider a setup in which the researcher observes $(y_s^l, w_{g(s)}^l, z_{2g(s)}^l)$ for every worker s and alternative l ; that is, the researcher observes worker-specific choices but group-level wage measures and wage shifters. Second, to further study the impact of data aggregation on our moment inequality estimates, we consider a setup in which the researcher observes $(y_s^l, w_s^l, z_{2g(s)}^l)$ for every worker s and alternative l ; that is, the researcher observes worker-specific choices and wage measures, but group-level wage shifters.

Estimation Approach and Simulation Results. We obtain confidence intervals for the parameters of interest using the moment inequalities in Appendix C.1, but with $\tilde{w}_{g(s)}^l$ or $z_{2g(s)}^l$ in place of w_s^l and z_s^l , respectively, in equations (C.1) to (C.8). The results are in Table C.7. Panel A reports estimates computed when the researcher observes worker-specific choices but group-level wage measures and wage shifters; Panel B reports estimates computed when the researcher observes worker-specific choices and wage measures, but group-level wage

shifters.

Panel A shows that aggregating both the wage measure and the instrument yields biased estimates when the number of groups is small. In particular, when aggregating the data into a number of groups $G \leq 20$, we observe a downward bias in our estimate of the wage coefficient α , and this downward bias is larger for smaller G , that is, when the measurement error due to aggregation of both the wage measure and the instrument is larger. We similarly observe a bias in the estimates of the amenity parameter κ_2 implied by the bounding moment inequalities, while the confidence set implied by the odds-based moment inequalities is generally empty. The maximum likelihood estimates exhibit a similar bias whenever the number of groups G is large enough for computation to be feasible; for $G = 10$ and $G = 6$, the limited variation in the predicted wage measure makes the maximum likelihood estimates infeasible to compute.

A comparison of the results reported in Panel A to those reported in Panel B illustrates that all the bias in the moment inequality estimates observed in Panel A is due exclusively to the aggregation of the wage measure. When worker-specific wage measures are used in our moment inequalities, the aggregation of the instrument into groups does not by itself yield biased moment inequality estimates. More specifically, the confidence intervals implied by the bounding moment inequalities always include the true parameter value, but these confidence intervals become wider as the number of groups G decreases.

C.6.3 Heterogeneity in Information Sets Across Individuals

Setup. We consider a setup that differs from that in Section 4.1 only in the information set assigned to each of the $s = 1, \dots, S$ workers in the simulated dataset. More specifically, we depart from the setting described in Section 4.1 in that, instead of assuming that every worker s in the data observes (z_{1s}^l, z_{2s}^l) for each alternative l , we assume that a randomly selected fraction sh of all workers in the data only observes the dummy variable $m_{2s}^l \equiv \mathbb{1}\{z_{2s}^l \geq \mu_2^l\}$ for each alternative l , where μ_2^l is both the mean and median of the distribution of z_{2s}^l . That is, for a fraction $1 - sh$ of workers in the simulated data, it is the case that $\mathbb{E}[w_s^l | \mathcal{W}_s] = z_{1s}^l + z_{2s}^l$ for every alternative l ; for the remaining fraction sh of workers, it is the case that $\mathbb{E}[w_s^l | \mathcal{W}_s] = \mathbb{E}[z_{2s}^l | m_{2s}^l] = 0.5((1 - m_{2s}^l)(2\mu_2^l - \sigma_2) + m_{2s}^l(2\mu_2^l + \sigma_2))$, where the first equality uses the property that z_{1s}^l and z_{2s}^l are independent of each other, and the second equality uses the property that, according to the setup described in Section 4.1, the random variable z_{2s}^l is uniformly distributed in the interval $[\mu_2^l - \sigma_2, \mu_2^l + \sigma_2]$. To simplify the interpretation of the results below, we focus on a parametrization with $\sigma_1 = 0.5$ and $\sigma_3 = 0$.

Estimation Approach and Simulation Results. We obtain confidence sets for the parameter vector of interest using the moment inequalities in Appendix C.1 with z_{2s}^l as the wage

Table C.7: Estimation with Group-specific Wage Data

Case	G	Moment Inequalities				Max. Likelihood		
		First Step	Second Step			θ_α	θ_2	θ_3
		θ_α	Mom. Ineq.	θ_2	θ_3			
<i>Panel A: Group-specific Wage Measure & Instrument</i>								
1	50	[0.99, 1.00]	Bounding Odds-based	$[-0.02, 0.00]$ $[-0.01, 0.00]$	$[0.98, 1.00]$ $\emptyset$	1	0	0.99
2	40	[0.99, 1.00]	Bounding Odds-based	$[-0.02, 0.00]$ $[-0.01, 0.00]$	$[0.97, 1.00]$ $\emptyset$	0.99	0	0.99
3	30	[0.99, 1.00]	Bounding Odds-based	$[-0.04, 0.02]$ $[-0.01, -0.01]$	$[0.95, 1.01]$ $\emptyset$	0.99	-0.01	0.98
4	20	[0.98, 0.99]	Bounding Odds-based	$[-0.03, 0.00]$ $[-0.02, -0.01]$	$[0.95, 0.98]$ $\emptyset$	0.98	-0.02	0.96
5	10	[0.94, 0.94]	Bounding Odds-based	$[-0.05, 0.00]$ $\emptyset$	$[0.88, 0.93]$ $\emptyset$	-	-	-
6	6	[0.86, 0.88]	Bounding Odds-based	$[-0.08, 0.02]$ $\emptyset$	$[0.80, 0.90]$ $\emptyset$	-	-	-
<i>Panel B: Group-specific Instrument</i>								
1	50	[0.99, 1.01]	Bounding Odds-based	$[-0.04, 0.04]$ $[-0.01, 0.01]$	$[0.95, 1.04]$ $\emptyset$	1	0	1
2	40	[0.99, 1.02]	Bounding Odds-based	$[-0.05, 0.04]$ $[-0.01, 0.01]$	$[0.95, 1.05]$ $\emptyset$	0.99	0	0.99
3	30	[0.98, 1.03]	Bounding Odds-based	$[-0.09, 0.07]$ $[-0.01, 0.02]$	$[0.92, 1.09]$ $\emptyset$	0.99	0	0.99
4	20	[0.97, 1.04]	Bounding Odds-based	$[-0.13, 0.11]$ $[-0.01, 0.03]$	$[0.88, 1.13]$ $\emptyset$	0.98	0	0.98
5	10	[0.91, 1.18]	Bounding Odds-based	$[-0.50^*, 0.39]$ $[-0.02, 0.03]$	$[0.65, 1.41]$ $[1.00, 1.00]$	-	-	-
6	6	[0.82, 1.50*]	Bounding Odds-based	$[-0.50^*, 0.50^*]$ $[-0.02, 0.04]$	$[0.50^*, 1.50^*]$ $[1.00, 1.00]$	-	-	-

Note: This table contains 95% CIs computed using the inequalities described in Appendix C.1 with the only exception that aggregated wage measures and an aggregated instrument are used instead of the worker-specific wage measure w_s^l and instrument z_s^l , respectively. The variable G denotes the number of groups. All results are computed setting $\sigma_1 = \sigma_3 = 0$ and $\sigma_2 = 8$, and with the wage shifter z_{2s}^l as the instrument z_s^l .

predictor z_s^l in equations (C.1) to (C.8). The resulting moment inequality estimates are in Table C.8, with the corresponding maximum likelihood estimates in the last three columns.

Note that the moment inequalities used in estimation hold at the true parameter value when the variable z_{2s}^l belongs to the information set $\mathcal{W}_s$ of every worker $s = 1, \dots, S$. It is thus expected that the confidence sets reported in case 1, which correspond to a case in which $sh = 0$, implying that every worker's information set includes the variable z_{2s}^l , do

Table C.8: Heterogeneous Information Set

Case	sh	Moment Inequalities				Max. Likelihood		
		First Step	Second Step					
		θ_α	Mom. Ineq.	θ_2	θ_3	θ_α	θ_2	θ_3
1	0	[0.98, 1.08]	Bounding Odds-based	$[-0.08, 0.08]$ $[0.00, 0.00]$	$[0.92, 1.08]$ $[1.00, 1.00]$	0.98	0	0.98
2	10%	[0.92, 0.98]	Bounding Odds-based	$[-0.04, 0.02]$ $\emptyset$	$[0.94, 0.96]$ $[0.92, 0.92]$	0.91	-0.02	0.91
3	20%	[0.88, 0.90]	Bounding Odds-based	$\emptyset$ $\emptyset$	$\emptyset$ $\emptyset$	0.85	-0.04	0.86
4	30%	$\emptyset$	Bounding Odds-based	$\emptyset$ $\emptyset$	$\emptyset$ $\emptyset$	0.81	-0.05	0.81
5	40%	$\emptyset$	Bounding Odds-based	$\emptyset$ $\emptyset$	$\emptyset$ $\emptyset$	0.76	-0.07	0.76

Note: This table contains 95% CIs computed using the inequalities described in Appendix C.1 with z_{2s}^l as the wage predictor z_s^l . The variable sh denotes the fraction of simulated workers whose information set only includes the variable $m_s^l \equiv \mathbb{1}\{z_{2s}^l \geq \mu_s^l\}$. All results are computed setting $\sigma_1 = 0.5$, $\sigma_2 = 4$ and $\sigma_3 = 0$.

indeed include the true value of the parameter vector. As we consider cases with a larger share of workers whose information set does not include the variable z_{2s}^l (that is, larger values of sh), Table C.8 shows that the bounding moment inequalities first tend to provide confidence sets that are non-empty but both narrower than those obtained when $sh = 0$ and biased towards values of α that are smaller than the true one. However, once the share of workers whose true information set does not include the value z_{2s}^l is sufficiently large (30% or above), the confidence interval for α computed using the first-step moment inequalities becomes empty.

D Data and Summary Statistics

D.1 Data Sources and Sample Construction

The RAIS data. Our main data source is the *Relação Anual de Informações Sociais* (RAIS), an administrative dataset maintained by Brazil’s Ministry of Labor. It includes the universe of formal employment spells in the private and public sectors. Individual workers are identified by government-issued identification numbers (PIS/PASEP and CPF), allowing us to track them as they change employers. For all spells observed between 1993 and 2011, we use information on their start and end dates, average monthly wage, number of work hours stipulated in the contract, 2-digit sector (according to the *Classificação Nacional de Atividades*

Econômicas, CNAE), and information on the worker’s gender, age, race, and education level. All information is reported by the employers.

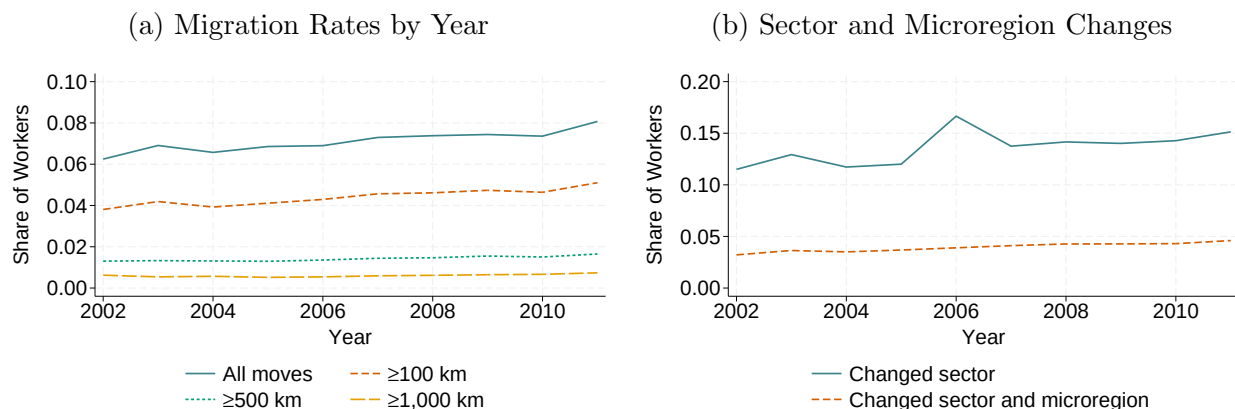
We have no information on workers without formal jobs. These workers may be employed in the informal sector, self-employed, unemployed, or out of the labor force. Based on the 2010 Census, 51% of the Brazilian labor force was in the formal sector. The implied total number of formal workers in the Census closely matches the number of workers in RAIS.

Geography and wage definitions. To determine workers’ location, we use the microregion of the *establishment* at which the worker is employed. Microregions are groups of municipalities that span the entirety of the Brazilian territory. They are defined by the *Instituto Brasileiro de Geografia e Estatística* (IBGE). During our sample period, Brazil had 558 microregions. While RAIS does not contain information on the residence of workers, [Dix-Carneiro and Kovak \(2017\)](#) use 2000 Census data to show that only 3.4% of individuals lived and worked in different microregions. Previous research has used microregions as local labor markets (e.g., [Dix-Carneiro, 2014](#); [Dix-Carneiro and Kovak, 2017, 2019](#); [Felix, 2022](#); [Szerman, 2024](#)).

Workers may hold multiple employment spells (jobs) in a year. To obtain a dataset in which each observation corresponds to a worker and a year, we assign to each worker-year pair the microregion and sector corresponding to the job that the worker held for the longest period during that year. We compute the wage of a worker in a year by adding the labor income earned in every job this worker had in the corresponding year. We calculate the labor income of a worker in each of their jobs by transforming the average monthly wages reported for that job into a measure of average daily wages and multiplying this by the total number of days worked in the job reported in the data. If no start or end dates are provided, we assume that these are January 1 and December 31, respectively.

Sample restrictions and sampling. We limit our data to workers aged 25 to 64, as these are the workers most likely to have completed their education and not yet retired. The sample period is 2002-2011. We use 1993-2001 information to measure each worker’s experience in each sector and microregion. To limit our data to workers with a sufficiently close labor relationship with the formal sector, we restrict our sample to workers observed in RAIS for at least seven years in the sample period. We also restrict our sample to workers with similar demographic characteristics; specifically, we focus on workers identified as male and white with at least a high school degree. For computational reasons, we focus on a sample of 10 million worker-year pairs. To ensure we observe a large enough number of individuals per market, we focus on 1,000 labor markets consisting of all combinations of the 50 microregions (out of 558) and 20 sectors (out of 51) with the largest total employment reported in RAIS. We then obtain our sample by randomly sampling 1 million individuals per sample year among those employed in the 1,000 labor markets of interest.

Figure D.1: Migration and Sector Changes, by Year



Panel (a) shows yearly migration rates, both overall and by minimum migration distance. The distance-specific lines report the share of workers who moved at least 100 km, 500 km, and 1,000 km, respectively. Panel (b) shows the share of workers who changed sectors from the previous year and the share who both changed sectors and changed microregions. Data include all white male workers aged 25–64 with at least a high school education.

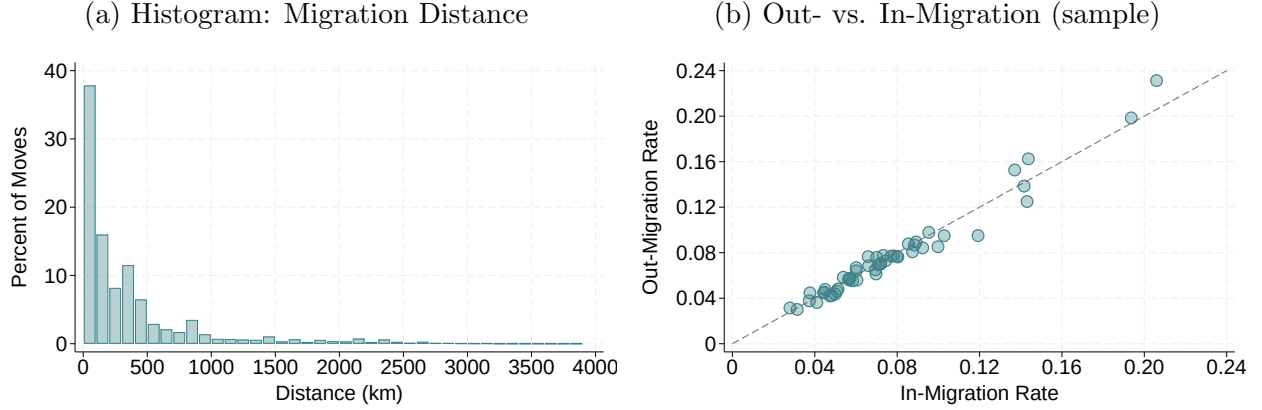
Additional data sources. We measure distances between microregions by the geodesic distance between their population centroids. Data on internet connections are from the *Agência Nacional de Telecomunicações* (ANATEL), which provides the number of broadband connections by municipality and year from 2007 onwards. We define internet access at the microregion level as the average share of households with broadband internet access in the 2007–2011 period.

D.2 Summary Statistics: Migration

Consistent with our estimation sample, this section focuses on white male workers with at least a high school education in the 2002–2011 period. However, statistics are computed using information on all workers with those demographic characteristics; i.e., not only those linked to the 50 largest microregions and 20 largest sectors.

Figure D.1a provides yearly migration rates; i.e., the share of workers who change microregion of employment between years t and $t - 1$. It shows an upward trend over the sample period, from 6.2% in 2002 to 8.1% in 2011. Migration over longer distances is substantially less common: in 2011, 5.1% of workers moved at least 100 km, 1.6% moved at least 500 km, and 0.7% moved at least 1,000 km. Figure D.1b provides a similar figure for sectoral changes, which are more common. The share of workers changing sector ranges from 11.5% to 16.7% over the sample period, while the share changing both sector and microregion rises from 3.2% in 2002 to 4.6% in 2011. Thus, most changes in employment sector are not accompanied by

Figure D.2: Migration Patterns



Panel (a) provides a histogram of distances between origin and destination microregions for observed migrations. In panel (b), each marker represents one of the 50 microregions in our estimation sample. The y-axis measures the out-migration rate, while the x-axis measures the in-migration rate. These rates include moves to and from all microregions, including microregions outside the 50 largest ones. The dashed line represents the 45-degree line. In both panels, data include all white male workers aged 25–64 with at least a high school education.

migration.

Figure D.2a depicts the distribution of distances between origin and destination microregions for those who migrate. Moves are highly concentrated at short distances: 37.8% of moves are below 100 km, 53.8% are below 200 km, and 79.9% are below 500 km. Nonetheless, a non-negligible share of moves occurs over long distances, with 8.6% of moves covering at least 1,000 km. Figure D.2b provides a scatter plot depicting the distribution of in-migration and out-migration rates for the 50 microregions in our sample. Each marker represents one of these microregions. In- and out-migration rates are strongly correlated across microregions, with a correlation of 0.98 and rates varying close to one-for-one. The bulk of microregions have migration rates in the 3 to 12% range, although a few locations have substantially higher rates.

E Appendix to Empirical Analysis

E.1 Implementation of Moment Inequalities

Approximation points. Computing the moments in equations (12) and (25) requires specifying the approximation points $e_{is}^{ll'}$ and $e_{isjr}^{ll'}$, respectively. Equations (14) and (27) provide functional forms for these approximation points that, according to corollaries 1 and 3, re-

sult in second-step bounding inequalities and first-step inequalities, respectively, that can point identify the parameters of interest. Furthermore, as discussed in Appendices B.1 and B.5, the approximation points in equations (14) and (27) yield the tightest identified sets among all approximation points in a family of functions described in detail in those sections. The expressions in equations (14) and (27), however, depend on the expectation of specific wage differences conditional on the wage predictor used to build the corresponding inequality. Since we do not know the true value of those expectations, we must approximate them in some way. In our empirical application, we use approximation points that are simple to compute and that, importantly, do not depend on any regression estimate.³⁰

Specifically, we compute the approximation points entering the bounding moment inequalities in equation (12) as

$$e_{is}^{ll'} = \psi_0 + \psi_1 \Delta z_{st}^{ll'}, \quad (\text{E.1})$$

where ψ_0 and ψ_1 are constants the researcher chooses. In practice, we simultaneously use in estimation several different bounding inequalities of the type in equation (12) computed using the approximation points in equation (E.1) with different values of the constants ψ_0 and ψ_1 . Similarly, we compute the approximation points entering the first-step moment inequalities in equation (25) as

$$e_{is}^{ll'} = \psi_2 0.5(\Delta z_{st}^{ll'} + \Delta z_{rt}^{ll'}), \quad (\text{E.2})$$

where ψ_2 is a constant. In practice, we also use simultaneously in estimation several different first-step moment inequalities of the type in equation (25) computed using the approximation points in equation (E.2) with different values of the constant ψ_2 .³¹

Aggregating across pairs of locations in the first-step moment inequalities. The moment inequality in equation (25) holds for any two locations l and l' . As we consider 50 possible

³⁰This stands in contrast with the approach we follow in our simulation setting when computing the CIs reported in Table 1, where we approximate the conditional expectations entering equations (14) and (27) using linear regressions; see Appendix C.1 for more details. While the large number of observations we use in our simulation setting implies that any noise in the linear regression estimates will have a minimal impact on our estimated CIs, this may not be true in our empirical application. Thus, to simplify the computation of the moment inequality CIs in our application, we restrict ourselves to using approximation points that are not functions of prior estimates.

³¹The formulas for the approximation points in equations (E.1) and (E.2) are similar to those used to compute the simulation results discussed in Appendix C.5. However, the formulas in equations (E.1) and (E.2) differ from those in equations (C.10) and (C.11) in that they do not depend on the structural parameters to be estimated, but on constants chosen by the researcher. The fact that the researcher can explore the identifying power of a large set of possible values for the constants (ψ_0, ψ_1, ψ_2) has the advantage of generating tighter CIs, and the computational disadvantage that the number of moment inequalities used in estimation increases in the set of values of (ψ_0, ψ_1, ψ_2) the researcher uses.

locations in our empirical application, there is a very large number of potential location pairs. Instead of using a correspondingly large number of different moment inequalities of the type in equation (25), we use a smaller number of inequalities that aggregate across location pairs. Our choice of which location pairs to combine is guided by Corollary 3. This corollary indicates that a requisite for the inequality in equation (25) to point identify the wage parameter θ_α is that the locations l and l' being compared offer the same amenity level in the reference population; that is, using the notation in our empirical application, $\kappa_{nt}^l - \kappa_{nt}^{l'} = 0$. Enforcing this condition is infeasible as these amenity levels are continuous variables that are only estimated later in our estimation procedure. However, as κ_{nt}^l accounts for migration costs in our setting, it is reasonable to expect it will vary with the distance between locations n and l . Thus, we hypothesize that locations l and l' that are at a similar distance to an origin n are more likely to have similar amenity levels from the perspective of workers located in n and, when combined in the context of the inequality in equation (25), should yield smaller identified sets. Consequently, we form the sample analogue of the moment in equation (25) aggregating only across location pairs l and l' for which the difference between the distance from n to l and the distance from n to l' is in the lower tercile of all pairwise differences in distance to n . Given such location pairs, we form the moment function in equation (25) by further aggregating across all sector pairs s and r , and across worker pairs within those sectors.

Instrument vectors. Given a wage predictor z_{st}^l for every l , s , and t , we construct moment inequalities using instruments of the type in equation (30) for every $d_k \in \{-1, 1\}$ and $[\underline{z}_k, \bar{z}_k] \in \{[-\infty, Q_{25}(\Delta z_{st}^{ll'})], [Q_{25}(\Delta z_{st}^{ll'}), Q_{50}(\Delta z_{st}^{ll'})], [Q_{50}(\Delta z_{st}^{ll'}), Q_{75}(\Delta z_{st}^{ll'})], [Q_{75}(\Delta z_{st}^{ll'}), \infty]\}$, where $Q_q(\Delta z_{st}^{ll'})$ denotes the percentile q of the distribution of $\Delta z_{st}^{ll'}$. Thus, we use 8 different instrument vectors.

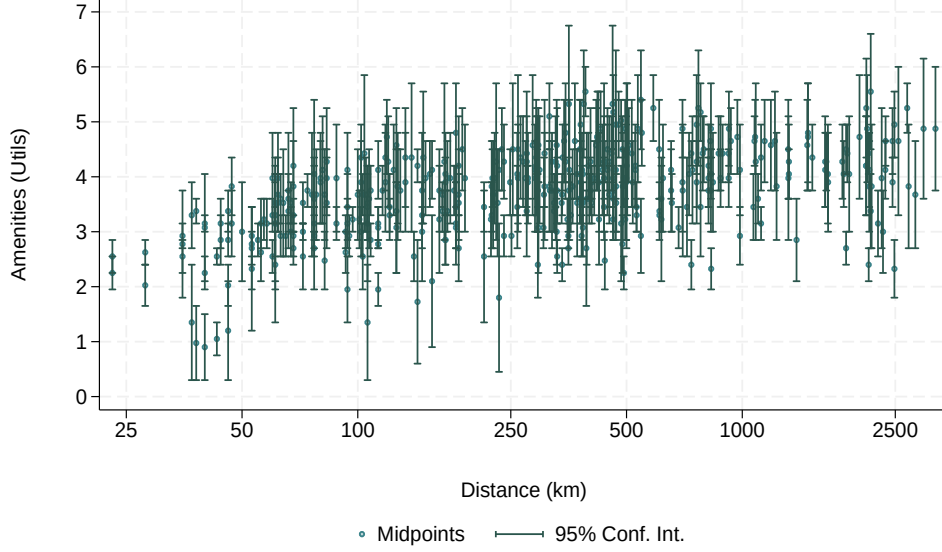
E.2 Additional Result

Figure E.1 reports confidence intervals for the estimated bilateral non-wage utility components κ_{nt}^l described in Section 5.3.2.

E.3 PPML-IV Estimator

We describe here our implementation of the estimator in Artuç and McLaren (2015). To rationalize the implementation of this estimator in our setting, we must assume all workers employed in the same sector s have the same information set in any given period t , regardless of their location of residence. Thus, $\mathcal{J}_{ist} = \mathcal{J}_{i'st}$ for any sector s , period t , and any two workers i and i' employed in s at t . Given this assumption, we can write the model-implied number

Figure E.1: Amenities from Moment Inequalities with Confidence Intervals



Each point shows the midpoint of the 95% confidence interval for a given bilateral non-wage utility component κ_{nt}^l , with the associated 95% confidence interval.

of sector s workers that migrate between locations n and l at t as:

$$\begin{aligned}
 M_{nst}^l &= \frac{\exp(\alpha \mathbb{E}[w_{st}^l | \mathcal{J}_{st}] - \kappa_{nt}^l)}{\sum_k \exp(\alpha \mathbb{E}[w_{st}^k | \mathcal{J}_{st}] - \kappa_{nt}^k)} L_{nst-1} \\
 &= \exp(\alpha \mathbb{E}[w_{st}^l | \mathcal{J}_{st}] - \kappa_{nt}^l + \Gamma_{nst}) \\
 &= \exp(\alpha(w_{st}^l - \nu_{st}^l) - \kappa_{nt}^l + \Gamma_{nst}) \\
 &= \exp(\Lambda_{st}^l + \Psi_{nt}^l + \Gamma_{nst}), \tag{E.3}
 \end{aligned}$$

where L_{nst-1} is the total number of workers in location n and sector s at period $t-1$; $\nu_{st}^l \equiv w_{st}^l - \mathbb{E}[w_{st}^l | \mathcal{J}_{st}]$ is the expectational error of these workers when predicting wages in sector s , period t , and location l ; and

$$\Lambda_{st}^l \equiv \alpha(w_{st}^l - \nu_{st}^l), \quad \Gamma_{nst} \equiv -\ln \left(\sum_k \exp(\alpha \mathbb{E}[w_{st}^k | \mathcal{J}_{st}] - \kappa_{nt}^k) \right) + \ln L_{nst}, \quad \Psi_{nt}^l \equiv -\kappa_{nt}^l.$$

Using information on $\{M_{nst}^l\}_{n=1, l=1}^{L, L}$ for a period t , sector s , and L origin and destination locations, the procedure in [Artuç and McLaren \(2015\)](#) recovers estimates of α and $\{\kappa_{nt}^l\}_{n=1, l=1}^{L, L}$ in three steps. First, it computes PPML estimates of $\{\Lambda_{st}^l\}_{l=1}^L$, $\{\Gamma_{nst}\}_{n=1}^L$ and $\{\Psi_{nt}^l\}_{n=1, l=1}^{L, L}$ using the expression in the last line in equation (E.3). Second, under the assumption that a variable z_{st}^l correlated with w_{st}^l belongs to the information set $\mathcal{J}_{st}$, it computes an IV

Table E.1: Robustness of First-Step Estimates of the Migration Elasticity

Specification	95% CI for α	Width
<i>Panel A: Alternative wage-information coarsenings</i>		
Quartiles (4 bins)	[1.21, 1.84]	0.64
Terciles (3 bins)	[1.15, 1.74]	0.59
Quartiles and terciles stacked	[1.20, 1.81]	0.61
<i>Panel B: Random 20% subsamples, quartile information</i>		
Subsample 1 (20.0%)	[1.18, 1.75]	0.57
Subsample 2 (20.0%)	[1.10, 1.72]	0.62
Subsample 3 (20.0%)	[1.14, 1.85]	0.71
Subsample 4 (20.0%)	[1.17, 1.73]	0.56
Subsample 5 (20.0%)	[1.21, 1.79]	0.58

Panel A reports first-step moment inequality confidence intervals for α under alternative coarsenings of lagged wages. “Quartiles and terciles stacked” combines the moments from both coarsenings. Panel B reports the same quartile-based estimator on five independent random subsamples of the baseline sample, each of them including 20% of the observations in the original sample. All intervals are 95% confidence intervals.

estimate of α by regressing $\hat{\Lambda}_{st}^l$ on w_{st}^l using z_{st}^l as an instrument, with $\hat{\Lambda}_{st}^l$ denoting the first-step estimate of Λ_{st}^l . Third, it recovers $\hat{\kappa}_{nt}^l = -\hat{\Psi}_{nt}^l$ for every origin n and destination l .

E.4 Additional Robustness Checks for the First-Step Estimate

This subsection reports two additional robustness checks for the first-step moment inequality confidence interval of α . The preferred specification in Figure 1 uses quartiles of lagged wages as the wage-information proxy and is estimated on the full sample. We first explore the robustness of our baseline confidence interval for α to the particular bins in which we classify lagged wages; in particular, we replace the quartiles by terciles as a wage-information proxy, and also combine the tercile- and quartile-based moments in a single specification. We then ask whether the preferred quartile result is stable in smaller samples by re-estimating it on five independent random subsamples, each of them including 20% of the observations included in the original sample. In each subsample, workers are drawn within year-by-origin-by-sector cells, so that the subsamples preserve the composition of the original estimation sample across years and, for each year, across prior origins and sectors.

Table E.1 reports the results. Panel A shows that the estimated confidence set is similar when we use terciles instead of quartiles: the tercile specification yields a 95% confidence interval for α equal to [1.15, 1.74], similar to the baseline confidence interval of [1.21, 1.84] obtained when using quartiles to build the moment inequality. Stacking the tercile and quartile moments gives a confidence interval of [1.20, 1.81], also similar to the baseline one.

Panel B shows that the quartile-based estimate is also stable across random subsets of

the original estimation sample, with each subsample including a fifth of the observations in the original sample. All five subsamples yield confidence intervals that are non-empty. The lower bound ranges from 1.10 to 1.21 and the upper bound ranges from 1.72 to 1.85.

E.5 Time-Variation in Estimated Amenities

As discussed in Section 5.2, our empirical model allows the worker’s utility of choosing a labor market in a particular prior location at a particular period to depend on a parameter that varies flexibly across locations indexed by l , periods indexed by t , and the prior location of the worker, which we index by n . Making explicit the dependence of the parameter κ_t^l in equation (34b) on the prior location of the worker, we denote it here as κ_{nt}^l . As discussed in Section 2, this term is broader than what is often called amenities in the spatial literature. In our empirical application, κ_{nt}^l captures all non-wage payoff components attached to choosing destination l at year t for a worker initially in origin n : destination-year amenities, prices, and bilateral migration costs. We refer to this term by the shorthand “amenity” for simplicity.

Given a value of the wage coefficient α , we use the second-step moment inequalities introduced in Section 3.1 to compute a confidence interval for κ_{nt}^l for each sample period t and each pair of origin and destination locations n and l , respectively. In practice, we compute such confidence intervals at each value of α in a confidence interval for this parameter obtained using the first-step moment inequalities introduced in Section 3.2. Formally, denoting as $[\underline{\kappa}_{nt,0.96}^l(\alpha), \bar{\kappa}_{nt,0.96}^l(\alpha)]$ the 96% confidence interval for κ_{nt}^l computed given a value of α , and denoting as $[\underline{\alpha}_{0.99}, \bar{\alpha}_{0.99}]$ the 99% confidence interval for α , we compute a 95% confidence interval for κ_{nt}^l as follows:

$$\left[\underline{\kappa}_{nt,0.95}^l, \bar{\kappa}_{nt,0.95}^l \right] = \left[\min_{\alpha \in [\underline{\alpha}_{0.99}, \bar{\alpha}_{0.99}]} \underline{\kappa}_{nt,0.96}^l(\alpha), \max_{\alpha \in [\underline{\alpha}_{0.99}, \bar{\alpha}_{0.99}]} \bar{\kappa}_{nt,0.96}^l(\alpha) \right].$$

Denoting the mid-point of each of these intervals as $\hat{\kappa}_{nt}^l$ (i.e., $\hat{\kappa}_{nt}^l = 0.5(\underline{\kappa}_{nt,0.95}^l + \bar{\kappa}_{nt,0.95}^l)$), we provide here summary statistics on the distribution of $\hat{\kappa}_{nt}^l$. In particular, we compute two complementary statistics. First, for each origin–destination pair (n, l) , we calculate the standard deviation of $\hat{\kappa}_{nt}^l$ over time and then average this object across pairs. Second, for each year t , we calculate the cross-sectional standard deviation of $\hat{\kappa}_{nt}^l$ across origin-destination pairs and then average this object across years. We report these statistics both for all origin-destination pairs in the sample and for the subset of high-flow pairs, defined as pairs in the top half of the average annual origin-destination migration-flow distribution.

Table E.2 reports the results. Among all reported pairs, the average within-pair standard deviation of $\hat{\kappa}_{nt}^l$ over time is 0.565, whereas the average cross-sectional standard deviation across pairs within a year is 1.737. The ratio of these two moments is therefore 0.325. Among

Table E.2: Time Variation in Estimated Amenities

	All Pairs	High-Flow Pairs
Average within-pair s.d. over time	0.565	0.549
Average cross-pair s.d. within year	1.737	1.522
Ratio	0.325	0.360
Number of origin–destination pairs	1,646	845

This table uses the midpoint of the confidence interval for κ_{nt}^l obtained by taking, for each origin–destination pair and year, the minimum lower bound and maximum upper bound across the admissible values of α . “All pairs” includes all origin–destination pairs. “High-flow pairs” includes the subset of those pairs that are in the top half of the average annual migration-flow distribution. The sample period is 2002–2011.

high-flow pairs, the corresponding within-pair and cross-pair average standard deviations are 0.549 and 1.522, yielding a ratio of 0.360. These numbers imply that the estimated bilateral non-wage utility components move over time, but that this time variation remains smaller than the persistent cross-sectional dispersion in κ_{nt}^l across origin–destination pairs.

Economically, these results suggest that allowing κ_{nt}^l to vary by year is empirically meaningful, but not because bilateral non-wage utility components fluctuate so strongly from year to year that they overturn the cross-sectional structure of migration frictions. Rather, the estimates indicate a stable cross-sectional ranking of origin–destination pairs, overlaid with a smaller but non-negligible time-varying component. This pattern is consistent with a setting in which persistent bilateral migration frictions remain central for explaining workers’ location choices, while year-specific movements in amenities, prices, or other bilateral conditions introduce additional variation that would be ruled out by a time-invariant specification.

E.6 Adjustment Speed After a Permanent Wage Shock

Here, we study the speed with which migration flows adjust to a location-specific unexpected persistent local labor demand shock. More specifically, we measure the model-predicted change in the population share in the shocked region at different time horizons, and we compute such predicted change under different assumptions on the content of workers’ information set and on the value of the wage coefficient; that is, under different assumptions on the content of the set $\mathcal{W}_i$ and the parameter α in the model described in Section 5.2.

We consider a permanent positive wage shock in São Paulo, Brazil’s largest microregion. In particular, at some period t^* , we simulate the impact of an unexpected and permanent increase of 0.10 log points in wages in São Paulo in all sectors. Specifically, for each simulation draw, we compare two economies. First, a baseline economy in which there is no shock. Second, an economy in which the realized wage path is exactly the same in every location as in the baseline economy, except for the additive wage increase in São Paulo from a period

t^* onward. For each of the two economies, we compute the predictions of three different models, each of them averaged over 20 simulated wage paths: (a) a full-information model in which workers have perfect wage information and assign a relatively low weight to wages in utility (in particular we set $\alpha = 0.5$); (b) a partial-information model in which workers must forecast wages using only information on the quartile of lagged wages and wages again carry a relatively low weight in workers' utility (again, we set $\alpha = 0.5$); and, (c) a partial-information model in which workers have the same information sets as in model (b) above but assign a relatively larger weight to wages in utility (in particular, we set $\alpha = 1.5$).

Table E.3 reports the results. The first numeric column reports the long-run increase in São Paulo's population share, measured at period $t^* + 17$. The second numeric column reports the share of the long-run increase in São Paulo's population share that was already reached at period $t^* + 3$; that is, three periods after the shock. This share is a simple measure of the speed with which labor is predicted to be reallocated in reaction to the shock we study.

Several patterns emerge from the results in Table E.3. Focusing first on the long-run change in population in São Paulo, we reach two conclusions. First, a higher value of α results in a larger long-run population response. Under incomplete information, raising α from 0.5 to 1.5 more than triples the long-run population response, from 0.21% to 0.77%. Second, the long-run adjustment to a permanent wage shock also increases in the precision of workers' information: with complete information and $\alpha = 0.5$, the population share in São Paulo increases by 0.28%, whereas the corresponding increase under incomplete information is 0.21%.

Focusing next on the predicted speed of adjustment, we can again reach two conclusions. First, the low- α model that assumes that workers are only partially informed about wages predicts that only 20.2% of the long-run adjustment would have been reached three periods after the shock, while the analogous model that features fully-informed workers predicts this percentage to be 27.0%. Second, the relatively slow adjustment of the model featuring partially informed workers can be reversed if the value of α is sufficiently large, as the model displayed in scenario 3 in Table E.3 predicts that 33.6% of the long-run adjustment would have been reached three periods after the shock.

In sum, models that feature a higher value of the wage coefficient α , or more precise wage information on the part of workers, predict both a larger long-run adjustment and a faster adjustment to a location-specific unexpected permanent wage change.

Table E.3: Speed of Adjustment After a Permanent Wage Shock in São Paulo

Scenario	Worker Information	Value of α	Long-Run Change in Population	Percentage of Long-Run Change Reached by Period $t^* + 3$
1	Complete	0.5	0.28%	27.0%
2	Incomplete, 4 bins	0.5	0.21%	20.2%
3	Incomplete, 4 bins	1.5	0.77%	33.6%

This table reports responses to an unexpected permanent increase in wages in São Paulo in all sectors of 0.10 log points from period t^* onwards. Each entry is the prediction of the model described by the information in the columns “Worker Information” and “Value of α ” averaged across 20 simulated wage paths. The long-run change in São Paulo’s population share is measured 17 periods after the shock. The final column reports the ratio of the period $t^* + 3$ to the period $t^* + 17$ change in São Paulo’s population share.

F Extension: Dynamic Model of Location Choice

We describe here how to extend our estimation method to settings with forward-looking agents facing one-time migration costs. In Appendix F.1, we describe our dynamic migration model. In Appendix F.2, we show how to adapt the procedure in Section 3 to estimate the parameters of the dynamic model. Appendix F.3 provides additional details.

F.1 Theoretical Framework

Defining a dummy variable y_{ist}^l that equals one if worker i of type s chooses l at t , we assume

$$y_{ist}^l \equiv \mathbb{1}\{l = \operatorname{argmax}_{l'=1,\dots,L} \mathbb{E}[\mathcal{V}_{ist}^{l'} | \mathcal{J}_{ist}]\} \quad \text{for } l = 1, \dots, L, \quad (\text{F.1})$$

where $\mathcal{V}_{ist}^l$ is the choice-specific value function and $\mathbb{E}[\cdot]$ defined as in equation (2). We impose:

$$\mathcal{V}_{ist}^l = v_{ist}^l + \varepsilon_{ist}^l, \quad (\text{F.2a})$$

$$v_{ist}^l = \beta x_{nt}^l + \lambda_t^l + \alpha w_{ist}^l + \delta \mathcal{V}_{ist+1}^{(lt)}, \quad (\text{F.2b})$$

where n indexes the location of worker i of type s at period $t - 1$, and

$$\mathcal{V}_{ist+1}^{(lt)} \equiv \max_{l'=1,\dots,L} \mathbb{E}[\mathcal{V}_{ist+1}^{(lt)l'} | \mathcal{J}_{ist+1}^{(lt)}]. \quad (\text{F.3})$$

Equation (F.2a) splits the choice-specific value function into the idiosyncratic component ε_{ist}^l and a variable v_{ist}^l that equation (F.2b) defines as the sum of four terms. First, the migration costs between locations n and l , modeled as a function of observed covariates x_{nt}^l and a vector of parameters β . Second, a location- and period-specific term λ_t^l , which captures a location’s

amenities and (log) price index. Third, the wage component αw_{ist}^l . Fourth, the product of the discount factor δ and a variable $\mathcal{V}_{ist+1}^{(lt)}$ that, according to equation (F.3), equals the worker's period- $t + 1$ value function conditional on choosing alternative l at period t .³²

Defining $\lambda_t = (\lambda_t^1, \dots, \lambda_t^L)$ and $\varepsilon_{ist} = (\varepsilon_{ist}^1, \dots, \varepsilon_{ist}^L)$, we assume that

$$(\varepsilon_{ist}, \lambda_t) \subseteq \mathcal{J}_{ist}. \quad (\text{F.4})$$

Thus, when making choices at t , workers know the vectors of contemporaneous idiosyncratic preferences ε_{ist} and amenities λ_t . Equation (F.4) does not restrict the information workers have about wages $w_{ist'} = (w_{ist'}^1, \dots, w_{ist'}^L)$ for any $t' \geq t$ or amenities $\lambda_{t'}$ for any $t' > t$.

While we do not specify the full content of workers' information sets, we limit the processes that determine them and assume that, for any $t' > t$,

$$\mathcal{J}_{ist'} \perp\!\!\!\perp y_{ist} | \mathcal{J}_{ist}. \quad (\text{F.5})$$

Thus, conditional on the worker's information set at a period t , the worker's information set in subsequent periods does not depend on the worker's choice at t . Our framework thus does not allow for endogenous learning, understood as the process through which the worker's information set at t may depend on the history of locations visited by the worker.

Defining $\Delta v_{ist}^{ll'} \equiv v_{ist}^l - v_{ist}^{l'}$, we impose that for any period t , locations l and l' , types s and r , and workers i and j that share a common prior location n ,

$$\mathbb{E}[\Delta v_{ist}^{ll'} | \mathcal{J}_{ist}, \mathcal{J}_{jrt}] = \mathbb{E}[\Delta v_{ist}^{ll'} | \mathcal{J}_{ist}] = \mathbb{E}[\Delta v_{ist}^{ll'} | \mathcal{W}_{ist}]. \quad (\text{F.6})$$

The first equality imposes that every worker has at least as much information as any other worker of a different type r who shares the same prior location n about differences in that worker's own location-specific value function. The second equality imposes that, once we condition on all other elements of the information set of worker i of type s at period t , the idiosyncratic preferences in ε_{ist} do not contain any information on $\Delta v_{ist}^{ll'}$ for any two locations l and l' .³³

As in the static model, we restrict the information workers have about location-specific wages. For workers i and j of types s and r , respectively, and locations l and l' , we impose

³²A comparison of equations (3) and (F.2) shows that, at the expense of assuming $\delta = 0$, the static model allows for a more flexible specification of migration costs, which may vary freely between locations and periods.

³³The variable $\Delta v_{ist}^{ll'}$ depends on the worker's future choices, which will depend on $\varepsilon_{ist'}$ for $t' > t$; thus, equation (F.6) will generally not hold unless ε_{ist} is independent over time.

that

$$\mathbb{E}[\Delta w_{ist}^{ll'} | \mathcal{W}_{ist}, \mathcal{W}_{jrt}] = \mathbb{E}[\Delta w_{st}^{ll'} | \mathcal{W}_{ist}]. \quad (\text{F.7})$$

Thus, the worker's period- t expectation of the contemporaneous wage difference between two locations l and l' equals the expectation of terms that do not vary across workers of the same type s . We do not restrict the information workers have about the difference in type- and location-specific wages $\Delta w_{st}^{ll'}$ between any two locations l and l' and in any period t' .

Finally, as in equation (6), we assume that, for workers i and j of types s and r ,

$$F_{\varepsilon}(\varepsilon_{ist}, \varepsilon_{jrt} | \mathcal{W}_{ist}, \mathcal{W}_{jrt}) = F_{\varepsilon}(\varepsilon_{ist})F_{\varepsilon}(\varepsilon_{jrt}) = \exp\left(-\sum_{l=1}^L (\exp(-\varepsilon_{ist}^l) + \exp(-\varepsilon_{jrt}^l))\right). \quad (\text{F.8})$$

That is, the vectors ε_{ist} and ε_{jrt} are independent of $(\mathcal{W}_{ist}, \mathcal{W}_{jrt})$ and of each other, and the elements of each vector are *iid* according to a type I extreme value distribution.

The elements of λ_t are identified up to a common shifter. We normalize $\lambda_t^1 = 0$ for all t ; the model parameters are thus $(\lambda_t^2, \dots, \lambda_t^L)$ for all t , α , and β .

F.2 Estimation With Moment Inequalities

We provide a two-step estimator. In the first step, we compute a confidence set for (α, β) using inequalities that difference out the amenity term λ_t^l for all l and t . In the second step, for each $l = 2, \dots, L$ and sample period t , we derive inequalities that depend only on α , β , and λ_t^l , and combine these inequalities with the confidence set for (α, β) to compute a confidence interval for λ_t^l . We denote by $(\theta_{\alpha}, \theta_{\beta})$ the parameter vector with true value (α, β) , and by $\Theta_{(\alpha, \beta)}$ the set of possible values of $(\theta_{\alpha}, \theta_{\beta})$. We denote by θ_t^l the parameter with true value λ_t^l and by Θ_t^l the set of possible values of θ_t^l . In Appendix F.2.1, we discuss the estimation of $\{\theta_t^l\}_{l,t}$. In Appendix F.2.2, we describe the estimation of $(\theta_{\alpha}, \theta_{\beta})$.

F.2.1 Second-Step: Estimating Location-Specific Amenities

Denote by $\Delta\theta_t^{ll'} \equiv \theta_t^l - \theta_t^{l'}$ the parameter with true value $\Delta\lambda_t^{ll'} \equiv \lambda_t^l - \lambda_t^{l'}$, and by $\Theta_t^{ll'}$ the set of possible values of $\Delta\theta_t^{ll'}$. Then, for any pair of locations l and l' , z_{st} , and a scalar random variable $e_{ist}^{ll'}$, we define the moment

$$\tilde{m}^{ll'}(z_{st}, \Delta\theta_t^{ll'}) \equiv \mathbb{E}[y_{ist}^{l'} - y_{ist}^l \exp(-e_{ist}^{ll'}) (1 + e_{ist}^{ll'} - \Delta\tilde{v}_{ist}^{ll'}(\Delta\theta_t^{ll'})) | z_{st}], \quad (\text{F.9})$$

with

$$\Delta \tilde{v}_{ist}^{ll'}(\Delta \theta_t^{ll'}) = \beta \Delta x_{nt}^{ll'} + \Delta \theta_t^{ll'} + \alpha \Delta w_{ist}^{ll'} + \delta \beta \sum_{l''=1}^L y_{ist+1}^{(lt)l''} (x_{lt+1}^{l''} - x_{l't+1}^{l''}). \quad (\text{F.10})$$

Theorem 4 establishes a property of this moment when evaluated at $\Delta \theta_t^{ll'} = \Delta \lambda_t^{ll'}$.

Theorem 4 *Assume equations (F.1) to (F.8) hold. Then, $\tilde{m}^{ll'}(z_s, \Delta \lambda_t^{ll'}) \geq 0$ if $e_{ist}^{ll'} \subseteq \mathcal{J}_{ist}$ and $z_{st} \subseteq \mathcal{W}_{ist}$.*

We prove this theorem in Appendix F.3.1. Theorem 4 shows that, given knowledge of (α, β) , one may bound the amenity difference $\Delta \lambda_t^{ll'}$ for any sample period t and locations l and l' , and Appendix F.3.2 provides an expression for the optimal scalar $e_{ist}^{ll'}$.

To obtain the inequality $\tilde{m}^{ll'}(z_{st}, \Delta \lambda_t^{ll'}) \geq 0$, we first follow steps analogous to those taken to derive the static bounding inequality in equation (13).³⁴ In this way, we obtain

$$\mathbb{E}[y_{ist}^{l'} - y_{ist}^l \exp(-e_{ist}^{ll'})(1 + e_{ist}^{ll'} - (v_{ist}^l - v_{ist}^{l'})) | z_{st}] \geq 0. \quad (\text{F.11})$$

This inequality cannot be used for estimation as the value function difference $v_{ist}^l - v_{ist}^{l'}$ is not solely a function of observed covariates and parameters. We follow Morales et al. (2019) and implement a discrete analogue of Euler's perturbation method to derive an inequality that can be used for estimation. Specifically, we substitute $v_{ist}^{l'}$ in equation (F.11) by a function $\tilde{v}_{ist}^{l'}$, where $v_{ist}^{l'}$ and $\tilde{v}_{ist}^{l'}$ differ in that the latter conditions on the choices that, from period $t + 1$ onwards, would be optimal for worker i of type s if they had chosen alternative l at t . As our dynamic model exhibits one-period dependence, $v_{ist}^l - \tilde{v}_{ist}^{l'}$ is a function exclusively of the difference in static utilities at period t and the discounted difference in static utilities at $t + 1$ that are due to whether the worker chooses alternatives l or l' at t . Specifically,

$$v_{ist}^l - \tilde{v}_{ist}^{l'} = u_{ist}^l - u_{ist}^{l'} + \delta \beta \sum_{l''=1}^L y_{ist+1}^{(lt)l''} (x_{lt+1}^{l''} - x_{l't+1}^{l''}), \quad (\text{F.12})$$

where $y_{ist+1}^{(lt)l''}$ is the optimal choice at $t + 1$ of worker i of type s if they were to choose alternative l at t . The expression in equation (F.12) is a function of observed covariates and parameters. Moreover, $v_{ist}^{l'} \geq \tilde{v}_{ist}^{l'}$ for every worker, period, and pair of choices l and l' . Thus, the sign of the moment inequality in equation (F.11) is preserved if $\tilde{v}_{ist}^{l'}$ takes the place of $v_{ist}^{l'}$.

³⁴While it may be feasible to use the odds-based moment inequalities introduced in Section 3.1.2 in the context of our dynamic model, we have not found a way of doing so.

F.2.2 First-Step: Estimating Migration Costs and Wage Coefficient

For any period t , locations l and l' , worker i of type s and worker j of type r , vectors z_{st} and z_{rt} , and scalar random variable $e_{ijsrt}^{ll'}$, we define the moment

$$\begin{aligned} \tilde{\mathbb{M}}^{ll'}(z_{st}, z_{rt}, \theta_\alpha, \theta_\beta) \equiv \\ \mathbb{E}[y_{ist}^l y_{jrt}^l + y_{ist}^{l'} y_{jrt}^{l'} - y_{ist}^l y_{jrt}^{l'} \exp(-e_{ijsrt}^{ll'}) (2 + 2e_{ijsrt}^{ll'} - (\Delta \tilde{v}_{ist}^{ll'} + \Delta \tilde{v}_{jrt}^{l'l})) | z_s, z_r]. \end{aligned} \quad (\text{F.13})$$

Theorem 5 establishes a property of this moment when evaluated at $\theta_\alpha = \alpha$ and $\theta_\beta = \beta$.

Theorem 5 *Assume equations (F.1) to (F.8) hold. Then, $\mathbb{M}^{ll'}(z_{st}, z_{rt}, \alpha, \beta) \geq 0$ if $e_{ijsrt}^{ll'} \subseteq \mathcal{J}_{ist} \cup \mathcal{J}_{jrt}$, $z_{st} \subseteq \mathcal{W}_{ist}$, and $z_{rt} \subseteq \mathcal{W}_{jrt}$.*

We prove Theorem 5 in Appendix F.3.3. Theorem 5 states that, given equations (F.1) to (F.8), the assumption that z_{st} belongs to the information set of worker i of type s at period t , and the assumption that z_{rt} belongs to the information set of worker j of type r at period t , the moment in equation (F.13) is positive when evaluated at $(\theta_\alpha, \theta_\beta) = (\alpha, \beta)$. Furthermore, this is true regardless of the period t , the two locations l and l' we compare, the worker-type pairs (i, s) and (j, r) we consider, the vectors z_{st} and z_{rt} on which we condition, and scalar random variables $e_{ijsrt}^{ll'}$ we use to form the moment. We thus may compute the set of values of $(\theta_\alpha, \theta_\beta)$ that satisfy

$$\tilde{\mathbb{M}}^{ll'}(z_{st}, z_{rt}, \theta_\alpha, \theta_\beta) \geq 0, \quad (\text{F.14})$$

and, if equations (F.1) to (F.8) hold, $z_{st} \subseteq \mathcal{W}_{ist}$, $z_{rt} \subseteq \mathcal{W}_{jrt}$, and $e_{ijsrt}^{ll'} \subseteq \mathcal{J}_{ist} \cup \mathcal{J}_{jrt}$, (α, β) will belong to this set.

F.3 Proofs and Additional Details

F.3.1 Second-Step Bounding Inequalities: Proof of Theorem 4

Equation (F.1) implies that, for any worker i of type s , period t , and locations l and l' ,

$$(y_{ist}^l + y_{ist}^{l'}) (\mathbb{1}\{\mathbb{E}[\mathcal{V}_{ist}^l - \mathcal{V}_{ist}^{l'} | \mathcal{J}_{ist}] \geq 0\} - y_{ist}^l) = 0.$$

Equations (F.2a), (F.4), and (F.6) imply we can rewrite this equality as

$$(y_{ist}^l + y_{ist}^{l'}) (\mathbb{1}\{\mathbb{E}[\Delta v_{ist}^{ll'} | \mathcal{W}_{ist}] + \Delta \varepsilon_{ist}^{ll'} \geq 0\} - y_{ist}^l) = 0, \quad (\text{F.15})$$

where $\Delta v_{ist}^{ll'} = v_{ist}^l - v_{ist}^{l'}$ and $\Delta \varepsilon_{ist}^{ll'} = \varepsilon_{ist}^l - \varepsilon_{ist}^{l'}$. This equality holds for any worker i of any type s , any period t , and any two locations l and l' . Thus, computing a conditional expectation of both sides of this equality, we obtain

$$\mathbb{E}[\mathbb{1}\{\mathbb{E}[\Delta v_{ist}^{ll'} | \mathcal{W}_{ist}] + \Delta \varepsilon_{ist}^{ll'} \geq 0\} - y_{ist}^l | \mathcal{W}_{ist}, y_{ist}^l + y_{ist}^{l'} = 1] = 0.$$

Following steps analogous to those described in Appendix A.1, we can derive the following moment inequality

$$\mathbb{E}[y_{ist}^{l'} + y_{ist}^l \exp(-e_{ist}^{ll'})(-(1 + e_{ist}^{ll'}) + \Delta v_{ist}^{ll'}) | z_{st}] \geq 0. \quad (\text{F.16})$$

This moment differs from that in equation (12) only in that the difference in the static utility between alternatives l and l' entering equation (12) (i.e., $\Delta \theta_t^{ll'} + \alpha \Delta w_s^{ll'}$) is replaced by the difference in the corresponding choice-specific value functions (i.e., $\Delta v_{ist}^{ll'}$).

The inequality in equation (F.16) is not immediately useful for the identification of the parameters of the dynamic model described in Appendix F.1. The term $\Delta v_{ist}^{ll'}$ depends on the optimal choices of worker i of type s in every period $t' > t$ both conditional on choosing l at period t (which matters for the value of v_{ist}^l) and conditional on choosing l' at period t (which matters for the value of $v_{ist}^{l'}$). To derive a moment inequality that can be used to partially identify the parameters of the model described in Appendix F.1, we follow the approach in Morales et al. (2019). Specifically, we substitute $\Delta v_{ist}^{ll'}$ by the variable

$$\Delta \tilde{v}_{ist}^{ll'} \equiv v_{ist}^l - \tilde{v}_{ist}^{l'}, \quad (\text{F.17})$$

where $\tilde{v}_{ist}^{l'}$ is the discounted sum of static utilities from period t onwards (that is, in every period $t' \geq t$) if worker i of type s chooses location l' at period t but follows in every subsequent period $t' > t$ the path of choices that would be optimal if they had instead chosen location l at t . To define v_{ist}^l , $v_{ist}^{l'}$, and $\tilde{v}_{ist}^{l'}$, denote by

$$y_{ist'}^{(lt)} = (y_{ist'}^{(lt)1}, \dots, y_{ist'}^{(lt)L}) \quad (\text{F.18})$$

the choices of worker i of type s at t' if they were to choose alternative l at t . Then,

$$\begin{aligned} v_{ist}^l = & \kappa_{nt}^l + \alpha w_t^{sl} + \delta \sum_{l''=1}^L y_{ist+1}^{(lt)l''} (\kappa_{lt+1}^{l''} + \alpha w_{st+1}^{l''} + \varepsilon_{ist+1}^{l''}) \\ & + \sum_{t'>t+1} \delta^{t'-t} \sum_{n'=1}^L \sum_{l''=1}^L y_{ist'-1}^{(lt)n'} y_{ist'}^{(lt)l''} (\kappa_{n't'}^{l''} + \alpha w_{st'}^{l''} + \varepsilon_{ist'}^{l''}), \end{aligned} \quad (\text{F.19a})$$

$$v_{ist}^{l'} = \kappa_{nt}^{l'} + \alpha w_{st}^{l'} + \delta \sum_{l''=1}^L y_{ist+1}^{(l't)l''} (\kappa_{l't+1}^{l''} + \alpha w_{st+1}^{l''} + \varepsilon_{ist+1}^{l''}) \quad (\text{F.19b})$$

$$+ \sum_{t'>t+1} \delta^{t'-t} \sum_{n'=1}^L \sum_{l''=1}^L y_{ist'-1}^{(l't)n'} y_{ist'}^{(l't)l''} (\kappa_{n't'}^{l''} + \alpha w_{st'}^{l''} + \varepsilon_{ist'}^{l''}),$$

$$\tilde{v}_{ist}^{l'} = \kappa_{nt}^{l'} + \alpha w_{st}^{l'} + \delta \sum_{l''=1}^L y_{ist+1}^{(lt)l''} (\kappa_{l't+1}^{l''} + \alpha w_{st+1}^{l''} + \varepsilon_{ist+1}^{l''}) \quad (\text{F.19c})$$

$$+ \sum_{t'>t+1} \delta^{t'-t} \sum_{n'=1}^L \sum_{l''=1}^L y_{ist'-1}^{(lt)n'} y_{ist'}^{(lt)l''} (\kappa_{n't'}^{l''} + \alpha w_{st'}^{l''} + \varepsilon_{ist'}^{l''}).$$

Equations (F.1) and (F.5) imply that $\mathbb{E}[v_{ist}^{l'} | \mathcal{W}_{ist}] \geq \mathbb{E}[\tilde{v}_{ist}^{l'} | \mathcal{W}_{ist}]$, and, consequently,

$$\mathbb{E}[\Delta v_{ist}^{ll'} | \mathcal{W}_{ist}] \geq \mathbb{E}[\Delta \tilde{v}_{ist}^{ll'} | \mathcal{W}_{ist}]. \quad (\text{F.20})$$

Equations (F.16) and (F.20) imply the following moment inequality

$$\mathbb{E}[y_{ist}^{l'} + y_{ist}^l \exp(-h_{ist}^{ll'}(z_{st}, \Delta \lambda_t^{ll'}))(-1 + h_{ist}^{ll'}(z_{st}, \Delta \lambda_t^{ll'})) + \Delta \tilde{v}_{ist}^{ll'} | z_{st}] \geq 0. \quad (\text{F.21})$$

Comparing the expressions for v_{ist}^l and $\tilde{v}_{ist}^{l'}$ in equations (F.19a) and (F.19c), we can write

$$\begin{aligned} \Delta \tilde{v}_{ist}^{ll'} &= v_{ist}^l - \tilde{v}_{ist}^{l'} = (\kappa_{nt}^l - \kappa_{nt}^{l'}) + \alpha(w_{st}^l - w_{st}^{l'}) + \delta \sum_{l''=1}^L y_{ist+1}^{(lt)l''} (\kappa_{lt+1}^{l''} - \kappa_{l't+1}^{l''}) \\ &= \beta(x_{nt}^l - x_{nt}^{l'}) + (\lambda_t^l - \lambda_t^{l'}) + \alpha(w_{st}^l - w_{st}^{l'}) + \delta \beta \sum_{l''=1}^L y_{ist+1}^{(lt)l''} (x_{lt+1}^{l''} - x_{l't+1}^{l''}). \end{aligned} \quad (\text{F.22})$$

Combining equations (F.22) and (F.21), we obtain an inequality whose moment equals that in equation (F.9) for $\Delta \theta_t^{ll'} = \Delta \lambda_t^{ll'}$. Equations (F.21) and (F.22) thus imply Theorem 4. ■

F.3.2 Second-Step Bounding Inequalities: Additional Derivations

Derivation of optimal approximation points. Given $z_{st} \in \mathcal{Z}_{st}$, compute $e_{ist}^{ll'} = h_{ist}^{ll'}(z_{st}, \Delta \theta_t^{ll'})$ that minimizes the moment in equation (F.9) at each value of $\Delta \theta_t^{ll'}$. Given z_{st} and $\Delta \theta_t^{ll'}$, the first-order condition of the moment in equation (F.9) with respect to $h_{ist}^{ll'}(z_{st}, \Delta \theta_t^{ll'})$ is

$$\mathbb{E}[y_{ist}^l (h_{ist}^{ll'}(z_{st}, \Delta \theta_t^{ll'}) - \Delta \tilde{v}_{ist}^{ll'}) | z_{st}] = 0;$$

equivalently, $\mathbb{E}[h_{ist}^{ll'}(z_{st}, \Delta \theta_t^{ll'}) - \Delta \tilde{v}_{ist}^{ll'} | z_{st}, y_{ist}^l = 1] = 0$. Solving for $h_{ist}^{ll'}(z_{st}, \Delta \theta_t^{ll'})$, we obtain:

$$h_{ist}^{ll'}(z_{st}, \Delta \theta_t^{ll'}) = \mathbb{E}[\Delta \tilde{v}_{ist}^{ll'} | z_{st}, y_{ist}^l = 1], \quad (\text{F.23})$$

F.3.3 First-Step Moment Inequalities: Proof of Theorem 5

For choices l and l' , worker i of type s , worker j of type r , and period t , equation (F.15) implies

$$y_{jrt}'(y_{ist}^l + y_{ist}^{l'}) (\mathbb{1}\{\mathbb{E}[\Delta v_{ist}^{l'} | \mathcal{W}_{ist}] + \Delta \varepsilon_{ist}^{l'} \geq 0\} - y_{ist}^l) = 0. \quad (\text{F.24})$$

Taking the expectation of both sides of this equality conditional on $\mathcal{W}_{ist}$, $\mathcal{W}_{jrt}$, and a dummy variable that equals one if worker i of type s chooses either l or l' at t , we obtain

$$\mathbb{E}[y_{jrt}'(y_{ist}^l + y_{ist}^{l'}) (\mathbb{1}\{\mathbb{E}[\Delta v_{ist}^{l'} | \mathcal{W}_{ist}] + \Delta \varepsilon_{ist}^{l'} \geq 0\} - y_{ist}^l) | \mathcal{W}_{ist}, \mathcal{W}_{jrt}, y_{ist}^l + y_{ist}^{l'} = 1] = 0.$$

Given equations (F.1) and (F.8), we can rewrite this moment equality as

$$\mathbb{E}\left[y_{jrt}' \left(\frac{\exp(\mathbb{E}[\Delta v_{ist}^{l'} | \mathcal{W}_{ist}])}{1 + \exp(\mathbb{E}[\Delta v_{ist}^{l'} | \mathcal{W}_{ist}])} - y_{ist}^l \right) \middle| \mathcal{W}_{ist}, \mathcal{W}_{jrt}, y_{ist}^l + y_{ist}^{l'} = 1 \right] = 0,$$

or, equivalently, after multiplying by $1 + \exp(-\mathbb{E}[\Delta v_{ist}^{l'} | \mathcal{W}_{ist}])$, and rearranging,

$$\mathbb{E}[y_{jrt}'(1 - y_{ist}^l - y_{ist}^l \exp(-\mathbb{E}[\Delta v_{ist}^{l'} | \mathcal{W}_{ist}])) | \mathcal{W}_{ist}, \mathcal{W}_{jrt}, y_{ist}^l + y_{ist}^{l'} = 1] = 0.$$

Given that this expectation conditions on the event $y_{ist}^l + y_{ist}^{l'} = 1$, we can further rewrite

$$\mathbb{E}[y_{ist}^{l'} y_{jrt}' + y_{ist}^l y_{jrt}' (-\exp(-\mathbb{E}[\Delta v_{ist}^{l'} | \mathcal{W}_{ist}])) | \mathcal{W}_{ist}, \mathcal{W}_{jrt}] = 0.$$

As $-\exp(-x)$ is concave, we derive the following inequality given any scalar $e_{ijsrt}^{l'}$,

$$\mathbb{E}[y_{ist}^{l'} y_{jrt}' + y_{ist}^l y_{jrt}' \exp(-e_{ijsrt}^{l'}) (- (1 + e_{ijsrt}^{l'}) + \mathbb{E}[\Delta v_{ist}^{l'} | \mathcal{W}_{ist}]) | \mathcal{W}_{ist}, \mathcal{W}_{jrt}] \geq 0. \quad (\text{F.25})$$

Consider the alternative moment

$$\mathbb{E}[y_{ist}^{l'} y_{jrt}' + y_{ist}^l y_{jrt}' \exp(-e_{ijsrt}^{l'}) (- (1 + e_{ijsrt}^{l'}) + \Delta v_{ist}^{l'}) | \mathcal{W}_{ist}, \mathcal{W}_{jrt}]. \quad (\text{F.26})$$

Equation (F.6) implies $\nu_{ist}^{l'} = \Delta w_{ist}^{l'} - \mathbb{E}[\Delta w_{st}^{l'} | \mathcal{W}_{ist}, \mathcal{W}_{jrt}]$ and, thus, we can conclude that

$$\mathbb{E}[\nu_{ist}^{l'} | \mathcal{W}_{ist}, \mathcal{W}_{jrt}] = 0. \quad (\text{F.27})$$

As $\mathcal{W}_{ist} \subseteq \mathcal{J}_{ist}$ and $\mathcal{W}_{jrt} \subseteq \mathcal{J}_{jrt}$, the LIE allows us to write the moment in equation (F.26) as

$$\mathbb{E}[\mathbb{E}[y_{ist}^{l'} y_{jrt}' + y_{ist}^l y_{jrt}' \exp(-e_{ijsrt}^{l'}) (- (1 + e_{ijsrt}^{l'}) + \mathbb{E}[\Delta v_{ist}^{l'} | \mathcal{W}_{ist}] + \nu_{ist}^{l'}) | \mathcal{J}_{ist}, \mathcal{J}_{jrt}] | \mathcal{W}_{ist}, \mathcal{W}_{jrt}].$$

Equation (F.1) implies $\mathbb{E}[y_{ist}^l y_{jrt}^{l'} | \mathcal{J}_{ist}, \mathcal{J}_{jrt}] = y_{ist}^l y_{jrt}^{l'}$. Consequently, if $z_{st} \subseteq \mathcal{W}_{ist}$ and $z_{rt} \subseteq \mathcal{W}_{jrt}$, then $z_{st} \subseteq \mathcal{J}_{ist}$ and $z_{rt} \subseteq \mathcal{J}_{jrt}$, and we can rewrite the moment in equation (F.26) as

$$\mathbb{E}[y_{ist}^{l'} y_{jrt}^{l'} + y_{ist}^l y_{jrt}^{l'} \exp(-e_{ijsrt}^{ll'})(-(1 + e_{ijsrt}^{ll'}) + \mathbb{E}[\mathbb{E}[\Delta v_{ist}^{ll'} | \mathcal{W}_{ist}] + \nu_{ist}^{ll'} | \mathcal{W}_{ist}, \mathcal{W}_{jrt}]) | \mathcal{W}_{ist}, \mathcal{W}_{jrt}],$$

and equation (F.27) implies we can rewrite the moment in equation (F.26) as

$$\mathbb{E}[y_{ist}^{l'} y_{jrt}^{l'} + y_{ist}^l y_{jrt}^{l'} \exp(-e_{ijsrt}^{ll'})(-(1 + e_{ijsrt}^{ll'}) + \mathbb{E}[\Delta v_{ist}^{ll'} | \mathcal{W}_{ist}]) | \mathcal{W}_{ist}, \mathcal{W}_{jrt}].$$

This moment is the same as that in equation (F.25); hence, we have shown:

$$\mathbb{E}[y_{ist}^{l'} y_{jrt}^{l'} + y_{ist}^l y_{jrt}^{l'} \exp(-e_{ijsrt}^{ll'})(-(1 + e_{ijsrt}^{ll'}) + \Delta v_{ist}^{ll'}) | \mathcal{W}_{ist}, \mathcal{W}_{jrt}] \geq 0. \quad (\text{F.28})$$

Following steps analogous to those we follow to derive the inequality in equation (F.21) from that in equation (F.16) (see Appendix F.3.1), we derive the following inequality from that in equation (F.28), where $\Delta \tilde{v}_{ist}^{ll'}$ is defined as in equation (F.22):

$$\mathbb{E}[y_{ist}^{l'} y_{jrt}^{l'} + y_{ist}^l y_{jrt}^{l'} \exp(-e_{ijsrt}^{ll'})(-(1 + e_{ijsrt}^{ll'}) + \Delta \tilde{v}_{ist}^{ll'}) | \mathcal{W}_{ist}, \mathcal{W}_{jrt}] \geq 0. \quad (\text{F.29})$$

This inequality is one of the two we combine to obtain the inequality that we use to bound the parameters θ_α and θ_β . To obtain the other one, we start with the following expression

$$y_{ist}^l (y_{jrt}^l + y_{jrt}^{l'}) (\mathbb{1}\{\mathbb{E}[\Delta v_{jrt}^{ll'} | \mathcal{W}_{jrt}] + \Delta \varepsilon_{jrt}^{ll'} \geq 0\} - y_{jrt}^{l'}) = 0. \quad (\text{F.30})$$

Following the same steps we use to go from equation (F.24) to (F.29), we derive from equation (F.30) the following inequality

$$\mathbb{E}[y_{ist}^{l'} y_{jrt}^{l'} + y_{ist}^l y_{jrt}^{l'} \exp(-e_{ijsrt}^{ll'})(-(1 + e_{ijsrt}^{ll'}) + \Delta \tilde{v}_{jrt}^{ll'}) | \mathcal{W}_{ist}, \mathcal{W}_{jrt}] \geq 0. \quad (\text{F.31})$$

As the moments in equations (F.28) and (F.31) share the same function $g_{ijsrt}^{ll'}: \mathcal{Z}_{st} \times \mathcal{Z}_{rt} \times \Theta_{(\alpha, \beta)} \rightarrow \mathbb{R}$ and the same conditioning set, we add them to obtain:

$$\mathbb{E}[y_{ist}^l y_{jrt}^l + y_{ist}^{l'} y_{jrt}^{l'} - y_{ist}^l y_{jrt}^{l'} \exp(-e_{ijsrt}^{ll'}) (2 + 2e_{ijsrt}^{ll'} - (\Delta \tilde{v}_{ist}^{ll'} + \Delta \tilde{v}_{jrt}^{ll'})) | \mathcal{W}_{ist}, \mathcal{W}_{jrt}] \geq 0,$$

with

$$\Delta \tilde{v}_{ist}^{ll'} = \beta(x_{nt}^l - x_{nt}^{l'}) + (\lambda_t^l - \lambda_t^{l'}) + \alpha(w_{st}^l - w_{st}^{l'}) + \delta\beta \sum_{l''=1}^L y_{ist+1}^{(lt)l''} (x_{lt+1}^{l''} - x_{l't+1}^{l''}),$$

$$\Delta \tilde{v}_{jrt}^{l'l} = \beta(x_{nt}^{l'} - x_{nt}^l) + (\lambda_t^{l'} - \lambda_t^l) + \alpha(w_{rt}^{l'} - w_{rt}^l) + \delta\beta \sum_{l''=1}^L y_{ist+1}^{(lt)l''} (x_{l't+1}^{l''} - x_{lt+1}^{l''}).$$

Finally, if $z_{st} \subseteq \mathcal{W}_{ist}$ and $z_{rt} \subseteq \mathcal{W}_{jrt}$, we can use the LIE and conclude that

$$\mathbb{E}[y_{ist}^l y_{jrt}^l + y_{ist}^{l'} y_{jrt}^{l'} - y_{ist}^l y_{jrt}^{l'} \exp(-e_{ijsrt}^{ll'}) (2 + 2e_{ijsrt}^{ll'} - (\Delta \tilde{v}_{ist}^{ll'} + \Delta \tilde{v}_{jrt}^{l'l})) | z_{st}, z_{rt}] \geq 0, \quad (\text{F.33})$$

with

$$\begin{aligned} \Delta \tilde{v}_{ist}^{ll'} + \Delta \tilde{v}_{jrt}^{l'l} = & \alpha(w_{ist}^l - w_{ist}^{l'} + w_{jrt}^{l'} - w_{jrt}^l) \\ & + \delta\beta \sum_{l''=1}^L y_{ist+1}^{(lt)l''} (x_{lt+1}^{l''} - x_{l't+1}^{l''} + x_{l't+1}^{l''} - x_{lt+1}^{l''}). \end{aligned} \quad (\text{F.34})$$

Plugging equation (F.34) into equation (F.33), we obtain a moment inequality whose moment equals that in equation (F.13) when evaluated at $(\theta_\alpha, \theta_\beta) = (\alpha, \beta)$. Equations (F.33) and (F.34) thus imply Theorem 5. $\blacksquare$

F.3.4 First-Step Moment Inequalities: Additional Derivations

Derivation of optimal approximation points. Consider approximation points expressed as $e_{ijsrt}^{ll'} = g_{ijsrt}^{ll'}(z_{st}, z_{rt}, \theta_\alpha, \theta_\beta)$. We find the value of $g_{ijsrt}^{ll'}(z_{st}, z_{rt}, \theta_\alpha, \theta_\beta)$ that, given $z_{st} \in \mathcal{Z}_{st}$ and $z_{rt} \in \mathcal{Z}_{rt}$, minimizes the moment in equation (F.33) at each value of $(\theta_\alpha, \theta_\beta)$. Specifically, given z_{st} , z_{rt} , θ_α , and θ_β , the first-order condition of the moment in equation (F.33) with respect to the scalar $g_{ijsrt}^{ll'}(z_{st}, z_{rt}, \theta_\alpha, \theta_\beta)$ is

$$\mathbb{E}[y_{ist}^l y_{jrt}^{l'} (2g_{ijsrt}^{ll'}(z_{st}, z_{rt}, \theta_\alpha, \theta_\beta) - (\Delta \tilde{v}_{ist}^{ll'} + \Delta \tilde{v}_{jrt}^{l'l})) | z_s, z_r] = 0,$$

or, equivalently, $\mathbb{E}[2g_{ijsrt}^{ll'}(z_{st}, z_{rt}, \theta_\alpha, \theta_\beta) - (\Delta \tilde{v}_{ist}^{ll'} + \Delta \tilde{v}_{jrt}^{l'l}) | z_{st}, z_{rt}, y_{ist}^l y_{jrt}^{l'} = 1] = 0$. Solving for $g_{ijsrt}^{ll'}(z_{st}, z_{rt}, \theta_\alpha, \theta_\beta)$, we obtain:

$$g_{ijsrt}^{ll'}(z_{st}, z_{rt}, \theta_\alpha, \theta_\beta) = \mathbb{E}[\Delta \tilde{v}_{ist}^{ll'} + \Delta \tilde{v}_{jrt}^{l'l} | z_{st}, y_{ist}^l = 1], \quad (\text{F.35})$$

with $\Delta \tilde{v}_{ist}^{ll'} + \Delta \tilde{v}_{jrt}^{l'l}$ defined as in equation (F.34).

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